STAR RX72N - rx_comm_manager Library

1.0.0

Generated by Doxygen 1.16.1

Chapter 1

Directory Hierarchy

1.1 Directories

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2.1 File List

Here is a list of all files with brief descriptions:

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Directory Documentation

3.1 libs/rx_comm_manager/inc Directory Reference

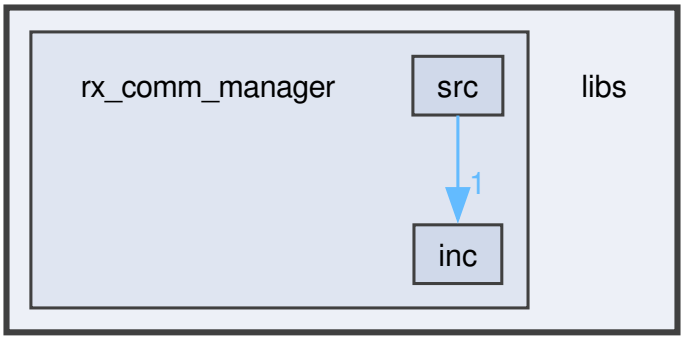
Directory dependency graph for inc:

Files

- file rx_comm_manager.h
  Unified Multi-Channel Communication Manager for USB CDC and SPI Protocols.

3.2 libs Directory Reference

Directory dependency graph for libs:

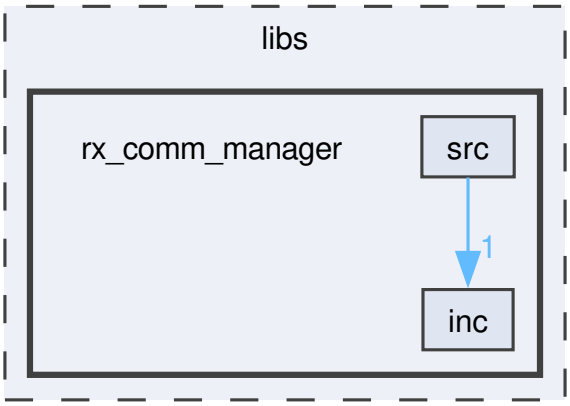


Directories

- directory rx_comm_manager

3.3 libs/rx_comm_manager Directory Reference

Directory dependency graph for rx_comm_manager:

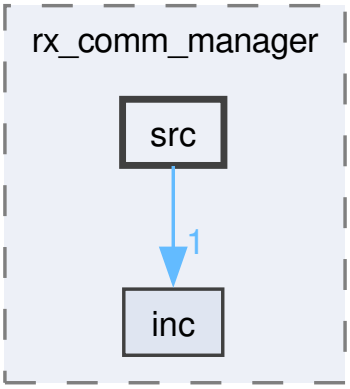


Directories

- directory inc
- directory src

3.4 libs/rx_comm_manager/src Directory Reference

Directory dependency graph for src:



Files

- file rx_comm_manager.c
  Unified Multi-Channel Communication Manager Implementation.

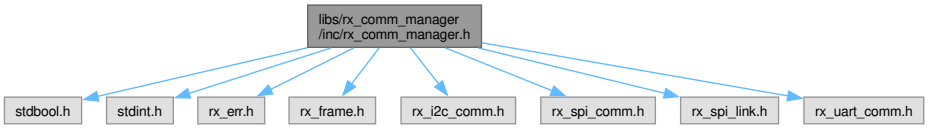
Chapter 4

File Documentation

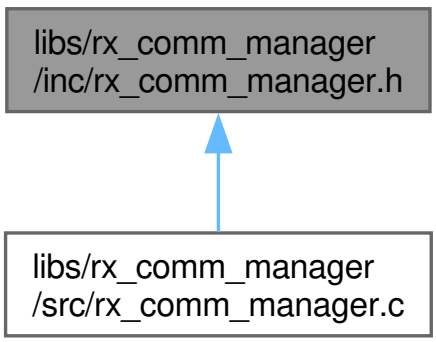
4.1 libs/rx_comm_manager/inc/rx_comm_manager.h File Reference

Unified Multi-Channel Communication Manager for USB CDC and SPI Protocols.

#include <stdbool.h>
#include <stdint.h>
#include "rx_err.h"
#include "rx_frame.h"
#include "rx_i2c_comm.h"
#include "rx_spi_comm.h"
#include "rx_spi_link.h"
#include "rx_uart_comm.h"
Include dependency graph for rx_comm_manager.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct rx_comm_event_entry_t

  Single entry in the static event FIFO queue.
- struct rx_comm_heartbeat_state_t

  Per-transport heartbeat tracking state.
- struct rx_comm_manager_config_t

  Communication manager configuration.
- struct rx_comm_manager_t

  Communication manager handle.
- struct rx_comm_send_params_t

  Parameters for rx_comm_manager_send.

Typedefs

- typedef void(* rx_comm_frame_callback_t) (rx_comm_channel_t channel, const rx_frame_t *frame, void *ctx)

  Frame received callback function type.
- typedef void(* rx_comm_link_status_callback_t) (rx_comm_channel_t channel, rx_comm_link_status_t new_status, void *ctx)

  Callback invoked when a link status changes.

Enumerations

- enum rx_comm_manager_constants_t : uint16_t { k_comm_manager_ascii_buf_size = 2048 , k_comm_event_queue_depth = 8 }

  Communication manager compile-time constants.
- enum rx_comm_heartbeat_constants_t : uint32_t { k_heartbeat_implicit_timeout_ms = 200 , k_heartbeat_ping_interval_ms = 1000 , k_heartbeat_check_interval_ms = 50 }

  Heartbeat timing constants.
- enum rx_comm_event_retry_constants_t : uint16_t { k_event_max_retries = 3 , k_event_initial_backoff_ms = 10 , k_event_max_backoff_ms = 15 }

  Event queue retry constants for SPI HARQ.
- enum rx_comm_channel_t : uint8_t { k_comm_channel_uart = 0 , k_comm_channel_spi = 1 , k_comm_channel_i2c = 2 , k_comm_channel_count = 3 }

  Communication channel identifiers.
- enum rx_comm_link_status_t : uint8_t { k_link_status_unknown = 0 , k_link_status_healthy = 1 , k_link_status_dead = 2 }

  Link health status from heartbeat monitoring.
- enum rx_comm_channel_mask_t : uint8_t { k_comm_channel_mask_none = 0x00U , k_comm_channel_mask_uart = 0x01U , k_comm_channel_mask_spi = 0x02U , k_comm_channel_mask_i2c = 0x04U , k_comm_channel_mask_all }

  Bitmask selecting which comm channels to initialize and use.

Functions

- rx_err_t rx_comm_manager_init (rx_comm_manager_t *mgr, const rx_comm_manager_config_t *cfg)

  Initialize communication manager.
- rx_err_t rx_comm_manager_deinit (rx_comm_manager_t *mgr)

  Deinitialize communication manager.
- rx_err_t rx_comm_manager_poll (rx_comm_manager_t *mgr)

  Poll all enabled channels for incoming frames.
- rx_err_t rx_comm_manager_send (rx_comm_manager_t *mgr, const rx_comm_send_params_t *params)

  Send frame on specific channel.
- rx_err_t rx_comm_manager_respond (rx_comm_manager_t *mgr, rx_comm_channel_t channel, const uint8_t *payload, uint32_t payload_len)

  Send response on same channel as received frame.
- rx_err_t rx_comm_manager_stream_send (rx_comm_manager_t *mgr, const rx_comm_send_params_t *params)

  Send data on the stream path (fire-and-forget).
- rx_err_t rx_comm_manager_event_send (rx_comm_manager_t *mgr, const rx_comm_send_params_t *params)

  Queue data on the event path (reliable, SPI retry).
- rx_err_t rx_comm_manager_link_status (const rx_comm_manager_t *mgr, rx_comm_channel_t channel, rx_comm_link_status_t *status)

  Query link health status for a transport channel.
- rx_err_t rx_comm_manager_channel_ready (rx_comm_manager_t *mgr, rx_comm_channel_t channel, bool *ready)

  Check if channel is ready for communication.
- const char * rx_comm_manager_channel_name (rx_comm_channel_t channel)

  Get channel name string.
- rx_err_t rx_comm_manager_process_retransmits (rx_comm_manager_t *mgr, uint32_t current_time_ms)

  Process pending retransmissions on SPI channel.
- rx_err_t rx_comm_manager_set_auto_retransmit (rx_comm_manager_t *mgr, bool enabled, const rx_spi_comm_retransmit_config_t *config)

  Enable or disable automatic retransmission on SPI channel.

4.1.1 Detailed Description

Unified Multi-Channel Communication Manager for USB CDC and SPI Protocols.

4.1.1.1 Overview

Production-quality communication manager providing unified abstraction over multiple simultaneous communication channels (USB CDC and SPI) with automatic frame protocol handling, non-blocking polling architecture, optional decoded ASCII output for debugging, and callback-based event notification.

Purpose: Coordinate independent, simultaneous communication channels from RX72N MCU to Raspberry Pi 5 host, enabling command/response messaging over both USB (via CDC class) and SPI (direct hardware connection).

Protocol: Both channels use identical frame protocol (rx_frame) allowing host to send commands on either channel. Manager automatically handles:

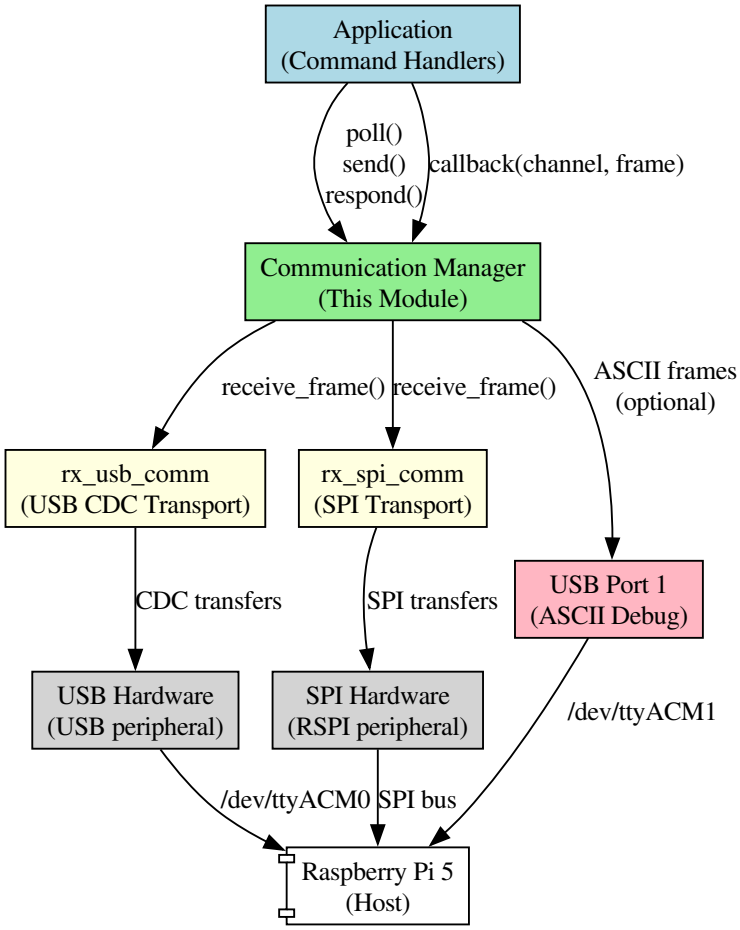
- Frame reception and validation
- ASCII decoding output (debugging via USB Port 1)
- Channel-aware response routing
- User callback invocation

Key Features:

- Multi-channel support: Simultaneous USB and SPI communication
- Non-blocking architecture: Poll-based, no busy-waiting or blocking calls
- Protocol-agnostic: Both channels use same frame format (type, flags, payload, CRC)
- Automatic ASCII output: Optional decoded frames to USB Port 1 (human-readable debugging)
- Callback-driven: User-defined handler invoked on frame reception
- Channel awareness: Responses automatically routed to originating channel
- Zero-copy where possible: Direct access to frame buffers
- Thread-safe design: Can be called from RTOS task or main loop

4.1.1.2 System Architecture

Communication manager sits between application logic and low-level transport drivers:



4.1.1.3 Channel Configuration

The manager supports two independent communication channels:

Channel	Transport	Device	Usage	RPi5 Interface
USB	CDC class	USB Port 0	Primary protocol channel	/dev/ttyACM0
SPI	RSPI peripheral	RSPI channel	High-speed backup channel	/dev/spidev0.0
Debug	CDC class	USB Port 1	ASCII decoded output (optional)	/dev/ttyACM1

USB Channel (k_comm_channel_usb):

- Plug-and-play connectivity (no additional wiring)
- Full-speed USB 2.0 (up to 12 Mbps theoretical, ∼1 MB/s practical)
- Automatic enumeration and driver loading on RPi5
- Suitable for command/response and moderate-bandwidth telemetry

SPI Channel (k_comm_channel_spi):

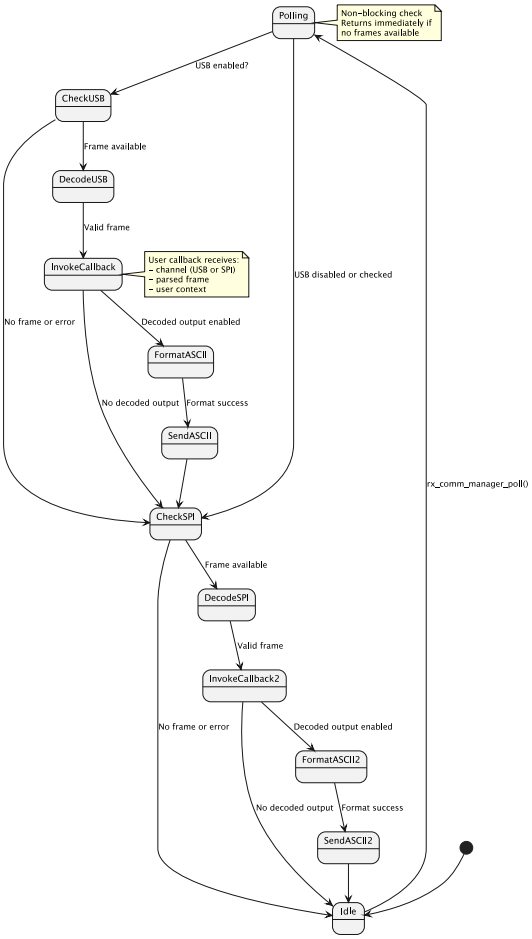
- Dedicated hardware connection (4-6 wires: COPI, CIPO, SCK, CS, optional handshake)
- Configurable speed (typically 10-20 Mbps)
- Lower latency than USB (no protocol overhead)
- Suitable for high-frequency control loops and real-time data

Debug Output (USB Port 1):

- Human-readable ASCII representation of all frames
- Format: [USB/SPI] TYPE=cmd FLAGS=0x00 LEN=42 DATA=...
- Enabled via enable_decoded_output config flag
- Zero impact on protocol channels (independent USB endpoint)

4.1.1.4 Communication Flow State Machine

The manager operates as a stateless polling architecture with per-frame state:



State Descriptions:

- Idle: Manager initialized, waiting for poll() call
- Polling: Actively checking enabled channels for frames
- CheckUSB: Query USB channel for available frame
- CheckSPI: Query SPI channel for available frame
- DecodeUSB/DecodeSPI: Validate and parse received frame
- InvokeCallback: Call user-provided frame handler
- FormatASCII: Convert frame to human-readable string
- SendASCII: Transmit ASCII representation to debug port

4.1.1.5 Hardware Requirements

Resource	Requirement	Usage
USB Peripheral	1 instance (full-speed)	CDC class with 2 ports (protocol + debug)
RSPI Peripheral	1 channel (optional)	SPI controller mode, 10-20 Mbps
GPIO	4-6 pins (if SPI)	COPI, CIPO, SCK, CS, optional handshake/interrupt
RAM	~2.5 KB per instance	Handle + ASCII buffer (2048 bytes)
Flash	~4 KB	Code + string constants

Resource	Requirement	Usage
RTOS Resources	None	Stateless, no threads/mutexes/semaphores

4.1.1.6 Performance Characteristics

Operation	Execution Time	Notes
rx_comm_manager_init()	~50 us	Handle initialization, no hardware config
rx_comm_manager_poll()	~15-200 us	Depends on channels enabled and frames available
- No frames available	~15 us	Quick check both channels, early return
- USB frame received	~120 us	Frame validation + callback + ASCII (if enabled)
- SPI frame received	~80 us	Faster than USB (less overhead)
- Both frames received	~200 us	Sequential processing of both channels
rx_comm_manager_send()	~100-150 us	Frame encoding + transport layer send
rx_comm_manager_respond()	~100-150 us	Same as send (convenience wrapper)
Callback overhead	~5-50 us	User-defined, depends on handler complexity

All timing measured @ 240 MHz CPU clock, -O2 optimization, typical frame size 64 bytes.

4.1.1.7 Memory Usage

Allocation	Size	Lifetime
rx_comm_manager_t	~12 KB	Application-managed (typically static/global)
- Pointers + state	~80 bytes	Handle metadata, heartbeat, queue indices
- ASCII buffer	2048 bytes	Decoded output formatting (if enabled)
- Event queue	8 x ~1028 bytes	Static FIFO (k_comm_event_queue_depth entries)
- Heartbeat state	~20 bytes	Per-transport heartbeat tracking
Stack (poll)	~120 bytes	Function call duration only
Stack (send)	~80 bytes	Function call duration only
Static data (.rodata)	~400 bytes	Channel names, format strings
Code (.text)	~6 KB	All functions

Total RAM: ~12 KB per manager instance (zero dynamic allocation). The event queue dominates memory: 8 entries x (k_frame_max_payload + metadata).

Memory Layout (rx_comm_manager_t structure):

Offset	Size	Field	Notes
0x00	8	usb_handle	Pointer to USB transport
0x08	8	spi_handle	Pointer to SPI transport

Offset	Size	Field	Notes
0x10	8	callback	Function pointer
0x18	8	callback_ctx	User context pointer
0x20	2048	ascii_buffer	ASCII formatting buffer
-	1	enable_decoded_output	Boolean flag
-	1	initialized	Boolean flag
-	~20	heartbeat[2]	Per-transport heartbeat state
-	~16	link_status_cb/ctx	Callback + context
-	~8224	event_queue[8]	8 x rx_comm_event_entry_t
-	3	queue head/tail/count	Queue management
Total	**~10.5 KB**		(actual size may vary by compiler)

4.1.1.8 Thread Safety Analysis

Conditional Thread Safety - Safe if following rules are observed:

1. Single poll context: Only call poll() from ONE thread/task/loop
2. Callback restrictions: User callback must not call manager functions (no recursion)
3. Send synchronization: If calling send()/respond() from multiple threads, use external mutex
4. Transport thread safety: Underlying USB/SPI transports must be thread-safe (typically are)

Not safe for:

- Concurrent poll() calls from multiple threads
- Calling manager functions from within frame callback
- Simultaneous send() calls without external synchronization

Recommendation: Use dedicated communication task:

```
void comm_task_entry(ULONG ctx) {
    rx_comm_manager_t* mgr = (rx_comm_manager_t*)ctx;
    while (1) {
        rx_comm_manager_poll(mgr);
        tx_thread_sleep(1);  // 1ms polling period = 1kHz
    }
}
```

4.1.1.9 Integration Example

Complete Setup and Usage:

```
#include "rx_comm_manager.h"
#include "rx_usb_comm.h"
#include "rx_spi_comm.h"

// Frame handler callback
void on_frame_received(rx_comm_channel_t channel, const rx_frame_t* frame, void* ctx) {
    rx_comm_manager_t* mgr = (rx_comm_manager_t*)ctx;

    rx_log_info("COMM", "Frame from %s: type=0x%02X len=%u",
                rx_comm_manager_channel_name(channel),
                frame->type, frame->payload_len);

    // Parse frame based on type
    switch (frame->type) {
        case k_frame_type_command:
            // Handle command
            uint8_t response[128];
            uint32_t response_len = process_command(frame->payload,
                                                    frame->payload_len,
                                                    response, sizeof(response));

            // Respond on same channel
            rx_comm_manager_respond(mgr, channel, response, response_len);
            break;

        case k_frame_type_telemetry_req:
            // Send telemetry data
            send_telemetry(mgr, channel);
            break;

        default:
            rx_log_warn("COMM", "Unknown frame type: 0x%02X", frame->type);
            break;
    }
}

// Main application
int main(void) {
    // Initialize transports
    rx_usb_comm_handle_t usb_comm;
    rx_spi_comm_handle_t spi_comm;

    rx_usb_comm_init(&usb_comm, &usb_config);
    rx_spi_comm_init(&spi_comm, &spi_config);

    // Initialize communication manager
    rx_comm_manager_t comm_mgr;
    rx_comm_manager_config_t cfg = {
        .usb_handle = &usb_comm,
        .spi_handle = &spi_comm,
        .callback = on_frame_received,
        .callback_ctx = &comm_mgr,  // Pass manager to callback
        .enable_decoded_output = true,  // ASCII debug on Port 1
    };

    rx_err_t err = rx_comm_manager_init(&comm_mgr, &cfg);
    if (err != k_rx_ok) {
        rx_log_error("APP", "Comm manager init failed: 0x%X", err);
        return -1;
    }

    // Main communication loop
    while (1) {
        // Poll for incoming frames (non-blocking)
        err = rx_comm_manager_poll(&comm_mgr);

        // err == k_rx_ok: Frame(s) received and processed
        // err == k_rx_err_timeout: No frames available (normal)

        // Other application tasks...

        tx_thread_sleep(1);  // 1ms = 1kHz polling rate
    }

    return 0;
}
```

Sending Unsolicited Data (Telemetry):

```
void send_periodic_telemetry(rx_comm_manager_t* mgr) {
    // Gather telemetry data
    telemetry_data_t telemetry;
```

```
    gather_telemetry(&telemetry);

    // Serialize to buffer
    uint8_t payload[256];
    uint32_t payload_len = serialize_telemetry(&telemetry, payload, sizeof(payload));

    // Send on USB channel (more reliable for telemetry)
    rx_comm_send_params_t params = {
        .channel = k_comm_channel_usb,
        .type = k_frame_type_telemetry,
        .flags = 0,
        .payload = payload,
        .payload_len = payload_len,
    };

    rx_err_t err = rx_comm_manager_send(mgr, &params);
    if (err != k_rx_ok) {
        rx_log_warn("TELEM", "Failed to send telemetry: 0x%X", err);
    }
}
```

Module Dependencies:

- rx_usb_comm: USB CDC transport layer
- rx_spi_comm: SPI transport layer
- rx_frame: Frame protocol definition and encoding/decoding
- rx_err: Standardized error code system
- rx_log: Logging infrastructure

See also

rx_spi_comm.h SPI transport implementation

rx_uart_comm.h UART transport implementation

rx_i2c_comm.h I2C peripheral transport implementation

rx_frame.h Frame protocol specification

docs/sections/01_nanopb_protocol.tex Communication protocol documentation

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp/longjmp, or recursion used
- Rule 2: [OK] All loops have compile-time bounded iterations (channel count = 2)
- Rule 3: [OK] Zero dynamic memory allocation (all buffers static)
- Rule 4: [OK] All functions < 60 lines
- Rule 5: [OK] Minimum 2 assertions per function via parameter validation
- Rule 6: [OK] Variables declared at smallest scope
- Rule 7: [OK] All return values checked and propagated
- Rule 8: [OK] Uses C23 typed enums exclusively (no magic numbers)
- Rule 9: [WARN] INTENTIONAL DEVIATION: Uses function pointers for callback (event-driven architecture)
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror (zero warnings)

SOLID Principles:

- S (Single Responsibility): Only channel coordination and frame routing. Protocol parsing delegated to rx_frame, transport to rx_usb_comm/rx_spi_comm.
- O (Open/Closed): Extensible to new channels by adding enum value and transport handle. Closed for modification (no changes needed for new frame types).
- L (Liskov Substitution): USB and SPI transports interchangeable via common frame interface. Can substitute mock transports for testing.
- I (Interface Segregation): Clean separation: lifecycle (init/deinit), polling (poll), sending (send/respond), status (channel_ready). Users include only what they need.
- D (Dependency Inversion): Depends on abstractions (rx_frame protocol, generic transport interface) not concrete USB/SPI implementations.

Warning

Do NOT call manager functions from within frame callback (recursion risk)

ASCII buffer is 2048 bytes - ensure payload_len + overhead < 2000 bytes to avoid truncation

Polling from multiple threads without synchronization will cause corruption

Attention

Manager does not own USB/SPI handles - caller must initialize transports before manager

Callback is invoked from poll() context - keep processing minimal or defer to task queue

Decoded ASCII output adds ~80 us latency per frame - disable for high-frequency applications

Author

Locked, Inc.

Date

2026-01-27

Version

1.0.0

Copyright

Changelog:

- 1.0.0 (2026-01-26): Initial implementation with USB and SPI channels
- 1.1.0 (2026-01-27): Added comprehensive Doxygen documentation, state machine diagrams, performance data

Definition in file rx_comm_manager.h.

4.1.2 Data Structure Documentation

4.1.2.1 struct rx_comm_event_entry_t

Single entry in the static event FIFO queue.

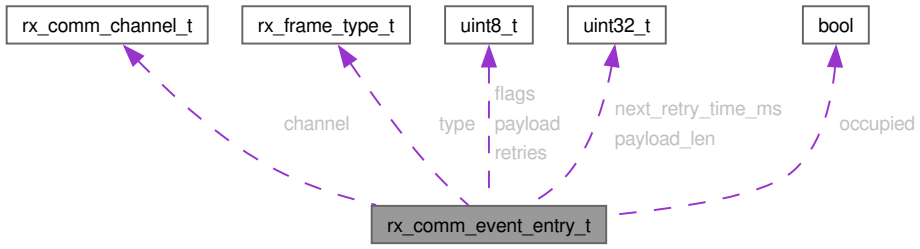
Events (commands) that require reliable delivery are queued here. Each entry contains a copy of the payload and metadata needed for retry logic. SPI entries retry with exponential backoff; USB entries are fire-and-forget.

Since

Version 1.0.0

Definition at line 713 of file rx_comm_manager.h.

Collaboration diagram for rx_comm_event_entry_t:



Data Fields

rx_comm_channel_t	channel	Target transport
uint8_t	flags	Frame flags
uint32_t	next_retry_time_ms	Earliest time (ms) for next retry
bool	occupied	Slot in use
uint8_t	payload[k_frame_max_payload]	Payload copy
uint32_t	payload_len	Payload length (bytes)
uint8_t	retries	Retry attempts so far
rx_frame_type_t	type	Frame type

4.1.2.2 struct rx_comm_heartbeat_state_t

Per-transport heartbeat tracking state.

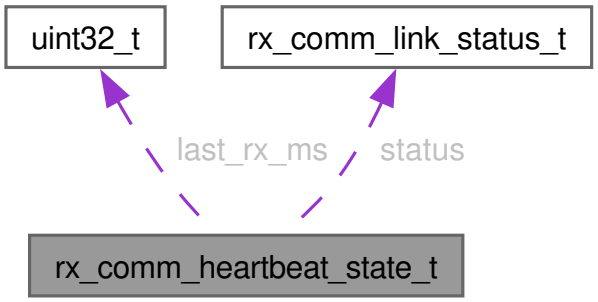
Tracks the last time a valid frame was received on each transport. Used by the poll loop to evaluate link health per the dual-detection heartbeat model.

Since

Version 1.0.0

Definition at line 735 of file rx_comm_manager.h.

Collaboration diagram for rx_comm_heartbeat_state_t:



Data Fields

uint32_t	last_rx_ms	Timestamp (ms) of last valid frame
rx_comm_link_status_t	status	Current link health

4.1.2.3 struct rx_comm_manager_config_t

Communication manager configuration.

Configuration structure passed to rx_comm_manager_init(). Defines which channels are enabled, callback function, and optional features.

Configuration Guidelines:

- At least one channel (usb_handle OR spi_handle) should be non-NULL
- Both channels can be enabled simultaneously
- Callback is optional (NULL = receive but don't notify)
- Decoded output recommended for development, disable for production

Memory Lifetime:

- Config structure only read during init, can be stack-allocated
- Transport handles must remain valid for manager lifetime
- Callback function must remain valid (typically static function)

Configuration Example:

```
rx_comm_manager_config_t cfg = {
    .usb_handle = &usb_comm,              // Enable USB channel
    .spi_handle = &spi_comm,              // Enable SPI channel
    .callback = on_frame_received,        // Frame handler
    .callback_ctx = &app_state,           // User context
    .enable_decoded_output = true,        // ASCII debug output
};
```

See also

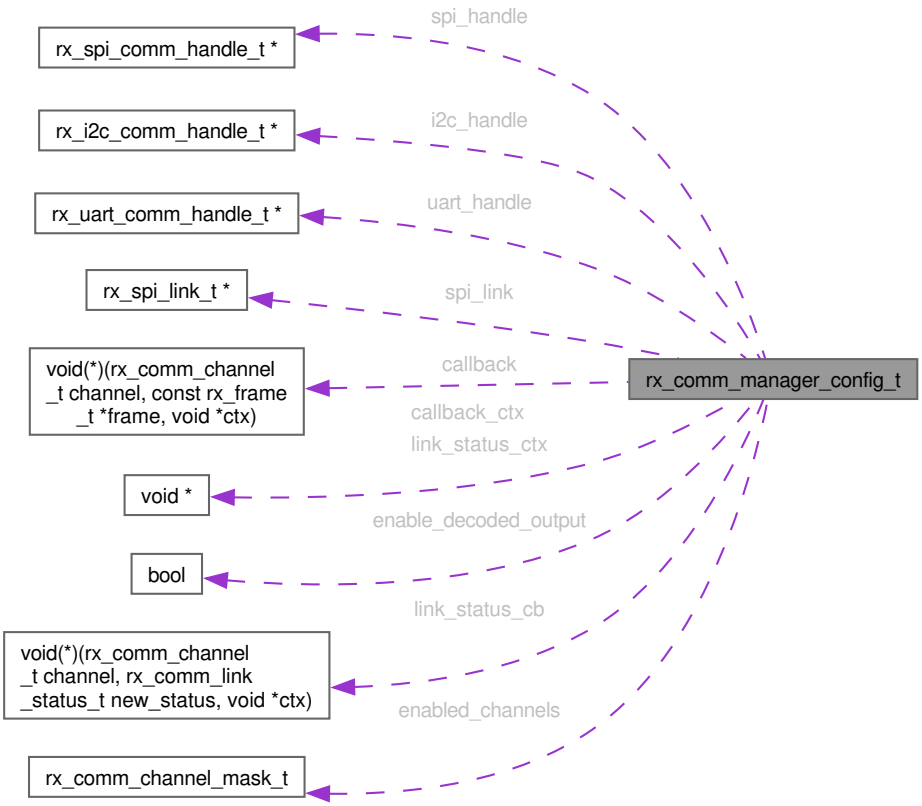
rx_comm_manager_init() Initialization function

Since

Version 1.0.0

Definition at line 813 of file rx_comm_manager.h.

Collaboration diagram for rx_comm_manager_config_t:



Data Fields

rx_comm_frame_callback_t	callback	Frame received callback function (NULL for no callback). User-defined function invoked when complete frame received on any channel. If NULL, frames are received but no notification occurs (not typical). Null handling: • NULL = Receive frames but don't notify (unusual, for testing only) • Non-NULL = Invoke callback on frame reception (normal operation) Type: rx_comm_frame_callback_t (function pointe Valid values: Non-NULL function pointer, or nullptr Signature: void callback(rx_comm_channel_t, cons See also rx_comm_frame_callback_t Callback function type definition

void *	callback_ctx	User context pointer passed to callback. Arbitrary pointer passed to callback as third parameter. Typically used to pass application state, manager pointer, or other context without requiring global variables. Common patterns: • Pointer to manager itself (enables response from callback) • Pointer to application state structure • NULL if no context needed Type: void* (application-defined) Valid values: Any pointer value, or nullptr Lifetime: Must remain valid if callback uses it

bool	enable_decoded_output	Enable decoded ASCII output to USB Port 1. When true, all received and sent frames are converted to human-readable ASCII format and sent to USB Port 1 (/dev/ttyACM1 on RPi5). ASCII Format: [USB] TYPE=cmd FLAGS=0x00 LEN=42 DATA=48656C6C6F... [SPI] TYPE=resp FLAGS=0x01 LEN=128 DATA=... Performance Impact: • Adds ∼80 us latency per frame (formatting + USB send) • Negligible for command/response (< 100 Hz) • May impact high-frequency telemetry (> 1 kHz) Use Cases: • Development: Enable for debugging and protocol analysis • Production: Disable for maximum performance Type: bool Value: true = enable ASCII output, false = disable Default: Recommend true for development, false for

rx_comm_channel_mask_t	enabled_channels	Bitmask of channels to initialize and poll (default: all channels). When a bit is clear the corresponding transport is skipped during internal_init_transports() and its handle is left NULL. Set to k_comm_channel_mask_all to attempt all four transports (legacy behaviour). Use comm_task_apply_system_config() to set this alongside the log backend so that SCI9 conflicts between UART comm and UART log are caught early. Valid values: Any combination of rx_comm_channe Default: k_comm_channel_mask_all (attempt all c See also rx_comm_channel_mask_t Bit definitions Since Version 1.0.0
rx_i2c_comm_handle_t *	i2c_handle	I2C communication handle (NULL to disable I2C channel). Pointer to initialized I2C peripheral transport handle. If NULL, I2C channel is disabled and all I2C operations are skipped. The RX72N acts as I2C peripheral (device) and the RPi5 acts as I2C controller. Type: rx_i2c_comm_handle_t* (pointer to I2C tra Valid values: Non-NULL initialized handle, or nullpt Lifetime: Must outlive manager instance
rx_comm_link_status_callback_t	link_status_cb	Callback invoked when a transport link status changes. Optional. Called when heartbeat monitoring detects a link transition (e.g., healthy -> dead). NULL to disable link status notifications.

void *	link_status_ctx	User context for link_status_cb.
rx_spi_comm_handle_t *	spi_handle	SPI communication handle (NULL to disable SPI channel). Pointer to initialized SPI transport handle. If NULL, SPI channel is disabled and all SPI operations are skipped. Requirements: • Must be initialized via rx_spi_comm_init() before passing to manager • Must remain valid for lifetime of manager • Requires hardware SPI connection between RX72N and RPi5 Null handling: • NULL = SPI channel disabled (valid configuration) • Non-NULL = SPI channel enabled Type: rx_spi_comm_handle_t* (pointer to SPI tra Valid values: Non-NULL initialized handle, or nullpt Lifetime: Must outlive manager instance

rx_spi_link_t *	spi_link	Optional SPI link layer with HARQ (NULL to use raw SPI). When non-NULL, SPI send/receive operations route through the link layer, which provides FEC encoding/decoding, Chase Combining, and ACK/NACK handshake. When NULL, SPI operates in raw mode (frame + CRC-32 only, no FEC/HARQ). Requirements: • Must be initialized via rx_spi_link_init() before passing to manager • spi_link->spi_handle MUST be the same as config->spi_handle • Must remain valid for lifetime of manager Invariant If spi_link is non-NULL: spi_link->spi_handle == config->spi_handle If spi_link is non-NULL: spi_link->initialized == true If spi_link is non-NULL: spi_link must outlive manager instance Valid values: Initialized rx_spi_link_t*, or nullptr f Lifetime: Must outlive manager instance See also rx_spi_link.h SPI link layer API rx_spi_link_init() Must be called before passing spi_link to manager Since Version 1.0.0

rx_uart_comm_handle_t *	uart_handle	UART communication handle (NULL to disable UART channel). Pointer to initialized UART (SCI9) transport handle. If NULL, UART channel is disabled and all UART operations are skipped. Type: rx_uart_comm_handle_t* (pointer to UART Valid values: Non-NULL initialized handle, or nullpt Lifetime: Must outlive manager instance

4.1.2.4 struct rx_comm_manager_t

Communication manager handle.

Opaque handle structure containing manager state and resources. Application must allocate this structure (typically static or global) and pass to all manager functions.

Initialization: Zero-initialize structure before passing to rx_comm_manager_init():

```
rx_comm_manager_t mgr = {};  // Or use memset
```

Memory Layout: See file-level documentation for detailed breakdown.

Thread Safety: Not thread-safe - protect with external mutex if accessed from multiple threads.

Warning

Do not access fields directly - use API functions only

Structure size ~2.5 KB - consider allocation carefully

See also

rx_comm_manager_init() Initialize handle

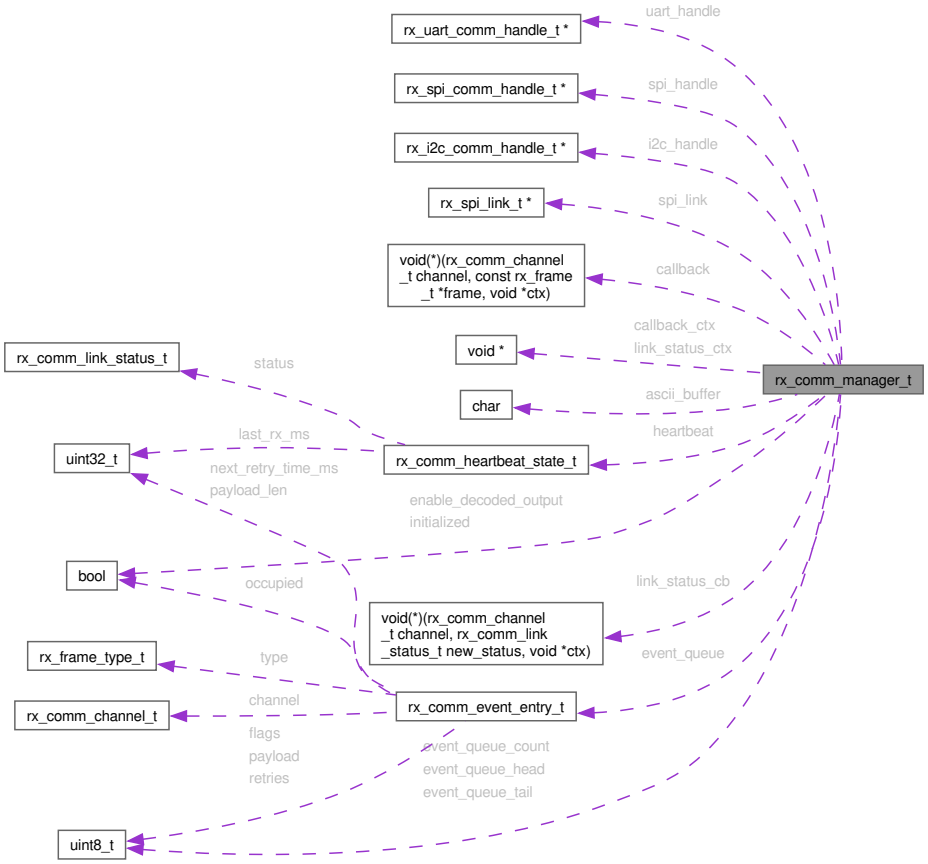
rx_comm_manager_deinit() Cleanup handle

Since

Version 1.0.0

Definition at line 1007 of file rx_comm_manager.h.

Collaboration diagram for rx_comm_manager_t:



Data Fields

char	ascii_buffer[k_comm_manager_ascii_buf_size]	ASCII buffer
rx_comm_frame_callback_t	callback	Frame callback
void *	callback_ctx	Callback context
bool	enable_decoded_output	Enable ASCII output
rx_comm_event_entry_t	event_queue[k_comm_event_queue_depth]	Static event FIFO queue for reliable command delivery.
uint8_t	event_queue_count	Number of occupied event queue slots.

uint8_t	event_queue_head	Event queue write index (next slot to write).
uint8_t	event_queue_tail	Event queue read index (next slot to process).
rx_comm_heartbeat_state_t	heartbeat[k_comm_channel_count]	Per-transport heartbeat tracking (indexed by rx_comm_channel_t).

rx_i2c_comm_handle_t *	i2c_handle	Optional SPI link layer with HARQ (NULL if disabled). I2C peripheral comm handle (NULL disables I2C channel) When non-NULL, points to an rx_spi_link_t handle providing FEC encoding/decoding, Chase Combining, and ACK/NACK handshake for SPI. When NULL, SPI operates in raw mode (frame + CRC-32 only, no FEC/HARQ). Ownership and Lifetime: • Pointer copied from config during rx_comm_manager_in • Manager does NOT own the link object - caller must

bool	initialized	Initialization flag
rx_comm_link_status_callback_t	link_status_cb	Link status change callback (optional).
void *	link_status_ctx	Link status callback context.
rx_spi_comm_handle_t *	spi_handle	SPI comm handle (NULL disables SPI channel)
rx_spi_link_t *	spi_link	
rx_uart_comm_handle_t *	uart_handle	UART (SCI9 via CY7C65213 bridge) comm handle – primary

4.1.2.5 struct rx_comm_send_params_t

Parameters for rx_comm_manager_send.

Wrapper structure providing type safety for send operations. Prevents parameter confusion when passing multiple uint8_t/uint32_t values.

Design Rationale: Without this struct, function signature would be error-prone:

```
// BAD: Easy to mix up channel, type, flags (all similar types)
rx_err_t send(mgr, uint8_t ch, uint8_t type, uint8_t flags, ...);
```

With struct, parameters are self-documenting:

```
// GOOD: Clear naming prevents mistakes
rx_comm_send_params_t params = {
    .channel = k_comm_channel_usb,
    .type = k_frame_type_command,
    .flags = 0x00,
    // ...
};
```

Usage Example:

```
rx_comm_send_params_t params = {
    .channel = k_comm_channel_spi,
    .type = k_frame_type_telemetry,
    .flags = 0,
    .payload = telemetry_data,
    .payload_len = sizeof(telemetry_data),
};
rx_comm_manager_send(&mgr, &params);
```

See also

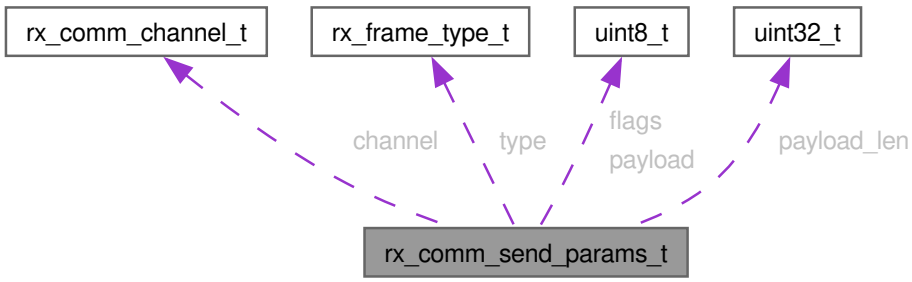
rx_comm_manager_send() Send function using this structure

Since

Version 1.0.0

Definition at line 1132 of file rx_comm_manager.h.

Collaboration diagram for rx_comm_send_params_t:



Data Fields

rx_comm_channel_t	channel	Channel to send on (USB or SPI)
uint8_t	flags	Frame flags (protocol-specific)
const uint8_t *	payload	Payload data pointer
uint32_t	payload_len	Payload length in bytes
rx_frame_type_t	type	Frame type (command, telemetry, etc.)

4.1.3 Typedef Documentation

4.1.3.1 rx_comm_frame_callback_t

typedef void(* rx_comm_frame_callback_t) (rx_comm_channel_t channel, const rx_frame_t *frame, void *ctx)

Frame received callback function type.

User-defined function invoked by communication manager when a complete, valid frame is received on any enabled channel. The callback receives:

- Channel identifier (which channel the frame arrived on)
- Parsed frame structure (type, flags, payload)
- User context pointer (passed during initialization)

4.1.3.2 Callback Contract

Responsibilities:

- Parse frame payload based on frame type
- Handle command execution or data processing
- Optionally send response via rx_comm_manager_respond()
- Return quickly to avoid blocking communication loop

Restrictions:

- Do NOT call rx_comm_manager_poll() from callback (recursion!)
- Keep processing minimal (~50 us recommended)
- For slow operations, queue to task and return immediately
- Do NOT modify frame structure (it may be const or recycled)

Thread Safety:

- Callback executed in context of poll() caller
- If poll() called from RTOS task, callback runs in that task context
- Synchronization required if accessing shared resources

Parameters

in	channel	The communication channel the frame was received on • Type: rx_comm_channel_t • Valid values: k_comm_channel_usb or k_comm_channel_spi • Use to determine response routing
in	frame	Pointer to the received and validated frame • Type: const rx_frame_t* • Valid until callback returns • Contains: type, flags, payload, payload_len (CRC already validated) • Do NOT modify or store pointer (frame may be recycled)

in	ctx	User context pointer • Type: void* (application-defined) • Value passed during rx_comm_manager_init() • Typically pointer to manager, application state, or nullptr • Use to access application resources without globals

Usage Example - Basic Command Handler:

```
void on_frame(rx_comm_channel_t channel, const rx_frame_t* frame, void* ctx) {
    rx_comm_manager_t* mgr = (rx_comm_manager_t*)ctx;

    // Log reception
    rx_log_info("COMM", "RX from %s: type=0x%02X len=%u",
                rx_comm_manager_channel_name(channel),
                frame->type, frame->payload_len);

    // Dispatch based on type
    switch (frame->type) {
        case k_frame_type_command:
            handle_command(mgr, channel, frame);
            break;
        case k_frame_type_telemetry_req:
            send_telemetry(mgr, channel);
            break;
        default:
            rx_log_warn("COMM", "Unknown type: 0x%02X", frame->type);
            break;
    }
}
```

Usage Example - Deferred Processing:

```
// For time-consuming operations, defer to queue
void on_frame(rx_comm_channel_t channel, const rx_frame_t* frame, void* ctx) {
    app_state_t* app = (app_state_t*)ctx;

    // Copy frame to queue (frame pointer only valid during callback)
    command_msg_t msg = {
        .channel = channel,
        .type = frame->type,
        .payload_len = frame->payload_len,
    };
    memcpy(msg.payload, frame->payload, frame->payload_len);

    // Queue for processing by command handler task
    UINT status = tx_queue_send(&app->cmd_queue, &msg, TX_NO_WAIT);
    if (status != TX_SUCCESS) {
        rx_log_error("COMM", "Command queue full!");
    }
}
```

Warning

Callback must return quickly (<50 us recommended) to avoid blocking communication

Do NOT call rx_comm_manager_poll() from callback (causes recursion)

Frame pointer only valid during callback - copy data if needed beyond callback scope

See also

rx_comm_manager_respond() Send response on same channel

rx_comm_manager_send() Send frame on specific channel

rx_frame_t Frame structure definition

Since

Version 1.0.0

Definition at line 681 of file rx_comm_manager.h.

4.1.3.3 rx_comm_link_status_callback_t

typedef void(* rx_comm_link_status_callback_t) (rx_comm_channel_t channel, rx_comm_link_status_t new_status, void *ctx)

Callback invoked when a link status changes.

Parameters

in	channel	Which transport changed status
in	new_status	New link status
in	ctx	User context pointer

Since

Version 1.0.0

Definition at line 750 of file rx_comm_manager.h.

4.1.4 Enumeration Type Documentation

4.1.4.1 rx_comm_channel_mask_t

enum rx_comm_channel_mask_t : uint8_t

Bitmask selecting which comm channels to initialize and use.

Each bit corresponds to one comm channel. Bit position matches the numeric value of the corresponding rx_comm_channel_t enumerator so that a single shift converts between the two representations.

Pass a bitmask in rx_comm_manager_config_t::enabled_channels to skip initialization of channels that are not needed (e.g. exclude UART when the log backend also uses SCI9).

See also

rx_comm_channel_t Channel enumerator

rx_comm_manager_config_t::enabled_channels Usage

Since

Version 1.0.0

Enumerator

k_comm_channel_mask_none	No channels enabled
k_comm_channel_mask_uart	UART (k_comm_channel_uart = 0)
k_comm_channel_mask_spi	SPI (k_comm_channel_spi = 1)
k_comm_channel_mask_i2c	I2C (k_comm_channel_i2c = 2)
k_comm_channel_mask_all	All three channels

Definition at line 771 of file rx_comm_manager.h.

```
00771             : uint8_t {
00772   k_comm_channel_mask_none = 0x00U, /**< No channels enabled */
00773   k_comm_channel_mask_uart = 0x01U, /**< UART (k_comm_channel_uart = 0) */
00774   k_comm_channel_mask_spi  = 0x02U, /**< SPI  (k_comm_channel_spi  = 1) */
00775   k_comm_channel_mask_i2c  = 0x04U, /**< I2C  (k_comm_channel_i2c  = 2) */
00776   k_comm_channel_mask_all  =        /**< All three channels */
00777   (k_comm_channel_mask_uart | k_comm_channel_mask_spi | k_comm_channel_mask_i2c),
00778 } rx_comm_channel_mask_t;
```

4.1.4.2 rx_comm_channel_t

enum rx_comm_channel_t : uint8_t

Communication channel identifiers.

Enumerates the available communication channels. The manager can handle multiple channels simultaneously, with each channel operating independently.

Channel Characteristics:

Channel	Bandwidth	Latency	Setup Complexity	Use Case
USB CDC	~1 MB/s	~1-5 ms	Low (plug-and-play)	Commands, telemetry
SPI	~2 MB/s	~0.1-1 ms	Medium (requires wiring)	High-speed data, control

Usage Pattern:

```
// Check which channel frame came from
void callback(rx_comm_channel_t channel, const rx_frame_t* frame, void* ctx) {
    if (channel == k_comm_channel_usb) {
        // Received via USB
    } else if (channel == k_comm_channel_spi) {
        // Received via SPI
    }
}
```

See also

rx_comm_manager_channel_name() Get human-readable channel name

rx_comm_frame_callback_t Callback receives channel identifier

Since

Version 1.0.0

Enumerator

k_comm_channel_uart	UART channel (SCI9 via CY7C65213 USB-UART bridge). Primary link to the Raspberry Pi 5 gateway. Carries protobuf command / response / telemetry frames AND k_frame_type_log_message frames with the RX72N runtime log stream, all multiplexed on the same byte stream. On the Pi5 the bridge enumerates as /dev/ttyACM0 at 115200 baud (CY7C65213 is a CDC-ACM class device; see k_uart_default_baudrate in libs/rx_hal/src/uart.c). Hardware: SCI9 (PB7 / TXD9, PB6 / RXD9) + CY7C65213 bridge IC Value: 0
k_comm_channel_spi	SPI peripheral channel. Dedicated SPI connection to the ROS2 spi_driver_node for low-latency, high-bandwidth motor control. 10 MHz clock, full-duplex. Hardware: RSPI peripheral (COPI / CIPO / SCK / CS) Value: 1
k_comm_channel_i2c	I2C peripheral channel (RIIC0). I2C connection where RX72N acts as I2C peripheral device and RPi5 is the I2C controller. Optional auxiliary channel. Hardware: RIIC0 peripheral (P16 / SCL0, P17 / SDA0) Value: 2
k_comm_channel_count	Total number of supported channels (3). Value: 3 (UART + SPI + I2C) Invariant: Must equal number of enum values (excluding this one)

Definition at line 531 of file rx_comm_manager.h.

```
00531             : uint8_t {
00532     /**
00533      * @brief UART channel (SCI9 via CY7C65213 USB-UART bridge)
```

```
00534   * @details
00535   * Primary link to the Raspberry Pi 5 gateway. Carries protobuf command /
00536   * response / telemetry frames AND k_frame_type_log_message frames with the
00537   * RX72N runtime log stream, all multiplexed on the same byte stream.
00538   *
00539   * On the Pi5 the bridge enumerates as /dev/ttyACM0 at 115200 baud (CY7C65213 is a CDC-ACM class device; see
        k_uart_default_baudrate in libs/rx_hal/src/uart.c).
00540   *
00541   * @par Hardware: SCI9 (PB7 / TXD9, PB6 / RXD9) + CY7C65213 bridge IC
00542   * @par Value: 0
00543   */
00544  k_comm_channel_uart = 0,
00545
00546  /**
00547   * @brief SPI peripheral channel
00548   * @details
00549   * Dedicated SPI connection to the ROS2 spi_driver_node for low-latency,
00550   * high-bandwidth motor control. 10 MHz clock, full-duplex.
00551   *
00552   * @par Hardware: RSPI peripheral (COPI / CIPO / SCK / CS)
00553   * @par Value: 1
00554   */
00555  k_comm_channel_spi = 1,
00556
00557  /**
00558   * @brief I2C peripheral channel (RIIC0)
00559   * @details
00560   * I2C connection where RX72N acts as I2C peripheral device and RPi5 is the
00561   * I2C controller. Optional auxiliary channel.
00562   *
00563   * @par Hardware: RIIC0 peripheral (P16 / SCL0, P17 / SDA0)
00564   * @par Value: 2
00565   */
00566  k_comm_channel_i2c = 2,
00567
00568  /**
00569   * @brief Total number of supported channels (3)
00570   *
00571   * @par Value: 3 (UART + SPI + I2C)
00572   * @par Invariant: Must equal number of enum values (excluding this one)
00573   */
00574  k_comm_channel_count = 3,
00575 } rx_comm_channel_t;
```

4.1.4.3 rx_comm_event_retry_constants_t

enum rx_comm_event_retry_constants_t : uint16_t

Event queue retry constants for SPI HARQ.

Events (commands) sent via SPI use retry with exponential backoff. USB sends are fire-and-forget (no retries).

Since

Version 1.0.0

Enumerator

k_event_max_retries	Maximum SPI retry attempts
k_event_initial_backoff_ms	Initial backoff delay (ms)
k_event_max_backoff_ms	Maximum backoff delay (ms): caps 10<<(retries-1) at retries=2

Definition at line 489 of file rx_comm_manager.h.

```
00489             : uint16_t {
00490  k_event_max_retries        = 3,  /**< Maximum SPI retry attempts */
00491  k_event_initial_backoff_ms = 10, /**< Initial backoff delay (ms) */
00492  k_event_max_backoff_ms = 15, /**< Maximum backoff delay (ms): caps 10«(retries-1) at retries=2 */
00493 } rx_comm_event_retry_constants_t;
```

4.1.4.4 rx_comm_heartbeat_constants_t

enum rx_comm_heartbeat_constants_t : uint32_t

Heartbeat timing constants.

Dual-detection heartbeat model:

- Implicit timeout (200ms): Any valid frame resets the timer. If no frames arrive within 200ms, the link is declared dead.
- Explicit PING (1000ms): Sent when link is idle for >1s as a probe. Firmware primarily receives PINGs from the gateway.
- Check interval (50ms): How often to evaluate heartbeat status.

Since

Version 1.0.0

Enumerator

k_heartbeat_implicit_timeout_ms	Declare link dead if no frame for 200ms
k_heartbeat_ping_interval_ms	Send PING probe when idle for 1s
k_heartbeat_check_interval_ms	Heartbeat evaluation interval (200ms / 4)

Definition at line 473 of file rx_comm_manager.h.

```
00473             : uint32_t {
00474   k_heartbeat_implicit_timeout_ms = 200,  /**< Declare link dead if no frame for 200ms */
00475   k_heartbeat_ping_interval_ms    = 1000, /**< Send PING probe when idle for 1s */
00476   k_heartbeat_check_interval_ms   = 50,   /**< Heartbeat evaluation interval (200ms / 4) */
00477 } rx_comm_heartbeat_constants_t;
```

4.1.4.5 rx_comm_link_status_t

enum rx_comm_link_status_t : uint8_t

Link health status from heartbeat monitoring.

Reported per-transport link health. Determined by the dual-detection heartbeat model (implicit 200ms timeout + explicit 1s PING).

Since

Version 1.0.0

Enumerator

k_link_status_unknown	No frames received yet
k_link_status_healthy	Frames received within implicit timeout
k_link_status_dead	No frames within implicit timeout (200ms)

Definition at line 695 of file rx_comm_manager.h.

```
00695             : uint8_t {
00696   k_link_status_unknown = 0, /**< No frames received yet */
00697   k_link_status_healthy = 1, /**< Frames received within implicit timeout */
00698   k_link_status_dead    = 2, /**< No frames within implicit timeout (200ms) */
00699 } rx_comm_link_status_t;
```

4.1.4.6 rx_comm_manager_constants_t

enum rx_comm_manager_constants_t : uint16_t

Communication manager compile-time constants.

Defines buffer sizes and limits for the communication manager. These constants are chosen based on typical frame sizes and ASCII formatting requirements.

Design Rationale:

- 2048-byte ASCII buffer: Accommodates maximum frame size (1024 bytes payload) plus formatting overhead (~1000 bytes for hex dump + headers)
- Buffer is stack-allocated in handle structure (not dynamically allocated)
- Sufficient for debugging without excessive memory usage

See also

rx_comm_manager_t.ascii_buffer

Since

Version 1.0.0

Enumerator

k_comm_manager_ascii_buf_size	ASCII formatting buffer size in bytes. Size of internal buffer used to format decoded ASCII output for debug port. Must accommodate: • Header line: ~80 bytes ([USB] TYPE=cmd FLAGS=0x00 LEN=1024) • Payload hex dump: payload_len * 3 bytes (XX per byte) • Footer + null terminator: ~20 bytes Maximum frame payload: 1024 bytes Required buffer: 80 + (1024 * 3) + 20 = 3172 bytes Allocated: 2048 bytes (limits payload display to ~650 bytes, which is acceptable) Value: 2048 bytes Rationale: Balance between functionality and memory usage

k_comm_event_queue_depth	Maximum number of entries in the event FIFO queue. Static FIFO queue depth for event (command) frames that require reliable delivery. Events are queued and sent with SPI-only retry + exponential backoff. Queue is statically allocated (zero dynamic memory). Value: 8 entries Rationale: Small queue prevents unbounded memory, sufficient for typical command burst scenarios

Definition at line 426 of file rx_comm_manager.h.

```
00426           : uint16_t {
00427   /**
00428    * @brief ASCII formatting buffer size in bytes
00429    * @details
00430    * Size of internal buffer used to format decoded ASCII output for debug port.
00431    * Must accommodate:
00432    * - Header line: ~80 bytes ('[USB] TYPE=cmd FLAGS=0x00 LEN=1024')
00433    * - Payload hex dump: payload_len * 3 bytes ('XX ` per byte)
00434    * - Footer + null terminator: ~20 bytes
00435    *
00436    * Maximum frame payload: 1024 bytes
00437    * Required buffer: 80 + (1024 * 3) + 20 = 3172 bytes
00438    * Allocated: 2048 bytes (limits payload display to ~650 bytes, which is acceptable)
00439    *
00440    * @par Value: 2048 bytes
00441    * @par Rationale: Balance between functionality and memory usage
00442    */
00443   k_comm_manager_ascii_buf_size = 2048,
00444
00445   /**
00446    * @brief Maximum number of entries in the event FIFO queue
00447    * @details
00448    * Static FIFO queue depth for event (command) frames that require reliable
00449    * delivery. Events are queued and sent with SPI-only retry + exponential
00450    * backoff. Queue is statically allocated (zero dynamic memory).
00451    *
00452    * @par Value: 8 entries
00453    * @par Rationale: Small queue prevents unbounded memory, sufficient for
00454    *   typical command burst scenarios
00455    */
00456   k_comm_event_queue_depth = 8,
00457 } rx_comm_manager_constants_t;
```

4.1.5 Function Documentation

4.1.5.1 rx_comm_manager_channel_name()

```
const char * rx_comm_manager_channel_name (
            rx_comm_channel_t channel)
```

Get channel name string.

Parameters

in	channel	Channel identifier

Returns

Human-readable channel name

Get channel name string.

Returns static string name for given channel enum value. Used for logging, debugging, and user interface display. Always returns a valid string (never NULL).

Returned Strings:

- k_comm_channel_usb -> "USB"
- k_comm_channel_spi -> "SPI"
- Invalid/unknown -> "UNKNOWN"

Use Cases:

- Logging channel activity
- Debugging frame traffic
- User interface display
- Error messages

Precondition

None (accepts any input)

Postcondition

Returns pointer to static string (valid for program lifetime)

Note

Returned string is static - do not free

Thread-safe (returns read-only static strings)

Never returns NULL - always safe to use in printf

Example - Logging:

```
static void on_frame_received(rx_comm_channel_t channel,
                              const rx_frame_t* frame,
                              void* ctx)
{
  printf("Frame received on %s: type=%d len=%d\n",
         rx_comm_manager_channel_name(channel),
         frame->header.type,
         frame->header.length);
}
```

Example - Error Messages:

```
rx_err_t err = rx_comm_manager_send(&g_comm_mgr, &params);
if (err != k_rx_ok) {
  fprintf(stderr, "Failed to send on %s: error %d\n",
          rx_comm_manager_channel_name(params.channel), err);
}
```

See also

rx_comm_channel_t Channel enumeration

Since

Version 1.0.0

Definition at line 1946 of file rx_comm_manager.c.

```
01947 {
01948   switch (channel) {
01949     case k_comm_channel_uart:
01950       return "UART";
01951     case k_comm_channel_spi:
01952       return "SPI";
01953     case k_comm_channel_i2c:
01954       return "I2C";
01955     default:
01956       return "UNKNOWN";
01957   }
01958 }
```

References k_comm_channel_i2c, k_comm_channel_spi, and k_comm_channel_uart.

4.1.5.2 rx_comm_manager_channel_ready()

```
rx_err_t rx_comm_manager_channel_ready (
                rx_comm_manager_t * mgr,
                rx_comm_channel_t channel,
                bool * ready)   [nodiscard]
```

Check if channel is ready for communication.

Parameters

in	mgr	Pointer to initialized manager
in	channel	Channel to check
out	ready	Set to true if channel is ready

Returns

k_rx_ok on success

k_rx_err_invalid_arg if mgr or ready is nullptr

Check if channel is ready for communication.

Checks if specified channel is configured and ready for communication. USB channel readiness depends on USB enumeration and configuration by host. SPI channel is ready immediately if handle is configured (no enumeration required).

Readiness Criteria:

- USB: usb_handle non-NULL AND USB device enumerated and configured by host
- SPI: spi_handle non-NULL (always ready if configured)

Use Cases:

- Check readiness before sending to avoid errors
- Implement fallback logic (try USB, fall back to SPI)
- Log channel availability status

Precondition

mgr must be non-NULL and initialized
ready must be non-NULL
channel must be valid (USB or SPI)

Postcondition

∗ready set to true or false based on channel state
On error, ∗ready set to false (safe default)

Note

USB readiness can change at runtime (host connects/disconnects)
SPI readiness is static (always ready if handle configured)
This function does NOT initiate communication - only queries state

Example - Send with Fallback:

```
uint8_t payload[16] = { ... };

// Try USB first
bool usb_ready = false;
rx_err_t err = rx_comm_manager_channel_ready(&g_comm_mgr,
                                             k_comm_channel_usb,
                                             &usb_ready);
if (err == k_rx_ok && usb_ready) {
  err = rx_comm_manager_respond(&g_comm_mgr,
                                k_comm_channel_usb,
                                payload,
                                sizeof(payload));
} else {
  // Fallback to SPI
  bool spi_ready = false;
  err = rx_comm_manager_channel_ready(&g_comm_mgr,
                                      k_comm_channel_spi,
                                      &spi_ready);
  if (err == k_rx_ok && spi_ready) {
    err = rx_comm_manager_respond(&g_comm_mgr,
                                  k_comm_channel_spi,
                                  payload,
                                  sizeof(payload));
  }
}
```

Example - Log Channel Status:

```
void log_channel_status(void)
{
  bool usb_ready = false, spi_ready = false;

  rx_comm_manager_channel_ready(&g_comm_mgr, k_comm_channel_usb, &usb_ready);
  rx_comm_manager_channel_ready(&g_comm_mgr, k_comm_channel_spi, &spi_ready);

  printf("USB: %s, SPI: %s\n",
         usb_ready ? "READY" : "NOT READY",
         spi_ready ? "READY" : "NOT READY");
}
```

See also

rx_usb_is_configured() USB enumeration status

rx_comm_manager_send() Send frame on channel

Since

Version 1.0.0

Definition at line 1857 of file rx_comm_manager.c.

```
01858 {
01859   /* Rule 5: Pre-condition validation */
01860   if (mgr == nullptr || ready == nullptr) {
01861     if (ready != nullptr) {
01862       *ready = false;
01863     }
01864     return k_rx_err_invalid_arg;
01865   }
01866   if (!mgr->initialized) {
01867     *ready = false;
01868     return k_rx_err_invalid_state;
01869   }
01870
01871   switch (channel) {
01872     case k_comm_channel_uart:
01873       *ready = (bool)(mgr->uart_handle != nullptr);
01874       break;
01875
01876     case k_comm_channel_spi:
01877       *ready = (bool)(mgr->spi_handle != nullptr);
01878       break;
01879
01880     case k_comm_channel_i2c:
01881       *ready = (bool)(mgr->i2c_handle != nullptr);
01882       break;
01883
01884     default:
01885       *ready = false;
01886       rx_log_error(s_tag, "Channel ready: invalid channel");
01887       return k_rx_err_invalid_arg;
01888   }
01889
01890   return k_rx_ok;
01891 }
```

References rx_comm_manager_t::i2c_handle, rx_comm_manager_t::initialized, k_comm_channel_i2c, k_comm_channel_spi, k_comm_channel_uart, s_tag, rx_comm_manager_t::spi_handle, and rx_comm_manager_t::uart_handle.

4.1.5.3 rx_comm_manager_deinit()

```
rx_err_t rx_comm_manager_deinit (
            rx_comm_manager_t * mgr)   [nodiscard]
```

Deinitialize communication manager.

Does not deinitialize the underlying USB/SPI handles.

Parameters

in,out	mgr	Pointer to manager handle

Returns

k_rx_ok on success

Deinitialize communication manager.

Marks communication manager as uninitialized, preventing further operations. This function does NOT deinitialize USB or SPI transport handles - caller must separately call rx_usb_comm_deinit() and rx_spi_comm_deinit() if needed.

Algorithm:

1. Validate mgr pointer (NULL check)
2. Validate initialized flag (must be initialized to deinitialize)
3. Clear initialized flag (marks handle as invalid)

Design Note: Minimal deinitialization - no dynamic resources to free (all memory is statically allocated). Function primarily serves as safety guard to prevent use-after-deinit.

Precondition

mgr must be non-NULL

mgr must be initialized (mgr->initialized == true)

Postcondition

mgr->initialized = false

Subsequent calls to rx_comm_manager_poll/send will return k_rx_err_invalid_state

Note

Does NOT free memory (handle is statically allocated)

Does NOT deinitialize transport handles (USB/SPI)

Thread-safe if no concurrent operations on same mgr handle

Example:

```
// Deinitialize manager
rx_err_t err = rx_comm_manager_deinit(&g_comm_mgr);
if (err == k_rx_ok) {
  // Manager no longer usable

  // Separately deinitialize transports if needed
  rx_usb_comm_deinit(&g_usb_handle);
  rx_spi_comm_deinit(&g_spi_handle);
}
```

See also

rx_comm_manager_init() Initialization

rx_spi_comm_deinit() SPI transport cleanup

Since

Version 1.0.0

Definition at line 1216 of file rx_comm_manager.c.

```
01217 {
01218   /* Rule 5: Pre-condition validation */
01219   if (mgr == nullptr) {
01220     rx_log_error(s_tag, "Deinit failed: NULL mgr");
01221     return k_rx_err_invalid_arg;
01222   }
01223   if (!mgr->initialized) {
01224     rx_log_warn(s_tag, "Deinit called on uninitialized mgr");
01225     return k_rx_err_invalid_state;
01226   }
01227
01228   mgr->initialized = false;
01229
01230   rx_log_info(s_tag, "Comm manager deinitialized");
01231   return k_rx_ok;
01232 }
```

References rx_comm_manager_t::initialized, and s_tag.

4.1.5.4 rx_comm_manager_event_send()

```
rx_err_t rx_comm_manager_event_send (
                rx_comm_manager_t * mgr,
                const rx_comm_send_params_t * params)   [nodiscard]
```

Queue data on the event path (reliable, SPI retry).

Event path is for commands and other data requiring reliable delivery. The payload is copied into a static FIFO queue and processed by the poll loop. Retry behavior depends on transport:

- SPI: Up to 3 retries with exponential backoff (10ms, 20ms, 40ms)
- USB: Fire-and-forget (USB HW reliability is sufficient)

Queue is processed in FIFO order by rx_comm_manager_poll().

Parameters

in,out	mgr	Pointer to initialized manager
in	params	Send parameters (channel, type, flags, payload, payload_len)

Returns

k_rx_ok on success (queued)

k_rx_err_invalid_arg if mgr or params is nullptr

k_rx_err_invalid_state if not initialized

k_rx_err_no_mem if event queue is full

Precondition

mgr must be initialized

params must be valid

params->payload_len <= k_frame_max_payload

Postcondition

Payload copied to queue, will be sent by next poll()

Note

Payload is COPIED into queue - caller can reuse buffer immediately

Warning

Queue depth is k_comm_event_queue_depth (8). If full, returns error.

See also

rx_comm_manager_stream_send() For fire-and-forget telemetry

Since

Version 1.0.0

Definition at line 1698 of file rx_comm_manager.c.

```
01699 {
01700   if (mgr == nullptr || params == nullptr) {
01701     rx_log_error(s_tag, "Event enqueue failed: NULL arg");
01702     return k_rx_err_invalid_arg;
01703   }
01704
01705   if (!mgr->initialized) {
01706     rx_log_error(s_tag, "Event enqueue failed: not initialized");
01707     return k_rx_err_invalid_state;
01708   }
01709
01710   if (params->payload_len > k_frame_max_payload) {
01711     rx_log_error_val(s_tag, "Event enqueue failed: payload too large", params->payload_len);
01712     return k_rx_err_invalid_arg;
01713   }
01714
01715   if (params->payload_len > 0 && params->payload == nullptr) {
01716     rx_log_error(s_tag, "Event enqueue failed: NULL payload with nonzero len");
01717     return k_rx_err_invalid_arg;
01718   }
01719
01720   /* Check queue capacity */
01721   if (mgr->event_queue_count >= k_comm_event_queue_depth) {
01722     rx_log_error_val(s_tag, "Event queue full, depth", (uint8_t)mgr->event_queue_count);
01723     return k_rx_err_no_mem;
01724   }
01725
01726   /* Enqueue: copy payload into static slot */
01727   rx_comm_event_entry_t* entry = &mgr->event_queue[mgr->event_queue_head];
01728   entry->channel     = params->channel;
01729   entry->type        = params->type;
01730   entry->flags       = params->flags;
01731   entry->payload_len = params->payload_len;
01732   entry->retries     = 0;
01733   entry->occupied    = true;
01734
01735   if (params->payload != nullptr && params->payload_len > 0) {
01736     for (uint32_t i = 0; i < params->payload_len; i++) {
01737       entry->payload[i] = params->payload[i];
```

```
01738     }
01739   }
01740
01741   mgr->event_queue_head = (mgr->event_queue_head + 1) % k_comm_event_queue_depth;
01742   mgr->event_queue_count++;
01743
01744   rx_log_debug_val(s_tag, "Event enqueued, queue depth", mgr->event_queue_count);
01745   return k_rx_ok;
01746 }
```

References rx_comm_event_entry_t::channel, rx_comm_send_params_t::channel, rx_comm_manager_t::event_queu rx_comm_manager_t::event_queue_count, rx_comm_manager_t::event_queue_head, rx_comm_event_entry_t::flag rx_comm_send_params_t::flags, rx_comm_manager_t::initialized, k_comm_event_queue_depth, rx_comm_event_entry_t::occupied, rx_comm_event_entry_t::payload, rx_comm_send_params_t::payload, rx_comm_event_entry_t::payload_len, rx_comm_send_params_t::payload_len, rx_comm_event_entry_t::retries, s_tag, rx_comm_event_entry_t::type, and rx_comm_send_params_t::type.

4.1.5.5 rx_comm_manager_init()

```
rx_err_t rx_comm_manager_init (
                rx_comm_manager_t * mgr,
                const rx_comm_manager_config_t * cfg)   [nodiscard]
```

Initialize communication manager.

Initializes the communication manager with the provided configuration. Validates configuration parameters and sets up internal state for USB and/or SPI channel operation.

Parameters

out	mgr	Pointer to manager handle
in	cfg	Configuration (NULL for defaults, both channels disabled)

Returns

k_rx_ok on success

k_rx_err_invalid_arg if mgr is nullptr

k_rx_err_invalid_arg if spi_link is non-NULL but not initialized

k_rx_err_invalid_arg if spi_link->spi_handle != cfg->spi_handle

Precondition

mgr must not be nullptr

If cfg->spi_link is non-NULL, spi_link must be initialized via rx_spi_link_init()

If cfg->spi_link is non-NULL, spi_link->spi_handle must equal cfg->spi_handle

cfg may be nullptr (results in default configuration with both channels disabled)

Postcondition

mgr->usb_handle == cfg->usb_handle (or nullptr if cfg is nullptr)

mgr->spi_handle == cfg->spi_handle (or nullptr if cfg is nullptr)

mgr->spi_link == cfg->spi_link (or nullptr if cfg is nullptr or cfg->spi_link is nullptr)

All internal state initialized (event queue empty, heartbeat timers reset)

Note

Thread-safe: May be called from any thread, but concurrent calls with same mgr are prohibited

Warning

Must be called before any other manager operations on this handle

See also

rx_spi_link_init() Must call this before passing spi_link to manager

rx_comm_manager_deinit() Cleanup function

Since

Version 1.0.0

Initialize communication manager.

Performs complete initialization of communication manager handle, configuring enabled channels, callback registration, and ASCII debugging output. This function must be called before any other communication manager operations.

Algorithm:

1. Validate parameters: Check mgr pointer for nullptr (NASA Rule 5)
2. Clear handle: Zero-fill entire structure via memset (328 bytes)
3. Apply configuration: Copy config fields if cfg is non-NULL
   - usb_handle: Pointer to initialized USB comm handle (may be NULL)
   - spi_handle: Pointer to initialized SPI comm handle (may be NULL)
   - callback: User frame handler function (may be NULL)
   - callback_ctx: User context pointer (may be NULL)
   - enable_decoded_output: ASCII debugging flag
4. Mark initialized: Set mgr->initialized = true
5. Post-condition validation: Verify ascii_buffer properly cleared

Configuration Options:

- cfg = NULL: All channels disabled, no callback, ASCII output off
- usb_handle = NULL: USB channel disabled (poll skips USB)
- spi_handle = NULL: SPI channel disabled (poll skips SPI)
- callback = NULL: No callback invoked (silent frame consumption)
- enable_decoded_output = true: ASCII debugging enabled (Port 1)

Precondition

mgr must point to allocated memory (uninitialized content OK)

USB/SPI handles (if provided) must already be initialized

Callback function (if provided) must be valid and reentrant

Postcondition

mgr->initialized = true on success

All mgr fields set to config values or zero

mgr->ascii_buffer is zero-filled (256 bytes)

Note

This function does NOT initialize USB or SPI transports - caller must initialize rx_usb_comm and rx_spi_comm separately before passing handles to this function

On success, at least one channel should be enabled (usb_handle or spi_handle non-NULL)

Thread-safe ONLY if each thread uses a different mgr handle

Performance:

- Execution time: ~5 us @ 240 MHz
- Dominated by memset (328 bytes)

Configuration Example - Both Channels Enabled:

```
static rx_comm_manager_t g_comm_mgr;
static rx_usb_comm_handle_t g_usb_handle;
static rx_spi_comm_handle_t g_spi_handle;

// Initialize transport layers first
rx_usb_comm_config_t usb_cfg = { .port = k_usb_port_proto };
rx_err_t err = rx_usb_comm_init(&g_usb_handle, &usb_cfg);
if (err != k_rx_ok) return err;

rx_spi_comm_config_t spi_cfg = { .spi_channel = 0 };
err = rx_spi_comm_init(&g_spi_handle, &spi_cfg);
if (err != k_rx_ok) return err;

// Initialize communication manager with both channels
rx_comm_manager_config_t cfg = {
  .usb_handle = &g_usb_handle,
  .spi_handle = &g_spi_handle,
  .callback = on_frame_received,
  .callback_ctx = nullptr,
  .enable_decoded_output = true,
};
err = rx_comm_manager_init(&g_comm_mgr, &cfg);
```

Configuration Example - USB Only, No Callback:

```
rx_comm_manager_config_t cfg = {
  .usb_handle = &g_usb_handle,
  .spi_handle = nullptr,              // SPI disabled
  .callback = nullptr,                // No callback
  .callback_ctx = nullptr,
  .enable_decoded_output = false,    // No ASCII debugging
};
rx_err_t err = rx_comm_manager_init(&g_comm_mgr, &cfg);
```

See also

rx_comm_manager_deinit() Cleanup and resource release

rx_comm_manager_poll() Poll for incoming frames

rx_spi_comm_init() Initialize SPI transport

rx_uart_comm_init() Initialize UART transport

rx_i2c_comm_init() Initialize I2C peripheral transport

Since

Version 1.0.0

Definition at line 1120 of file rx_comm_manager.c.

```
01121 {
01122   /* Rule 5: Pre-condition validation - nullptr check */
01123   if (mgr == nullptr) {
01124     rx_log_error(s_tag, "Init failed: NULL mgr");
01125     return k_rx_err_invalid_arg;
01126   }
01127
01128   /* Rule 5: Pre-condition validation - SPI link consistency checks */
01129   if (cfg != nullptr && cfg->spi_link != nullptr) {
01130     /* If spi_link is provided, spi_handle must also be provided */
01131     if (cfg->spi_handle == nullptr) {
01132       rx_log_error(s_tag, "Init failed: spi_link requires non-NULL spi_handle");
01133       return k_rx_err_invalid_arg;
01134     }
01135
01136     /* If spi_link is provided, it must be initialized */
01137     if (!cfg->spi_link->initialized) {
01138       rx_log_error(s_tag, "Init failed: spi_link not initialized (call rx_spi_link_init first)");
01139       return k_rx_err_invalid_arg;
01140     }
01141
01142     /* If spi_link is provided, its spi_handle must match config spi_handle */
01143     if (cfg->spi_link->spi_handle != cfg->spi_handle) {
01144       rx_log_error(s_tag, "Init failed: spi_link->spi_handle != cfg->spi_handle");
01145       return k_rx_err_invalid_arg;
01146     }
01147   }
01148
01149   *mgr = s_zero_mgr;
01150
01151   /* Apply configuration */
01152   if (cfg != nullptr) {
01153     mgr->uart_handle          = cfg->uart_handle;
01154     mgr->spi_handle           = cfg->spi_handle;
01155     mgr->i2c_handle           = cfg->i2c_handle;
01156     mgr->spi_link             = cfg->spi_link;
01157     mgr->callback             = cfg->callback;
01158     mgr->callback_ctx         = cfg->callback_ctx;
01159     mgr->enable_decoded_output = cfg->enable_decoded_output;
01160     mgr->link_status_cb       = cfg->link_status_cb;
01161     mgr->link_status_ctx      = cfg->link_status_ctx;
01162   }
01163
01164   mgr->initialized = true;
01165
01166   rx_log_info(s_tag, "Comm manager initialized");
01167   return k_rx_ok;
01168 }
```

References rx_comm_manager_config_t::callback, rx_comm_manager_t::callback, rx_comm_manager_config_t::callb
rx_comm_manager_t::callback_ctx, rx_comm_manager_config_t::enable_decoded_output, rx_comm_manager_t::e
rx_comm_manager_config_t::i2c_handle, rx_comm_manager_t::i2c_handle, rx_comm_manager_t::initialized,
rx_comm_manager_config_t::link_status_cb, rx_comm_manager_t::link_status_cb, rx_comm_manager_config_t::l
rx_comm_manager_t::link_status_ctx, s_tag, s_zero_mgr, rx_comm_manager_config_t::spi_handle,
rx_comm_manager_t::spi_handle, rx_comm_manager_config_t::spi_link, rx_comm_manager_t::spi_link,
rx_comm_manager_config_t::uart_handle, and rx_comm_manager_t::uart_handle.

4.1.5.6 rx_comm_manager_link_status()

```
rx_err_t rx_comm_manager_link_status (
                const rx_comm_manager_t * mgr,
                rx_comm_channel_t channel,
                rx_comm_link_status_t * status)   [nodiscard]
```

Query link health status for a transport channel.

Returns the current heartbeat-derived link status for the given channel. Status is updated by rx_comm_manager_poll() based on the dual-detection heartbeat model (200ms implicit timeout).

Parameters

in	mgr	Pointer to initialized manager
in	channel	Channel to query
out	status	Receives current link status

Returns

k_rx_ok on success

k_rx_err_invalid_arg if mgr or status is nullptr

Since

Version 1.0.0

Definition at line 1753 of file rx_comm_manager.c.

```
01756 {
01757   if (mgr == nullptr || status == nullptr) {
01758     rx_log_error(s_tag, "Link status query: NULL arg");
01759     return k_rx_err_invalid_arg;
01760   }
01761
01762   if (!mgr->initialized) {
01763     rx_log_error(s_tag, "Link status query: uninitialized manager");
01764     return k_rx_err_invalid_state;
01765   }
01766
01767   if (channel >= k_comm_channel_count) {
01768     rx_log_error(s_tag, "Link status query: invalid channel");
01769     return k_rx_err_invalid_arg;
01770   }
01771
01772   *status = mgr->heartbeat[channel].status;
01773   return k_rx_ok;
01774 }
```

References rx_comm_manager_t::heartbeat, rx_comm_manager_t::initialized, k_comm_channel_count, s_tag, and rx_comm_heartbeat_state_t::status.

4.1.5.7 rx_comm_manager_poll()

```
rx_err_t rx_comm_manager_poll (
                rx_comm_manager_t * mgr)   [nodiscard]
```

Poll all enabled channels for incoming frames.

Non-blocking check of USB and SPI channels. For each complete frame received:

1. If decoded output enabled, format and send ASCII to Port 1
2. Invoke user callback with frame data and channel

Call this from main loop or dedicated communication task.

Parameters

in,out	mgr	Pointer to initialized manager

Returns

k_rx_ok if at least one frame was received

k_rx_err_timeout if no frames available on any channel

k_rx_err_invalid_arg if mgr is nullptr

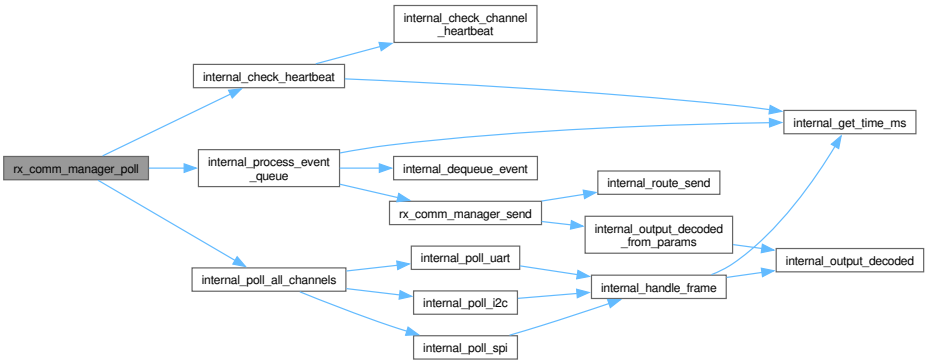
k_rx_err_invalid_state if not initialized

Definition at line 1383 of file rx_comm_manager.c.

```
01384 {
01385   /* Rule 5: Pre-condition validation */
01386   if (mgr == nullptr) {
01387     return k_rx_err_invalid_arg;
01388   }
01389   if (!mgr->initialized) {
01390     return k_rx_err_invalid_state;
01391   }
01392
01393   bool     received = false;
01394   rx_err_t poll_err = internal_poll_all_channels(mgr, &received);
01395   if (poll_err != k_rx_ok) {
01396     return poll_err;
01397   }
01398
01399   /* Process event queue: send one queued event per poll cycle */
01400   internal_process_event_queue(mgr);
01401
01402   /* Heartbeat: check implicit timeout on each transport */
01403   internal_check_heartbeat(mgr);
01404
01405   return (int)received ? k_rx_ok : k_rx_err_timeout;
01406 }
```

References rx_comm_manager_t::initialized, internal_check_heartbeat(), internal_poll_all_channels(), and internal_process_event_queue().

Here is the call graph for this function:



4.1.5.8 rx_comm_manager_process_retransmits()

```
rx_err_t rx_comm_manager_process_retransmits (
                rx_comm_manager_t * mgr,
                uint32_t current_time_ms)   [nodiscard]
```

Process pending retransmissions on SPI channel.

Thin pass-through to rx_spi_comm_process_retransmits() on the SPI handle. Call this from the communication task polling loop.

Parameters

in,out	mgr	Communication manager
in	current_time_ms	Current system time in milliseconds

Returns

rx_err_t Error code from rx_spi_comm_process_retransmits()

Return values

k_rx_ok	No action needed or retransmit sent
k_rx_err_invalid_arg	mgr is nullptr
k_rx_err_invalid_state	mgr not initialized
k_rx_err_retry_limit	Max retries exceeded

Since

Version 1.0.0

Definition at line 1965 of file rx_comm_manager.c.

```
01966 {
01967   if (mgr == nullptr) {
01968     return k_rx_err_invalid_arg;
01969   }
01970
01971   if (!mgr->initialized) {
01972     return k_rx_err_invalid_state;
01973   }
01974
01975   if (mgr->spi_handle == nullptr) {
01976     return k_rx_ok; /* No SPI channel configured */
01977   }
01978
01979   return rx_spi_comm_process_retransmits(mgr->spi_handle, current_time_ms);
01980 }
```

References rx_comm_manager_t::initialized, and rx_comm_manager_t::spi_handle.

4.1.5.9 rx_comm_manager_respond()

```
rx_err_t rx_comm_manager_respond (
                rx_comm_manager_t * mgr,
                rx_comm_channel_t channel,
                const uint8_t * payload,
                uint32_t payload_len)   [nodiscard]
```

Send response on same channel as received frame.

Convenience function for responding to commands. Sends a RESPONSE type frame on the specified channel.

Parameters

in,out	mgr	Pointer to initialized manager
in	channel	Channel to respond on (from callback)
in	payload	Response payload data
in	payload_len	Response payload length

Returns

k_rx_ok on success

Send response on same channel as received frame.

Convenience function for sending response frames. Automatically sets frame type to k_frame_type_response and flags to k_frame_flag_none. Simplifies common use case of responding to commands/requests.

Algorithm:

1. Build rx_comm_send_params_t with:
   - channel: User-specified
   - type: k_frame_type_response (fixed)
   - flags: k_frame_flag_none (fixed)
   - payload: User-specified
   - payload_len: User-specified
2. Call rx_comm_manager_send() with constructed params
3. Return result from rx_comm_manager_send()

When to Use:

- Responding to commands with telemetry data
- Sending acknowledgments
- Replying to requests

When NOT to Use:

- Need custom frame type (use rx_comm_manager_send instead)
- Need custom flags (use rx_comm_manager_send instead)

Precondition

Same preconditions as rx_comm_manager_send()

Postcondition

Response frame transmitted with type=response, flags=none

Note

Equivalent to calling rx_comm_manager_send() with type=k_frame_type_response

This is a thin wrapper - no additional validation performed

Example - Send Telemetry Response:

```
// Received a telemetry request on USB, send response
static void on_telemetry_request(rx_comm_channel_t channel, const rx_frame_t* frame)
{
  uint8_t telemetry[16];

  // Gather telemetry data
  telemetry[0] = get_temperature_celsius();
  telemetry[1] = get_current_ma();
  // ... fill remaining fields ...

  // Send response on same channel request came from
  rx_err_t err = rx_comm_manager_respond(&g_comm_mgr,
                                         channel,
                                         telemetry,
                                         sizeof(telemetry));
  if (err != k_rx_ok) {
    // Handle send error
  }
}
```

Example - Send Empty Acknowledgment:

```
// Send empty ACK response
rx_err_t err = rx_comm_manager_respond(&g_comm_mgr,
                                       k_comm_channel_usb,
                                       NULL,    // No payload
                                       0);      // Zero length
```

See also

rx_comm_manager_send() Full send function with all parameters

rx_comm_send_params_t Send parameters structure

Since

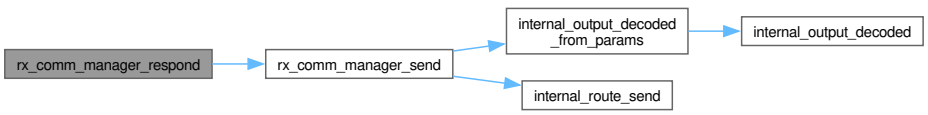
Version 1.0.0

Definition at line 1670 of file rx_comm_manager.c.

```
01674 {
01675   const rx_comm_send_params_t params = {
01676     .channel     = channel,
01677     .type        = k_frame_type_response,
01678     .flags       = k_frame_flag_none,
01679     .payload     = payload,
01680     .payload_len = payload_len,
01681   };
01682   return rx_comm_manager_send(mgr, &params);
01683 }
```

References rx_comm_manager_send().

Here is the call graph for this function:



4.1.5.10 rx_comm_manager_send()

```
rx_err_t rx_comm_manager_send (
                rx_comm_manager_t * mgr,
                const rx_comm_send_params_t * params)   [nodiscard]
```

Send frame on specific channel.

Encodes and sends a frame on the specified channel. If decoded output is enabled, also sends ASCII representation to Port 1.

Parameters

in,out	mgr	Pointer to initialized manager
in	params	Send parameters (channel, type, flags, payload, payload_len)

Returns

k_rx_ok on success

k_rx_err_invalid_arg if mgr or payload is nullptr

k_rx_err_invalid_state if not initialized or channel not enabled

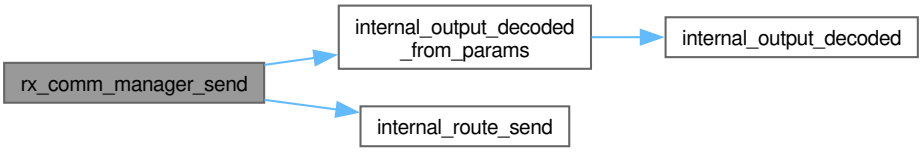
Definition at line 1560 of file rx_comm_manager.c.

```
01561 {
01562   /* Rule 5: Pre-condition validation */
01563   if (mgr == nullptr || params == nullptr) {
01564     rx_log_error(s_tag, "Send failed: NULL arg");
01565     return k_rx_err_invalid_arg;
01566   }
01567   if (!mgr->initialized) {
01568     rx_log_error(s_tag, "Send failed: not initialized");
01569     return k_rx_err_invalid_state;
01570   }
01571   if (params->payload == nullptr && params->payload_len > 0) {
01572     rx_log_error(s_tag, "Send failed: NULL payload with nonzero len");
01573     return k_rx_err_invalid_arg;
01574   }
01575   /* Validate payload length is within frame limits */
01576   if (params->payload_len > k_frame_max_payload) {
01577     rx_log_error_val(s_tag, "Send failed: payload too large", params->payload_len);
01578     return k_rx_err_invalid_arg;
01579   }
01580
01581   rx_err_t err = internal_route_send(mgr, params);
01582
01583   if (err != k_rx_ok) {
01584     rx_log_error_val(s_tag, "Transport send failed", err);
01585   }
01586
01587   /* Output decoded ASCII on success */
01588   if (err == k_rx_ok && (int)mgr->enable_decoded_output) {
01589     internal_output_decoded_from_params(mgr, params);
01590   }
01591
01592   return err;
01593 }
```

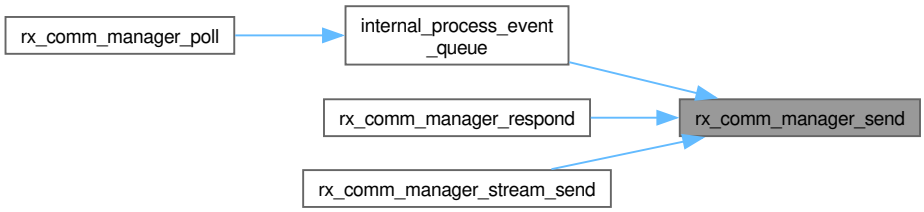
References rx_comm_manager_t::enable_decoded_output, rx_comm_manager_t::initialized, internal_output_decode
internal_route_send(), rx_comm_send_params_t::payload, rx_comm_send_params_t::payload_len,
and s_tag.

Referenced by internal_process_event_queue(), rx_comm_manager_respond(), and rx_comm_manager_stream_send(

Here is the call graph for this function:



Here is the caller graph for this function:



4.1.5.11 rx_comm_manager_set_auto_retransmit()

```
rx_err_t rx_comm_manager_set_auto_retransmit (
                rx_comm_manager_t * mgr,
                bool enabled,
                const rx_spi_comm_retransmit_config_t * config)   [nodiscard]
```

Enable or disable automatic retransmission on SPI channel.

Thin pass-through to rx_spi_comm_set_auto_retransmit() on the SPI handle.

Parameters

in,out	mgr	Communication manager
in	enabled	true to enable, false to disable
in	config	Retransmit config (NULL to keep current/defaults)

Returns

rx_err_t Error code from rx_spi_comm_set_auto_retransmit()

Since

Version 1.0.0

Definition at line 1982 of file rx_comm_manager.c.

```
01985 {
01986   if (mgr == nullptr) {
01987     return k_rx_err_invalid_arg;
01988   }
01989
01990   if (!mgr->initialized) {
01991     return k_rx_err_invalid_state;
01992   }
01993
01994   if (mgr->spi_handle == nullptr) {
01995     return k_rx_err_not_supported;
01996   }
01997
01998   return rx_spi_comm_set_auto_retransmit(mgr->spi_handle, enabled, config);
01999 }
```

References rx_comm_manager_t::initialized, and rx_comm_manager_t::spi_handle.

4.1.5.12 rx_comm_manager_stream_send()

```
rx_err_t rx_comm_manager_stream_send (
        rx_comm_manager_t * mgr,
        const rx_comm_send_params_t * params)   [nodiscard]
```

Send data on the stream path (fire-and-forget).

Stream path is for time-sensitive data like telemetry where stale data has no value. Sends immediately with NO retries on any transport. If the send fails, the data is simply dropped.

- USB CDC: Direct send, no retries (USB HW provides reliability)
- SPI: Direct send, no retries (stale telemetry is worthless)

Parameters

in,out	mgr	Pointer to initialized manager
in	params	Send parameters (channel, type, flags, payload, payload_len)

Returns

k_rx_ok on success

k_rx_err_invalid_arg if mgr or params is nullptr

k_rx_err_invalid_state if not initialized or channel not enabled

Precondition

mgr must be initialized

params must be valid

Postcondition

Data sent or silently dropped on failure

Note

This is a thin wrapper over rx_comm_manager_send() for semantic clarity

See also

rx_comm_manager_event_send() For reliable command delivery

Since

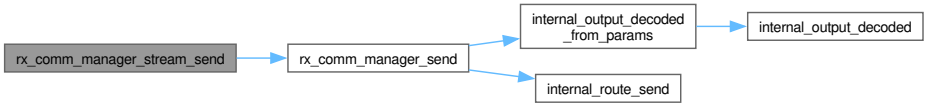
Version 1.0.0

Definition at line 1690 of file rx_comm_manager.c.

```
01691 {
01692   /* Stream path: fire-and-forget, no retries, no queuing.
01693    * Telemetry and other time-sensitive data goes here.
01694    * Stale data has no value, so failure is silently accepted. */
01695   return rx_comm_manager_send(mgr, params);
01696 }
```

References rx_comm_manager_send().

Here is the call graph for this function:



4.2 rx_comm_manager.h

Go to the documentation of this file.

```
00001 /**
00002  * @file rx_comm_manager.h
00003  * @brief Unified Multi-Channel Communication Manager for USB CDC and SPI Protocols
00004  *
00005  * @details
00006  * ## Overview
00007  * Production-quality communication manager providing unified abstraction over
00008  * multiple simultaneous communication channels (USB CDC and SPI) with automatic
00009  * frame protocol handling, non-blocking polling architecture, optional decoded
00010  * ASCII output for debugging, and callback-based event notification.
00011  *
00012  * **Purpose**: Coordinate independent, simultaneous communication channels from
00013  * RX72N MCU to Raspberry Pi 5 host, enabling command/response messaging over
00014  * both USB (via CDC class) and SPI (direct hardware connection).
00015  *
00016  * **Protocol**: Both channels use identical frame protocol (rx_frame) allowing
00017  * host to send commands on either channel. Manager automatically handles:
00018  * - Frame reception and validation
00019  * - ASCII decoding output (debugging via USB Port 1)
00020  * - Channel-aware response routing
00021  * - User callback invocation
00022  *
00023  * **Key Features:**
00024  * - **Multi-channel support**: Simultaneous USB and SPI communication
00025  * - **Non-blocking architecture**: Poll-based, no busy-waiting or blocking calls
00026  * - **Protocol-agnostic**: Both channels use same frame format (type, flags, payload, CRC)
00027  * - **Automatic ASCII output**: Optional decoded frames to USB Port 1 (human-readable debugging)
00028  * - **Callback-driven**: User-defined handler invoked on frame reception
00029  * - **Channel awareness**: Responses automatically routed to originating channel
00030  * - **Zero-copy where possible**: Direct access to frame buffers
00031  * - **Thread-safe design**: Can be called from RTOS task or main loop
00032  *
00033  * ## System Architecture
00034  *
00035  * Communication manager sits between application logic and low-level transport drivers:
00036  *
00037  * @dot
00038  * digraph comm_manager_arch {
00039  *   rankdir=TB;
00040  *   node [shape=box, style=rounded];
00041  *
00042  *   app [label="Application\n(Command Handlers)", fillcolor=lightblue, style=filled];
00043  *   comm_mgr [label="Communication Manager\n(This Module)", fillcolor=lightgreen, style=filled];
00044  *   usb_comm [label="rx_usb_comm\n(USB CDC Transport)", fillcolor=lightyellow, style=filled];
00045  *   spi_comm [label="rx_spi_comm\n(SPI Transport)", fillcolor=lightyellow, style=filled];
00046  *   usb_hw [label="USB Hardware\n(USB peripheral)", fillcolor=lightgray, style=filled];
00047  *   spi_hw [label="SPI Hardware\n(RSPI peripheral)", fillcolor=lightgray, style=filled];
00048  *   rpi5 [label="Raspberry Pi 5\n(Host)", shape=component];
00049  *
00050  *   app -> comm_mgr [label="poll()\nsend()\nrespond()"];
00051  *   comm_mgr -> app [label="callback(channel, frame)", dir=back];
00052  *   comm_mgr -> usb_comm [label="receive_frame()"];
00053  *   comm_mgr -> spi_comm [label="receive_frame()"];
00054  *   usb_comm -> usb_hw [label="CDC transfers"];
00055  *   spi_comm -> spi_hw [label="SPI transfers"];
00056  *   usb_hw -> rpi5 [label="/dev/ttyACM0"];
00057  *   spi_hw -> rpi5 [label="SPI bus"];
00058  *
00059  *   decoded [label="USB Port 1\n(ASCII Debug)", fillcolor=lightpink, style=filled];
00060  *   comm_mgr -> decoded [label="ASCII frames\n(optional)"];
00061  *   decoded -> rpi5 [label="/dev/ttyACM1"];
00062  * }
00063  * @enddot
00064  *
00065  * ## Channel Configuration
00066  *
00067  * The manager supports two independent communication channels:
00068  *
00069  * | Channel | Transport | Device | Usage | RPi5 Interface |
00070  * |---------|-----------|--------|-------|----------------|
00071  * | **USB** | CDC class | USB Port 0 | Primary protocol channel | /dev/ttyACM0 |
00072  * | **SPI** | RSPI peripheral | RSPI channel | High-speed backup channel | /dev/spidev0.0 |
00073  * | **Debug** | CDC class | USB Port 1 | ASCII decoded output (optional) | /dev/ttyACM1 |
00074  *
00075  * **USB Channel (k_comm_channel_usb):**
00076  * - Plug-and-play connectivity (no additional wiring)
00077  * - Full-speed USB 2.0 (up to 12 Mbps theoretical, ~1 MB/s practical)
00078  * - Automatic enumeration and driver loading on RPi5
00079  * - Suitable for command/response and moderate-bandwidth telemetry
00080  *
00081  * **SPI Channel (k_comm_channel_spi):**
00082  * - Dedicated hardware connection (4-6 wires: COPI, CIPO, SCK, CS, optional handshake)
```

```
00083  * - Configurable speed (typically 10-20 Mbps)
00084  * - Lower latency than USB (no protocol overhead)
00085  * - Suitable for high-frequency control loops and real-time data
00086  *
00087  * **Debug Output (USB Port 1):**
00088  * - Human-readable ASCII representation of all frames
00089  * - Format: `[USB/SPI] TYPE=cmd FLAGS=0x00 LEN=42 DATA=...`
00090  * - Enabled via `enable_decoded_output` config flag
00091  * - Zero impact on protocol channels (independent USB endpoint)
00092  *
00093  * ## Communication Flow State Machine
00094  *
00095  * The manager operates as a stateless polling architecture with per-frame state:
00096  *
00097  * @startuml
00098  * [*] --> Idle
00099  * Idle --> Polling : rx_comm_manager_poll()
00100  * Polling --> CheckUSB : USB enabled?
00101  * Polling --> CheckSPI : USB disabled or checked
00102  * CheckUSB --> DecodeUSB : Frame available
00103  * CheckUSB --> CheckSPI : No frame or error
00104  * DecodeUSB --> InvokeCallback : Valid frame
00105  * InvokeCallback --> FormatASCII : Decoded output enabled
00106  * InvokeCallback --> CheckSPI : No decoded output
00107  * FormatASCII --> SendASCII : Format success
00108  * SendASCII --> CheckSPI
00109  * CheckSPI --> DecodeSPI : Frame available
00110  * CheckSPI --> Idle : No frame or error
00111  * DecodeSPI --> InvokeCallback2 : Valid frame
00112  * InvokeCallback2 --> FormatASCII2 : Decoded output enabled
00113  * InvokeCallback2 --> Idle : No decoded output
00114  * FormatASCII2 --> SendASCII2 : Format success
00115  * SendASCII2 --> Idle
00116  *
00117  * note right of Polling
00118  *   Non-blocking check
00119  *   Returns immediately if
00120  *   no frames available
00121  * end note
00122  *
00123  * note right of InvokeCallback
00124  *   User callback receives:
00125  *   - channel (USB or SPI)
00126  *   - parsed frame
00127  *   - user context
00128  * end note
00129  * @enduml
00130  *
00131  * @par State Descriptions:
00132  * - **Idle**: Manager initialized, waiting for poll() call
00133  * - **Polling**: Actively checking enabled channels for frames
00134  * - **CheckUSB**: Query USB channel for available frame
00135  * - **CheckSPI**: Query SPI channel for available frame
00136  * - **DecodeUSB/DecodeSPI**: Validate and parse received frame
00137  * - **InvokeCallback**: Call user-provided frame handler
00138  * - **FormatASCII**: Convert frame to human-readable string
00139  * - **SendASCII**: Transmit ASCII representation to debug port
00140  *
00141  * ## Hardware Requirements
00142  *
00143  * | Resource | Requirement | Usage |
00144  * |----------|-------------|-------|
00145  * | **USB Peripheral** | 1 instance (full-speed) | CDC class with 2 ports (protocol + debug) |
00146  * | **RSPI Peripheral** | 1 channel (optional) | SPI controller mode, 10-20 Mbps |
00147  * | **GPIO** | 4-6 pins (if SPI) | COPI, CIPO, SCK, CS, optional handshake/interrupt |
00148  * | **RAM** | ~2.5 KB per instance | Handle + ASCII buffer (2048 bytes) |
00149  * | **Flash** | ~4 KB | Code + string constants |
00150  * | **RTOS Resources** | None | Stateless, no threads/mutexes/semaphores |
00151  *
00152  * ## Performance Characteristics
00153  *
00154  * | Operation | Execution Time | Notes |
00155  * |-----------|----------------|-------|
00156  * | **rx_comm_manager_init()** | ~50 us | Handle initialization, no hardware config |
00157  * | **rx_comm_manager_poll()** | ~15-200 us | Depends on channels enabled and frames available |
00158  * | - No frames available | ~15 us | Quick check both channels, early return |
00159  * | - USB frame received | ~120 us | Frame validation + callback + ASCII (if enabled) |
00160  * | - SPI frame received | ~80 us | Faster than USB (less overhead) |
00161  * | - Both frames received | ~200 us | Sequential processing of both channels |
00162  * | **rx_comm_manager_send()** | ~100-150 us | Frame encoding + transport layer send |
00163  * | **rx_comm_manager_respond()** | ~100-150 us | Same as send (convenience wrapper) |
00164  * | **Callback overhead** | ~5-50 us | User-defined, depends on handler complexity |
00165  *
00166  * All timing measured @ 240 MHz CPU clock, -O2 optimization, typical frame size 64 bytes.
00167  *
00168  * ## Memory Usage
00169  *
```

```
00170  * | Allocation | Size | Lifetime |
00171  * |------------|------|----------|
00172  * | **rx_comm_manager_t** | ~12 KB | Application-managed (typically static/global) |
00173  * | - Pointers + state | ~80 bytes | Handle metadata, heartbeat, queue indices |
00174  * | - ASCII buffer | 2048 bytes | Decoded output formatting (if enabled) |
00175  * | - Event queue | 8 x ~1028 bytes | Static FIFO (k_comm_event_queue_depth entries) |
00176  * | - Heartbeat state | ~20 bytes | Per-transport heartbeat tracking |
00177  * | **Stack (poll)** | ~120 bytes | Function call duration only |
00178  * | **Stack (send)** | ~80 bytes | Function call duration only |
00179  * | **Static data (.rodata)** | ~400 bytes | Channel names, format strings |
00180  * | **Code (.text)** | ~6 KB | All functions |
00181  *
00182  * Total RAM: ~12 KB per manager instance (zero dynamic allocation).
00183  * The event queue dominates memory: 8 entries x (k_frame_max_payload + metadata).
00184  *
00185  * **Memory Layout** (rx_comm_manager_t structure):
00186  *
00187  * | Offset | Size | Field | Notes |
00188  * |--------|------|-------|-------|
00189  * | 0x00 | 8 | usb_handle | Pointer to USB transport |
00190  * | 0x08 | 8 | spi_handle | Pointer to SPI transport |
00191  * | 0x10 | 8 | callback | Function pointer |
00192  * | 0x18 | 8 | callback_ctx | User context pointer |
00193  * | 0x20 | 2048 | ascii_buffer | ASCII formatting buffer |
00194  * | - | 1 | enable_decoded_output | Boolean flag |
00195  * | - | 1 | initialized | Boolean flag |
00196  * | - | ~20 | heartbeat[2] | Per-transport heartbeat state |
00197  * | - | ~16 | link_status_cb/ctx | Callback + context |
00198  * | - | ~8224 | event_queue[8] | 8 x rx_comm_event_entry_t |
00199  * | - | 3 | queue head/tail/count | Queue management |
00200  * | **Total** | **~10.5 KB** | | (actual size may vary by compiler) |
00201  *
00202  * ## Thread Safety Analysis
00203  *
00204  * **Conditional Thread Safety** - Safe if following rules are observed:
00205  *
00206  * 1. **Single poll context**: Only call `poll()` from ONE thread/task/loop
00207  * 2. **Callback restrictions**: User callback must not call manager functions (no recursion)
00208  * 3. **Send synchronization**: If calling `send()`/`respond()` from multiple threads, use external mutex
00209  * 4. **Transport thread safety**: Underlying USB/SPI transports must be thread-safe (typically are)
00210  *
00211  * **Not safe for:**
00212  * - Concurrent `poll()` calls from multiple threads
00213  * - Calling manager functions from within frame callback
00214  * - Simultaneous `send()` calls without external synchronization
00215  *
00216  * **Recommendation**: Use dedicated communication task:
00217  * ```c
00218  * void comm_task_entry(ULONG ctx) {
00219  *     rx_comm_manager_t* mgr = (rx_comm_manager_t*)ctx;
00220  *     while (1) {
00221  *         rx_comm_manager_poll(mgr);
00222  *         tx_thread_sleep(1);  // 1ms polling period = 1kHz
00223  *     }
00224  * }
00225  * ```
00226  *
00227  * ## Integration Example
00228  *
00229  * @par Complete Setup and Usage:
00230  * @code{.c}
00231  * #include "rx_comm_manager.h"
00232  * #include "rx_usb_comm.h"
00233  * #include "rx_spi_comm.h"
00234  *
00235  * // Frame handler callback
00236  * void on_frame_received(rx_comm_channel_t channel, const rx_frame_t* frame, void* ctx) {
00237  *     rx_comm_manager_t* mgr = (rx_comm_manager_t*)ctx;
00238  *
00239  *     rx_log_info("COMM", "Frame from %s: type=0x%02X len=%u",
00240  *                 rx_comm_manager_channel_name(channel),
00241  *                 frame->type, frame->payload_len);
00242  *
00243  *     // Parse frame based on type
00244  *     switch (frame->type) {
00245  *         case k_frame_type_command:
00246  *             // Handle command
00247  *             uint8_t response[128];
00248  *             uint32_t response_len = process_command(frame->payload,
00249  *                                                     frame->payload_len,
00250  *                                                     response, sizeof(response));
00251  *
00252  *             // Respond on same channel
00253  *             rx_comm_manager_respond(mgr, channel, response, response_len);
00254  *             break;
00255  *
00256  *         case k_frame_type_telemetry_req:
```

```
00257 *           // Send telemetry data
00258 *           send_telemetry(mgr, channel);
00259 *           break;
00260 *
00261 *       default:
00262 *           rx_log_warn("COMM", "Unknown frame type: 0x%02X", frame->type);
00263 *           break;
00264 *   }
00265 * }
00266 *
00267 * // Main application
00268 * int main(void) {
00269 *   // Initialize transports
00270 *   rx_usb_comm_handle_t usb_comm;
00271 *   rx_spi_comm_handle_t spi_comm;
00272 *
00273 *   rx_usb_comm_init(&usb_comm, &usb_config);
00274 *   rx_spi_comm_init(&spi_comm, &spi_config);
00275 *
00276 *   // Initialize communication manager
00277 *   rx_comm_manager_t comm_mgr;
00278 *   rx_comm_manager_config_t cfg = {
00279 *       .usb_handle = &usb_comm,
00280 *       .spi_handle = &spi_comm,
00281 *       .callback = on_frame_received,
00282 *       .callback_ctx = &comm_mgr,  // Pass manager to callback
00283 *       .enable_decoded_output = true,  // ASCII debug on Port 1
00284 *   };
00285 *
00286 *   rx_err_t err = rx_comm_manager_init(&comm_mgr, &cfg);
00287 *   if (err != k_rx_ok) {
00288 *       rx_log_error("APP", "Comm manager init failed: 0x%X", err);
00289 *       return -1;
00290 *   }
00291 *
00292 *   // Main communication loop
00293 *   while (1) {
00294 *       // Poll for incoming frames (non-blocking)
00295 *       err = rx_comm_manager_poll(&comm_mgr);
00296 *
00297 *       // err == k_rx_ok: Frame(s) received and processed
00298 *       // err == k_rx_err_timeout: No frames available (normal)
00299 *
00300 *       // Other application tasks...
00301 *
00302 *       tx_thread_sleep(1);  // 1ms = 1kHz polling rate
00303 *   }
00304 *
00305 *   return 0;
00306 * }
00307 * @endcode
00308 *
00309 * @par Sending Unsolicited Data (Telemetry):
00310 * @code{.c}
00311 * void send_periodic_telemetry(rx_comm_manager_t* mgr) {
00312 *   // Gather telemetry data
00313 *   telemetry_data_t telemetry;
00314 *   gather_telemetry(&telemetry);
00315 *
00316 *   // Serialize to buffer
00317 *   uint8_t payload[256];
00318 *   uint32_t payload_len = serialize_telemetry(&telemetry, payload, sizeof(payload));
00319 *
00320 *   // Send on USB channel (more reliable for telemetry)
00321 *   rx_comm_send_params_t params = {
00322 *       .channel = k_comm_channel_usb,
00323 *       .type = k_frame_type_telemetry,
00324 *       .flags = 0,
00325 *       .payload = payload,
00326 *       .payload_len = payload_len,
00327 *   };
00328 *
00329 *   rx_err_t err = rx_comm_manager_send(mgr, &params);
00330 *   if (err != k_rx_ok) {
00331 *       rx_log_warn("TELEM", "Failed to send telemetry: 0x%X", err);
00332 *   }
00333 * }
00334 * @endcode
00335 *
00336 * @par Module Dependencies:
00337 * - rx_usb_comm: USB CDC transport layer
00338 * - rx_spi_comm: SPI transport layer
00339 * - rx_frame: Frame protocol definition and encoding/decoding
00340 * - rx_err: Standardized error code system
00341 * - rx_log: Logging infrastructure
00342 *
00343 * @see rx_spi_comm.h SPI transport implementation
```

```
00344  * @see rx_uart_comm.h UART transport implementation
00345  * @see rx_i2c_comm.h I2C peripheral transport implementation
00346  * @see rx_frame.h Frame protocol specification
00347  * @see docs/sections/01_nanopb_protocol.tex Communication protocol documentation
00348  *
00349  * @par NASA Power of 10 Compliance:
00350  * - Rule 1: [OK] No goto, setjmp/longjmp, or recursion used
00351  * - Rule 2: [OK] All loops have compile-time bounded iterations (channel count = 2)
00352  * - Rule 3: [OK] Zero dynamic memory allocation (all buffers static)
00353  * - Rule 4: [OK] All functions < 60 lines
00354  * - Rule 5: [OK] Minimum 2 assertions per function via parameter validation
00355  * - Rule 6: [OK] Variables declared at smallest scope
00356  * - Rule 7: [OK] All return values checked and propagated
00357  * - Rule 8: [OK] Uses C23 typed enums exclusively (no magic numbers)
00358  * - Rule 9: [WARN] INTENTIONAL DEVIATION: Uses function pointers for callback (event-driven architecture)
00359  * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror (zero warnings)
00360  *
00361  * @par SOLID Principles:
00362  * - **S (Single Responsibility):** Only channel coordination and frame routing. Protocol parsing delegated to rx_frame, transport to rx_usb_comm/rx_spi_comm.
00363  * - **O (Open/Closed):** Extensible to new channels by adding enum value and transport handle. Closed for modification (no changes needed for new frame types).
00364  * - **L (Liskov Substitution):** USB and SPI transports interchangeable via common frame interface. Can substitute mock transports for testing.
00365  * - **I (Interface Segregation):** Clean separation: lifecycle (init/deinit), polling (poll), sending (send/respond), status (channel_ready). Users include only what they need.
00366  * - **D (Dependency Inversion):** Depends on abstractions (rx_frame protocol, generic transport interface) not concrete USB/SPI implementations.
00367  *
00368  * @warning Do NOT call manager functions from within frame callback (recursion risk)
00369  * @warning ASCII buffer is 2048 bytes - ensure payload_len + overhead < 2000 bytes to avoid truncation
00370  * @warning Polling from multiple threads without synchronization will cause corruption
00371  *
00372  * @attention Manager does not own USB/SPI handles - caller must initialize transports before manager
00373  * @attention Callback is invoked from poll() context - keep processing minimal or defer to task queue
00374  * @attention Decoded ASCII output adds ~80 us latency per frame - disable for high-frequency applications
00375  *
00376  * @author Locked, Inc.
00377  * @date 2026-01-27
00378  * @version 1.0.0
00379  * @copyright Copyright (c) 2026 Locked Inc.
00380  * SPDX-License-Identifier: MIT
00381  *
00382  * @par Changelog:
00383  * - 1.0.0 (2026-01-26): Initial implementation with USB and SPI channels
00384  * - 1.1.0 (2026-01-27): Added comprehensive Doxygen documentation, state machine diagrams, performance data
00385  */
00386
00387 #pragma once
00388
00389 #include <stdbool.h>
00390 #include <stdint.h>
00391
00392 #include "rx_err.h"
00393 #include "rx_frame.h"
00394 #include "rx_i2c_comm.h"
00395 #include "rx_spi_comm.h"
00396 #include "rx_spi_link.h"
00397 #include "rx_uart_comm.h"
00398
00399 #ifdef __cplusplus
00400 extern "C" {
00401 #endif
00402
00403 /*
     ================================================================================
00404  * Constants
00405  *
     ================================================================================
00406  */
00407
00408 /**
00409  * @enum rx_comm_manager_constants_t
00410  * @brief Communication manager compile-time constants
00411  *
00412  * @details
00413  * Defines buffer sizes and limits for the communication manager. These
00414  * constants are chosen based on typical frame sizes and ASCII formatting
00415  * requirements.
00416  *
00417  * **Design Rationale:**
00418  * - **2048-byte ASCII buffer**: Accommodates maximum frame size (1024 bytes payload)
00419  *   plus formatting overhead (~1000 bytes for hex dump + headers)
00420  * - Buffer is stack-allocated in handle structure (not dynamically allocated)
00421  * - Sufficient for debugging without excessive memory usage
00422  *
00423  * @see rx_comm_manager_t.ascii_buffer
```

```
00424  * @since Version 1.0.0
00425  */
00426 typedef enum : uint16_t {
00427   /**
00428    * @brief ASCII formatting buffer size in bytes
00429    * @details
00430    * Size of internal buffer used to format decoded ASCII output for debug port.
00431    * Must accommodate:
00432    * - Header line: ~80 bytes (‘[USB] TYPE=cmd FLAGS=0x00 LEN=1024‘)
00433    * - Payload hex dump: payload_len * 3 bytes (‘XX ` per byte)
00434    * - Footer + null terminator: ~20 bytes
00435    *
00436    * Maximum frame payload: 1024 bytes
00437    * Required buffer: 80 + (1024 * 3) + 20 = 3172 bytes
00438    * Allocated: 2048 bytes (limits payload display to ~650 bytes, which is acceptable)
00439    *
00440    * @par Value: 2048 bytes
00441    * @par Rationale: Balance between functionality and memory usage
00442    */
00443   k_comm_manager_ascii_buf_size = 2048,
00444
00445   /**
00446    * @brief Maximum number of entries in the event FIFO queue
00447    * @details
00448    * Static FIFO queue depth for event (command) frames that require reliable
00449    * delivery. Events are queued and sent with SPI-only retry + exponential
00450    * backoff. Queue is statically allocated (zero dynamic memory).
00451    *
00452    * @par Value: 8 entries
00453    * @par Rationale: Small queue prevents unbounded memory, sufficient for
00454    *   typical command burst scenarios
00455    */
00456   k_comm_event_queue_depth = 8,
00457 } rx_comm_manager_constants_t;
00458
00459 /**
00460  * @enum rx_comm_heartbeat_constants_t
00461  * @brief Heartbeat timing constants
00462  *
00463  * @details
00464  * Dual-detection heartbeat model:
00465  * - **Implicit timeout (200ms)**: Any valid frame resets the timer. If no
00466  *   frames arrive within 200ms, the link is declared dead.
00467  * - **Explicit PING (1000ms)**: Sent when link is idle for >1s as a probe.
00468  *   Firmware primarily receives PINGs from the gateway.
00469  * - **Check interval (50ms)**: How often to evaluate heartbeat status.
00470  *
00471  * @since Version 1.0.0
00472  */
00473 typedef enum : uint32_t {
00474   k_heartbeat_implicit_timeout_ms = 200,  /**< Declare link dead if no frame for 200ms */
00475   k_heartbeat_ping_interval_ms    = 1000, /**< Send PING probe when idle for 1s */
00476   k_heartbeat_check_interval_ms   = 50,   /**< Heartbeat evaluation interval (200ms / 4) */
00477 } rx_comm_heartbeat_constants_t;
00478
00479 /**
00480  * @enum rx_comm_event_retry_constants_t
00481  * @brief Event queue retry constants for SPI HARQ
00482  *
00483  * @details
00484  * Events (commands) sent via SPI use retry with exponential backoff.
00485  * USB sends are fire-and-forget (no retries).
00486  *
00487  * @since Version 1.0.0
00488  */
00489 typedef enum : uint16_t {
00490   k_event_max_retries        = 3,  /**< Maximum SPI retry attempts */
00491   k_event_initial_backoff_ms = 10, /**< Initial backoff delay (ms) */
00492   k_event_max_backoff_ms = 15, /**< Maximum backoff delay (ms): caps 10«(retries-1) at retries=2 */
00493 } rx_comm_event_retry_constants_t;
00494
00495 /*
      ================================================================================
00496  * Types
00497  *
      ================================================================================
00498  */
00499
00500 /**
00501  * @enum rx_comm_channel_t
00502  * @brief Communication channel identifiers
00503  *
00504  * @details
00505  * Enumerates the available communication channels. The manager can handle
00506  * multiple channels simultaneously, with each channel operating independently.
00507  *
00508  * **Channel Characteristics:**
```

```
00509  *
00510  * | Channel | Bandwidth | Latency | Setup Complexity | Use Case |
00511  * |---------|-----------|---------|------------------|----------|
00512  * | USB CDC | ~1 MB/s | ~1-5 ms | Low (plug-and-play) | Commands, telemetry |
00513  * | SPI | ~2 MB/s | ~0.1-1 ms | Medium (requires wiring) | High-speed data, control |
00514  *
00515  * @par Usage Pattern:
00516  * @code{.c}
00517  * // Check which channel frame came from
00518  * void callback(rx_comm_channel_t channel, const rx_frame_t* frame, void* ctx) {
00519  *     if (channel == k_comm_channel_usb) {
00520  *         // Received via USB
00521  *     } else if (channel == k_comm_channel_spi) {
00522  *         // Received via SPI
00523  *     }
00524  * }
00525  * @endcode
00526  *
00527  * @see rx_comm_manager_channel_name() Get human-readable channel name
00528  * @see rx_comm_frame_callback_t Callback receives channel identifier
00529  * @since Version 1.0.0
00530  */
00531 typedef enum : uint8_t {
00532     /**
00533      * @brief UART channel (SCI9 via CY7C65213 USB-UART bridge)
00534      * @details
00535      * Primary link to the Raspberry Pi 5 gateway. Carries protobuf command /
00536      * response / telemetry frames AND k_frame_type_log_message frames with the
00537      * RX72N runtime log stream, all multiplexed on the same byte stream.
00538      *
00539      * On the Pi5 the bridge enumerates as /dev/ttyACM0 at 115200 baud (CY7C65213 is a CDC-ACM class device; see
         k_uart_default_baudrate in libs/rx_hal/src/uart.c).
00540      *
00541      * @par Hardware: SCI9 (PB7 / TXD9, PB6 / RXD9) + CY7C65213 bridge IC
00542      * @par Value: 0
00543      */
00544     k_comm_channel_uart = 0,
00545
00546     /**
00547      * @brief SPI peripheral channel
00548      * @details
00549      * Dedicated SPI connection to the ROS2 spi_driver_node for low-latency,
00550      * high-bandwidth motor control. 10 MHz clock, full-duplex.
00551      *
00552      * @par Hardware: RSPI peripheral (COPI / CIPO / SCK / CS)
00553      * @par Value: 1
00554      */
00555     k_comm_channel_spi = 1,
00556
00557     /**
00558      * @brief I2C peripheral channel (RIIC0)
00559      * @details
00560      * I2C connection where RX72N acts as I2C peripheral device and RPi5 is the
00561      * I2C controller. Optional auxiliary channel.
00562      *
00563      * @par Hardware: RIIC0 peripheral (P16 / SCL0, P17 / SDA0)
00564      * @par Value: 2
00565      */
00566     k_comm_channel_i2c = 2,
00567
00568     /**
00569      * @brief Total number of supported channels (3)
00570      *
00571      * @par Value: 3 (UART + SPI + I2C)
00572      * @par Invariant: Must equal number of enum values (excluding this one)
00573      */
00574     k_comm_channel_count = 3,
00575 } rx_comm_channel_t;
00576
00577 /**
00578  * @typedef rx_comm_frame_callback_t
00579  * @brief Frame received callback function type
00580  *
00581  * @details
00582  * User-defined function invoked by communication manager when a complete,
00583  * valid frame is received on any enabled channel. The callback receives:
00584  * - Channel identifier (which channel the frame arrived on)
00585  * - Parsed frame structure (type, flags, payload)
00586  * - User context pointer (passed during initialization)
00587  *
00588  * ## Callback Contract
00589  *
00590  * **Responsibilities:**
00591  * - Parse frame payload based on frame type
00592  * - Handle command execution or data processing
00593  * - Optionally send response via `rx_comm_manager_respond()`
00594  * - Return quickly to avoid blocking communication loop
```

```
00595  *
00596  * **Restrictions:**
00597  * - Do NOT call `rx_comm_manager_poll()` from callback (recursion!)
00598  * - Keep processing minimal (~50 us recommended)
00599  * - For slow operations, queue to task and return immediately
00600  * - Do NOT modify frame structure (it may be const or recycled)
00601  *
00602  * **Thread Safety:**
00603  * - Callback executed in context of `poll()` caller
00604  * - If poll() called from RTOS task, callback runs in that task context
00605  * - Synchronization required if accessing shared resources
00606  *
00607  * @param[in] channel The communication channel the frame was received on
00608  *   - Type: rx_comm_channel_t
00609  *   - Valid values: k_comm_channel_usb or k_comm_channel_spi
00610  *   - Use to determine response routing
00611  *
00612  * @param[in] frame Pointer to the received and validated frame
00613  *   - Type: const rx_frame_t*
00614  *   - Valid until callback returns
00615  *   - Contains: type, flags, payload, payload_len (CRC already validated)
00616  *   - Do NOT modify or store pointer (frame may be recycled)
00617  *
00618  * @param[in] ctx User context pointer
00619  *   - Type: void* (application-defined)
00620  *   - Value passed during `rx_comm_manager_init()`
00621  *   - Typically pointer to manager, application state, or nullptr
00622  *   - Use to access application resources without globals
00623  *
00624  * @par Usage Example - Basic Command Handler:
00625  * @code{.c}
00626  * void on_frame(rx_comm_channel_t channel, const rx_frame_t* frame, void* ctx) {
00627  *     rx_comm_manager_t* mgr = (rx_comm_manager_t*)ctx;
00628  *
00629  *     // Log reception
00630  *     rx_log_info("COMM", "RX from %s: type=0x%02X len=%u",
00631  *                 rx_comm_manager_channel_name(channel),
00632  *                 frame->type, frame->payload_len);
00633  *
00634  *     // Dispatch based on type
00635  *     switch (frame->type) {
00636  *         case k_frame_type_command:
00637  *             handle_command(mgr, channel, frame);
00638  *             break;
00639  *         case k_frame_type_telemetry_req:
00640  *             send_telemetry(mgr, channel);
00641  *             break;
00642  *         default:
00643  *             rx_log_warn("COMM", "Unknown type: 0x%02X", frame->type);
00644  *             break;
00645  *     }
00646  * }
00647  * @endcode
00648  *
00649  * @par Usage Example - Deferred Processing:
00650  * @code{.c}
00651  * // For time-consuming operations, defer to queue
00652  * void on_frame(rx_comm_channel_t channel, const rx_frame_t* frame, void* ctx) {
00653  *     app_state_t* app = (app_state_t*)ctx;
00654  *
00655  *     // Copy frame to queue (frame pointer only valid during callback)
00656  *     command_msg_t msg = {
00657  *         .channel = channel,
00658  *         .type = frame->type,
00659  *         .payload_len = frame->payload_len,
00660  *     };
00661  *     memcpy(msg.payload, frame->payload, frame->payload_len);
00662  *
00663  *     // Queue for processing by command handler task
00664  *     UINT status = tx_queue_send(&app->cmd_queue, &msg, TX_NO_WAIT);
00665  *     if (status != TX_SUCCESS) {
00666  *         rx_log_error("COMM", "Command queue full!");
00667  *     }
00668  * }
00669  * @endcode
00670  *
00671  * @warning Callback must return quickly (<50 us recommended) to avoid blocking communication
00672  * @warning Do NOT call rx_comm_manager_poll() from callback (causes recursion)
00673  * @warning Frame pointer only valid during callback - copy data if needed beyond callback scope
00674  *
00675  * @see rx_comm_manager_respond() Send response on same channel
00676  * @see rx_comm_manager_send() Send frame on specific channel
00677  * @see rx_frame_t Frame structure definition
00678  *
00679  * @since Version 1.0.0
00680  */
00681 typedef void (*rx_comm_frame_callback_t)(rx_comm_channel_t channel,
```

```
00682                                   const rx_frame_t* frame,
00683                                   void*             ctx);
00684
00685 /**
00686  * @enum rx_comm_link_status_t
00687  * @brief Link health status from heartbeat monitoring
00688  *
00689  * @details
00690  * Reported per-transport link health. Determined by the dual-detection
00691  * heartbeat model (implicit 200ms timeout + explicit 1s PING).
00692  *
00693  * @since Version 1.0.0
00694  */
00695 typedef enum : uint8_t {
00696   k_link_status_unknown = 0, /**< No frames received yet */
00697   k_link_status_healthy = 1, /**< Frames received within implicit timeout */
00698   k_link_status_dead    = 2, /**< No frames within implicit timeout (200ms) */
00699 } rx_comm_link_status_t;
00700
00701 /**
00702  * @struct rx_comm_event_entry_t
00703  * @brief Single entry in the static event FIFO queue
00704  *
00705  * @details
00706  * Events (commands) that require reliable delivery are queued here.
00707  * Each entry contains a copy of the payload and metadata needed for
00708  * retry logic. SPI entries retry with exponential backoff; USB entries
00709  * are fire-and-forget.
00710  *
00711  * @since Version 1.0.0
00712  */
00713 typedef struct {
00714   rx_comm_channel_t channel;                    /**< Target transport */
00715   rx_frame_type_t   type;                       /**< Frame type */
00716   uint8_t           flags;                      /**< Frame flags */
00717   uint8_t           payload[k_frame_max_payload]; /**< Payload copy */
00718   uint32_t          payload_len;                /**< Payload length (bytes) */
00719   uint8_t           retries;                    /**< Retry attempts so far */
00720   uint32_t          next_retry_time_ms;         /**< Earliest time (ms) for next retry */
00721   bool              occupied;                   /**< Slot in use */
00722 } rx_comm_event_entry_t;
00723
00724 /**
00725  * @struct rx_comm_heartbeat_state_t
00726  * @brief Per-transport heartbeat tracking state
00727  *
00728  * @details
00729  * Tracks the last time a valid frame was received on each transport.
00730  * Used by the poll loop to evaluate link health per the dual-detection
00731  * heartbeat model.
00732  *
00733  * @since Version 1.0.0
00734  */
00735 typedef struct {
00736   uint32_t              last_rx_ms; /**< Timestamp (ms) of last valid frame */
00737   rx_comm_link_status_t status;     /**< Current link health */
00738 } rx_comm_heartbeat_state_t;
00739
00740 /**
00741  * @typedef rx_comm_link_status_callback_t
00742  * @brief Callback invoked when a link status changes
00743  *
00744  * @param[in] channel    Which transport changed status
00745  * @param[in] new_status New link status
00746  * @param[in] ctx        User context pointer
00747  *
00748  * @since Version 1.0.0
00749  */
00750 typedef void (*rx_comm_link_status_callback_t)(rx_comm_channel_t     channel,
00751                                                rx_comm_link_status_t new_status,
00752                                                void*                 ctx);
00753
00754 /**
00755  * @enum rx_comm_channel_mask_t
00756  * @brief Bitmask selecting which comm channels to initialize and use
00757  *
00758  * @details
00759  * Each bit corresponds to one comm channel.  Bit position matches the numeric
00760  * value of the corresponding rx_comm_channel_t enumerator so that a single
00761  * shift converts between the two representations.
00762  *
00763  * Pass a bitmask in rx_comm_manager_config_t::enabled_channels to skip
00764  * initialization of channels that are not needed (e.g. exclude UART when the
00765  * log backend also uses SCI9).
00766  *
00767  * @see rx_comm_channel_t  Channel enumerator
00768  * @see rx_comm_manager_config_t::enabled_channels  Usage
```

```
00769  * @since Version 1.0.0
00770  */
00771 typedef enum : uint8_t {
00772   k_comm_channel_mask_none = 0x00U, /**< No channels enabled */
00773   k_comm_channel_mask_uart = 0x01U, /**< UART (k_comm_channel_uart = 0) */
00774   k_comm_channel_mask_spi  = 0x02U, /**< SPI  (k_comm_channel_spi  = 1) */
00775   k_comm_channel_mask_i2c  = 0x04U, /**< I2C  (k_comm_channel_i2c  = 2) */
00776   k_comm_channel_mask_all  =        /**< All three channels */
00777   (k_comm_channel_mask_uart | k_comm_channel_mask_spi | k_comm_channel_mask_i2c),
00778 } rx_comm_channel_mask_t;
00779
00780 /**
00781  * @struct rx_comm_manager_config_t
00782  * @brief Communication manager configuration
00783  *
00784  * @details
00785  * Configuration structure passed to `rx_comm_manager_init()`. Defines which
00786  * channels are enabled, callback function, and optional features.
00787  *
00788  * **Configuration Guidelines:**
00789  * - At least one channel (usb_handle OR spi_handle) should be non-NULL
00790  * - Both channels can be enabled simultaneously
00791  * - Callback is optional (NULL = receive but don't notify)
00792  * - Decoded output recommended for development, disable for production
00793  *
00794  * **Memory Lifetime:**
00795  * - Config structure only read during init, can be stack-allocated
00796  * - Transport handles must remain valid for manager lifetime
00797  * - Callback function must remain valid (typically static function)
00798  *
00799  * @par Configuration Example:
00800  * @code{.c}
00801  * rx_comm_manager_config_t cfg = {
00802  *     .usb_handle = &usb_comm,              // Enable USB channel
00803  *     .spi_handle = &spi_comm,              // Enable SPI channel
00804  *     .callback = on_frame_received,        // Frame handler
00805  *     .callback_ctx = &app_state,           // User context
00806  *     .enable_decoded_output = true,        // ASCII debug output
00807  * };
00808  * @endcode
00809  *
00810  * @see rx_comm_manager_init() Initialization function
00811  * @since Version 1.0.0
00812  */
00813 typedef struct {
00814   /**
00815    * @brief SPI communication handle (NULL to disable SPI channel)
00816    * @details
00817    * Pointer to initialized SPI transport handle. If NULL, SPI channel
00818    * is disabled and all SPI operations are skipped.
00819    *
00820    * **Requirements:**
00821    * - Must be initialized via `rx_spi_comm_init()` before passing to manager
00822    * - Must remain valid for lifetime of manager
00823    * - Requires hardware SPI connection between RX72N and RPi5
00824    *
00825    * **Null handling:**
00826    * - NULL = SPI channel disabled (valid configuration)
00827    * - Non-NULL = SPI channel enabled
00828    *
00829    * @par Type: rx_spi_comm_handle_t* (pointer to SPI transport handle)
00830    * @par Valid values: Non-NULL initialized handle, or nullptr to disable
00831    * @par Lifetime: Must outlive manager instance
00832    */
00833   rx_spi_comm_handle_t* spi_handle;
00834
00835   /**
00836    * @brief I2C communication handle (NULL to disable I2C channel)
00837    * @details
00838    * Pointer to initialized I2C peripheral transport handle. If NULL, I2C channel
00839    * is disabled and all I2C operations are skipped. The RX72N acts as I2C
00840    * peripheral (device) and the RPi5 acts as I2C controller.
00841    *
00842    * @par Type: rx_i2c_comm_handle_t* (pointer to I2C transport handle)
00843    * @par Valid values: Non-NULL initialized handle, or nullptr to disable
00844    * @par Lifetime: Must outlive manager instance
00845    */
00846   rx_i2c_comm_handle_t* i2c_handle;
00847
00848   /**
00849    * @brief UART communication handle (NULL to disable UART channel)
00850    * @details
00851    * Pointer to initialized UART (SCI9) transport handle. If NULL, UART channel
00852    * is disabled and all UART operations are skipped.
00853    *
00854    * @par Type: rx_uart_comm_handle_t* (pointer to UART transport handle)
00855    * @par Valid values: Non-NULL initialized handle, or nullptr to disable
```

```
00856  * @par Lifetime: Must outlive manager instance
00857  */
00858 rx_uart_comm_handle_t* uart_handle;
00859
00860 /**
00861  * @brief Optional SPI link layer with HARQ (NULL to use raw SPI)
00862  * @details
00863  * When non-NULL, SPI send/receive operations route through the link
00864  * layer, which provides FEC encoding/decoding, Chase Combining, and
00865  * ACK/NACK handshake. When NULL, SPI operates in raw mode (frame +
00866  * CRC-32 only, no FEC/HARQ).
00867  *
00868  * **Requirements:**
00869  * - Must be initialized via rx_spi_link_init() before passing to manager
00870  * - spi_link->spi_handle MUST be the same as config->spi_handle
00871  * - Must remain valid for lifetime of manager
00872  *
00873  * @invariant If spi_link is non-NULL: spi_link->spi_handle == config->spi_handle
00874  * @invariant If spi_link is non-NULL: spi_link->initialized == true
00875  * @invariant If spi_link is non-NULL: spi_link must outlive manager instance
00876  *
00877  * @par Valid values: Initialized rx_spi_link_t*, or nullptr for raw SPI
00878  * @par Lifetime: Must outlive manager instance
00879  *
00880  * @see rx_spi_link.h SPI link layer API
00881  * @see rx_spi_link_init() Must be called before passing spi_link to manager
00882  * @since Version 1.0.0
00883  */
00884 rx_spi_link_t* spi_link;
00885
00886 /**
00887  * @brief Frame received callback function (NULL for no callback)
00888  * @details
00889  * User-defined function invoked when complete frame received on any channel.
00890  * If NULL, frames are received but no notification occurs (not typical).
00891  *
00892  * **Null handling:**
00893  * - NULL = Receive frames but don't notify (unusual, for testing only)
00894  * - Non-NULL = Invoke callback on frame reception (normal operation)
00895  *
00896  * @par Type: rx_comm_frame_callback_t (function pointer)
00897  * @par Valid values: Non-NULL function pointer, or nullptr to disable
00898  * @par Signature: `void callback(rx_comm_channel_t, const rx_frame_t*, void*)`
00899  *
00900  * @see rx_comm_frame_callback_t Callback function type definition
00901  */
00902 rx_comm_frame_callback_t callback;
00903
00904 /**
00905  * @brief User context pointer passed to callback
00906  * @details
00907  * Arbitrary pointer passed to callback as third parameter. Typically used
00908  * to pass application state, manager pointer, or other context without
00909  * requiring global variables.
00910  *
00911  * **Common patterns:**
00912  * - Pointer to manager itself (enables response from callback)
00913  * - Pointer to application state structure
00914  * - NULL if no context needed
00915  *
00916  * @par Type: void* (application-defined)
00917  * @par Valid values: Any pointer value, or nullptr
00918  * @par Lifetime: Must remain valid if callback uses it
00919  */
00920 void* callback_ctx;
00921
00922 /**
00923  * @brief Enable decoded ASCII output to USB Port 1
00924  * @details
00925  * When true, all received and sent frames are converted to human-readable
00926  * ASCII format and sent to USB Port 1 (/dev/ttyACM1 on RPi5).
00927  *
00928  * **ASCII Format:**
00929  * ```
00930  * [USB] TYPE=cmd FLAGS=0x00 LEN=42 DATA=48656C6C6F...
00931  * [SPI] TYPE=resp FLAGS=0x01 LEN=128 DATA=...
00932  * ```
00933  *
00934  * **Performance Impact:**
00935  * - Adds ~80 us latency per frame (formatting + USB send)
00936  * - Negligible for command/response (< 100 Hz)
00937  * - May impact high-frequency telemetry (> 1 kHz)
00938  *
00939  * **Use Cases:**
00940  * - Development: Enable for debugging and protocol analysis
00941  * - Production: Disable for maximum performance
00942  *
```

```
00943     * @par Type: bool
00944     * @par Value: true = enable ASCII output, false = disable
00945     * @par Default: Recommend true for development, false for production
00946     */
00947    bool enable_decoded_output;
00948
00949    /**
00950     * @brief Callback invoked when a transport link status changes
00951     * @details
00952     * Optional. Called when heartbeat monitoring detects a link transition
00953     * (e.g., healthy -> dead). NULL to disable link status notifications.
00954     */
00955    rx_comm_link_status_callback_t link_status_cb;
00956
00957    /**
00958     * @brief User context for link_status_cb
00959     */
00960    void* link_status_ctx;
00961
00962    /**
00963     * @brief Bitmask of channels to initialize and poll (default: all channels)
00964     * @details
00965     * When a bit is clear the corresponding transport is skipped during
00966     * internal_init_transports() and its handle is left NULL.  Set to
00967     * k_comm_channel_mask_all to attempt all four transports (legacy behaviour).
00968     *
00969     * Use comm_task_apply_system_config() to set this alongside the log backend
00970     * so that SCI9 conflicts between UART comm and UART log are caught early.
00971     *
00972     * @par Valid values: Any combination of rx_comm_channel_mask_t bits
00973     * @par Default: k_comm_channel_mask_all (attempt all channels)
00974     * @see rx_comm_channel_mask_t Bit definitions
00975     * @since Version 1.0.0
00976     */
00977    rx_comm_channel_mask_t enabled_channels;
00978 } rx_comm_manager_config_t;
00979
00980 /**
00981  * @struct rx_comm_manager_t
00982  * @brief Communication manager handle
00983  *
00984  * @details
00985  * Opaque handle structure containing manager state and resources. Application
00986  * must allocate this structure (typically static or global) and pass to all
00987  * manager functions.
00988  *
00989  * **Initialization:**
00990  * Zero-initialize structure before passing to `rx_comm_manager_init()`:
00991  * @code{.c}
00992  * rx_comm_manager_t mgr = {};  // Or use memset
00993  * @endcode
00994  *
00995  * **Memory Layout:** See file-level documentation for detailed breakdown.
00996  *
00997  * **Thread Safety:** Not thread-safe - protect with external mutex if accessed
00998  * from multiple threads.
00999  *
01000  * @warning Do not access fields directly - use API functions only
01001  * @warning Structure size ~2.5 KB - consider allocation carefully
01002  *
01003  * @see rx_comm_manager_init() Initialize handle
01004  * @see rx_comm_manager_deinit() Cleanup handle
01005  * @since Version 1.0.0
01006  */
01007 typedef struct {
01008    rx_uart_comm_handle_t*
01009                         uart_handle; /**< UART (SCI9 via CY7C65213 bridge) comm handle -- primary */
01010    rx_spi_comm_handle_t* spi_handle;  /**< SPI comm handle (NULL disables SPI channel) */
01011    rx_i2c_comm_handle_t* i2c_handle;  /**< I2C peripheral comm handle (NULL disables I2C channel) */
01012
01013    /**< @brief Optional SPI link layer with HARQ (NULL if disabled)
01014     * @details
01015     * When non-NULL, points to an rx_spi_link_t handle providing FEC
01016     * encoding/decoding, Chase Combining, and ACK/NACK handshake for SPI.
01017     * When NULL, SPI operates in raw mode (frame + CRC-32 only, no FEC/HARQ).
01018     *
01019     * **Ownership and Lifetime:**
01020     * - Pointer copied from config during rx_comm_manager_init()
01021     * - Manager does NOT own the link object - caller must allocate/deallocate
01022     * - Link object must remain valid for entire manager lifetime
01023     * - Link object must NOT be freed until after rx_comm_manager_deinit()
01024     * - If manager is deinitialized, spi_link remains valid but decoupled
01025     *
01026     * **Invariants/Constraints:**
01027     * - If non-NULL: Must point to initialized rx_spi_link_t (spi_link->initialized == true)
01028     * - If non-NULL: spi_link->spi_handle MUST equal manager's spi_handle
01029     * - If non-NULL: spi_link must outlive manager instance
```

```
01030  * - NULL is allowed to indicate no SPI HARQ (raw SPI mode)
01031  * - Fields inside spi_link object must remain stable while manager is active
01032  *
01033  * **Valid Ranges/Usage:**
01034  * - Set via config->spi_link during initialization
01035  * - NULL = raw SPI mode (valid configuration for testing or simple use cases)
01036  * - Non-NULL = HARQ mode (production mode with FEC and retransmission)
01037  * - Once set during init, this pointer is immutable (never modified by manager)
01038  *
01039  * **Thread Safety:**
01040  * - Manager does not perform synchronization on spi_link access
01041  * - If multiple threads call rx_comm_manager_poll(), caller must serialize
01042  * - spi_link internal state is protected by its own mechanisms (see rx_spi_link.h)
01043  * - Safe to read this pointer from multiple threads (read-only after init)
01044  *
01045  * @invariant If spi_link != NULL: spi_link->initialized == true
01046  * @invariant If spi_link != NULL: spi_link->spi_handle == this->spi_handle
01047  * @invariant If spi_link != NULL: spi_link must outlive this manager instance
01048  * @invariant Pointer value never changes after rx_comm_manager_init()
01049  *
01050  * @par Type: rx_spi_link_t* (pointer to HARQ link layer handle)
01051  * @par Valid values: Initialized rx_spi_link_t*, or nullptr for raw SPI mode
01052  * @par Lifetime: Must outlive manager instance (caller manages allocation)
01053  * @par Modification: Read-only after init (immutable pointer)
01054  * - Immutable after init (do not change while manager is active)
01055  * - Fields inside rx_spi_link_t must remain stable during use
01056  *
01057  * @see rx_spi_link.h SPI link layer API
01058  * @see rx_comm_manager_config_t::spi_link Configuration parameter
01059  * @since Version 1.0.0
01060  */
01061 rx_spi_link_t* spi_link;
01062
01063 rx_comm_frame_callback_t callback;                                  /**< Frame callback */
01064 void*                    callback_ctx;                              /**< Callback context */
01065 char                     ascii_buffer[k_comm_manager_ascii_buf_size]; /**< ASCII buffer */
01066 bool                     enable_decoded_output;                     /**< Enable ASCII output */
01067 bool                     initialized;                               /**< Initialization flag */
01068
01069 /** @brief Per-transport heartbeat tracking (indexed by rx_comm_channel_t) */
01070 rx_comm_heartbeat_state_t heartbeat[k_comm_channel_count];
01071
01072 /** @brief Link status change callback (optional) */
01073 rx_comm_link_status_callback_t link_status_cb;
01074
01075 /** @brief Link status callback context */
01076 void* link_status_ctx;
01077
01078 /** @brief Static event FIFO queue for reliable command delivery */
01079 rx_comm_event_entry_t event_queue[k_comm_event_queue_depth];
01080
01081 /** @brief Event queue write index (next slot to write) */
01082 uint8_t event_queue_head;
01083
01084 /** @brief Event queue read index (next slot to process) */
01085 uint8_t event_queue_tail;
01086
01087 /** @brief Number of occupied event queue slots */
01088 uint8_t event_queue_count;
01089 } rx_comm_manager_t;
01090
01091 /**
01092  * @struct rx_comm_send_params_t
01093  * @brief Parameters for rx_comm_manager_send
01094  *
01095  * @details
01096  * Wrapper structure providing type safety for send operations. Prevents
01097  * parameter confusion when passing multiple uint8_t/uint32_t values.
01098  *
01099  * **Design Rationale:**
01100  * Without this struct, function signature would be error-prone:
01101  * ```c
01102  * // BAD: Easy to mix up channel, type, flags (all similar types)
01103  * rx_err_t send(mgr, uint8_t ch, uint8_t type, uint8_t flags, ...);
01104  * ```
01105  *
01106  * With struct, parameters are self-documenting:
01107  * ```c
01108  * // GOOD: Clear naming prevents mistakes
01109  * rx_comm_send_params_t params = {
01110  *     .channel = k_comm_channel_usb,
01111  *     .type = k_frame_type_command,
01112  *     .flags = 0x00,
01113  *     // ...
01114  * };
01115  * ```
01116  *
```

```
01117  * @par Usage Example:
01118  * @code{.c}
01119  * rx_comm_send_params_t params = {
01120  *     .channel = k_comm_channel_spi,
01121  *     .type = k_frame_type_telemetry,
01122  *     .flags = 0,
01123  *     .payload = telemetry_data,
01124  *     .payload_len = sizeof(telemetry_data),
01125  * };
01126  * rx_comm_manager_send(&mgr, &params);
01127  * @endcode
01128  *
01129  * @see rx_comm_manager_send() Send function using this structure
01130  * @since Version 1.0.0
01131  */
01132 typedef struct {
01133     rx_comm_channel_t channel;     /**< Channel to send on (USB or SPI) */
01134     rx_frame_type_t   type;        /**< Frame type (command, telemetry, etc.) */
01135     uint8_t           flags;       /**< Frame flags (protocol-specific) */
01136     const uint8_t*    payload;     /**< Payload data pointer */
01137     uint32_t          payload_len; /**< Payload length in bytes */
01138 } rx_comm_send_params_t;
01139
01140 /*
      ============================================================================
01141  * Lifecycle API
01142  *
      ============================================================================
01143  */
01144
01145 /**
01146  * @brief Initialize communication manager
01147  *
01148  * @details
01149  * Initializes the communication manager with the provided configuration.
01150  * Validates configuration parameters and sets up internal state for
01151  * USB and/or SPI channel operation.
01152  *
01153  * @param[out] mgr Pointer to manager handle
01154  * @param[in]  cfg Configuration (NULL for defaults, both channels disabled)
01155  *
01156  * @return k_rx_ok on success
01157  * @return k_rx_err_invalid_arg if mgr is nullptr
01158  * @return k_rx_err_invalid_arg if spi_link is non-NULL but not initialized
01159  * @return k_rx_err_invalid_arg if spi_link->spi_handle != cfg->spi_handle
01160  *
01161  * @pre mgr must not be nullptr
01162  * @pre If cfg->spi_link is non-NULL, spi_link must be initialized via rx_spi_link_init()
01163  * @pre If cfg->spi_link is non-NULL, spi_link->spi_handle must equal cfg->spi_handle
01164  * @pre cfg may be nullptr (results in default configuration with both channels disabled)
01165  *
01166  * @post mgr->usb_handle == cfg->usb_handle (or nullptr if cfg is nullptr)
01167  * @post mgr->spi_handle == cfg->spi_handle (or nullptr if cfg is nullptr)
01168  * @post mgr->spi_link == cfg->spi_link (or nullptr if cfg is nullptr or cfg->spi_link is nullptr)
01169  * @post All internal state initialized (event queue empty, heartbeat timers reset)
01170  *
01171  * @note Thread-safe: May be called from any thread, but concurrent calls with same mgr are prohibited
01172  * @warning Must be called before any other manager operations on this handle
01173  *
01174  * @see rx_spi_link_init() Must call this before passing spi_link to manager
01175  * @see rx_comm_manager_deinit() Cleanup function
01176  *
01177  * @since Version 1.0.0
01178  */
01179 [[nodiscard]] rx_err_t rx_comm_manager_init(rx_comm_manager_t*              mgr,
01180                                             const rx_comm_manager_config_t* cfg);
01181
01182 /**
01183  * @brief Deinitialize communication manager
01184  *
01185  * Does not deinitialize the underlying USB/SPI handles.
01186  *
01187  * @param[in,out] mgr Pointer to manager handle
01188  * @return k_rx_ok on success
01189  */
01190 [[nodiscard]] rx_err_t rx_comm_manager_deinit(rx_comm_manager_t* mgr);
01191
01192 /*
      ============================================================================
01193  * Polling API
01194  *
      ============================================================================
01195  */
01196
01197 /**
01198  * @brief Poll all enabled channels for incoming frames
01199  *
```

```
01200  * Non-blocking check of USB and SPI channels. For each complete frame received:
01201  * 1. If decoded output enabled, format and send ASCII to Port 1
01202  * 2. Invoke user callback with frame data and channel
01203  *
01204  * Call this from main loop or dedicated communication task.
01205  *
01206  * @param[in,out] mgr Pointer to initialized manager
01207  *
01208  * @return k_rx_ok if at least one frame was received
01209  * @return k_rx_err_timeout if no frames available on any channel
01210  * @return k_rx_err_invalid_arg if mgr is nullptr
01211  * @return k_rx_err_invalid_state if not initialized
01212  */
01213 [[nodiscard]] rx_err_t rx_comm_manager_poll(rx_comm_manager_t* mgr);
01214
01215 /*
      ============================================================================
01216  * Send API
01217  *
      ============================================================================
01218  */
01219
01220 /**
01221  * @brief Send frame on specific channel
01222  *
01223  * Encodes and sends a frame on the specified channel. If decoded output
01224  * is enabled, also sends ASCII representation to Port 1.
01225  *
01226  * @param[in,out] mgr    Pointer to initialized manager
01227  * @param[in]     params Send parameters (channel, type, flags, payload, payload_len)
01228  *
01229  * @return k_rx_ok on success
01230  * @return k_rx_err_invalid_arg if mgr or payload is nullptr
01231  * @return k_rx_err_invalid_state if not initialized or channel not enabled
01232  */
01233 [[nodiscard]] rx_err_t rx_comm_manager_send(rx_comm_manager_t*           mgr,
01234                                             const rx_comm_send_params_t* params);
01235
01236 /**
01237  * @brief Send response on same channel as received frame
01238  *
01239  * Convenience function for responding to commands. Sends a RESPONSE type
01240  * frame on the specified channel.
01241  *
01242  * @param[in,out] mgr         Pointer to initialized manager
01243  * @param[in]     channel     Channel to respond on (from callback)
01244  * @param[in]     payload     Response payload data
01245  * @param[in]     payload_len Response payload length
01246  *
01247  * @return k_rx_ok on success
01248  */
01249 [[nodiscard]] rx_err_t rx_comm_manager_respond(rx_comm_manager_t* mgr,
01250                                                rx_comm_channel_t  channel,
01251                                                const uint8_t*     payload,
01252                                                uint32_t           payload_len);
01253
01254 /*
      ============================================================================
01255  * Stream / Event Send API
01256  *
      ============================================================================
01257  */
01258
01259 /**
01260  * @brief Send data on the stream path (fire-and-forget)
01261  *
01262  * @details
01263  * Stream path is for time-sensitive data like telemetry where stale data
01264  * has no value. Sends immediately with NO retries on any transport.
01265  * If the send fails, the data is simply dropped.
01266  *
01267  * - USB CDC: Direct send, no retries (USB HW provides reliability)
01268  * - SPI: Direct send, no retries (stale telemetry is worthless)
01269  *
01270  * @param[in,out] mgr    Pointer to initialized manager
01271  * @param[in]     params  Send parameters (channel, type, flags, payload, payload_len)
01272  *
01273  * @return k_rx_ok on success
01274  * @return k_rx_err_invalid_arg if mgr or params is nullptr
01275  * @return k_rx_err_invalid_state if not initialized or channel not enabled
01276  *
01277  * @pre mgr must be initialized
01278  * @pre params must be valid
01279  * @post Data sent or silently dropped on failure
01280  *
01281  * @note This is a thin wrapper over rx_comm_manager_send() for semantic clarity
01282  *
```

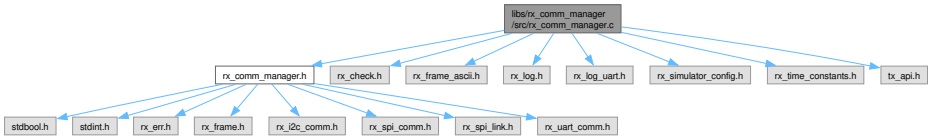
```
01283  * @see rx_comm_manager_event_send() For reliable command delivery
01284  * @since Version 1.0.0
01285  */
01286 [[nodiscard]] rx_err_t rx_comm_manager_stream_send(rx_comm_manager_t*           mgr,
01287                                                    const rx_comm_send_params_t* params);
01288
01289 /**
01290  * @brief Queue data on the event path (reliable, SPI retry)
01291  *
01292  * @details
01293  * Event path is for commands and other data requiring reliable delivery.
01294  * The payload is copied into a static FIFO queue and processed by the
01295  * poll loop. Retry behavior depends on transport:
01296  *
01297  * - **SPI**: Up to 3 retries with exponential backoff (10ms, 20ms, 40ms)
01298  * - **USB**: Fire-and-forget (USB HW reliability is sufficient)
01299  *
01300  * Queue is processed in FIFO order by rx_comm_manager_poll().
01301  *
01302  * @param[in,out] mgr     Pointer to initialized manager
01303  * @param[in]     params  Send parameters (channel, type, flags, payload, payload_len)
01304  *
01305  * @return k_rx_ok on success (queued)
01306  * @return k_rx_err_invalid_arg if mgr or params is nullptr
01307  * @return k_rx_err_invalid_state if not initialized
01308  * @return k_rx_err_no_mem if event queue is full
01309  *
01310  * @pre mgr must be initialized
01311  * @pre params must be valid
01312  * @pre params->payload_len <= k_frame_max_payload
01313  * @post Payload copied to queue, will be sent by next poll()
01314  *
01315  * @note Payload is COPIED into queue - caller can reuse buffer immediately
01316  * @warning Queue depth is k_comm_event_queue_depth (8). If full, returns error.
01317  *
01318  * @see rx_comm_manager_stream_send() For fire-and-forget telemetry
01319  * @since Version 1.0.0
01320  */
01321 [[nodiscard]] rx_err_t rx_comm_manager_event_send(rx_comm_manager_t*           mgr,
01322                                                   const rx_comm_send_params_t* params);
01323
01324 /*
      ============================================================================
01325  * Heartbeat API
01326  *
      ============================================================================
01327  */
01328
01329 /**
01330  * @brief Query link health status for a transport channel
01331  *
01332  * @details
01333  * Returns the current heartbeat-derived link status for the given channel.
01334  * Status is updated by rx_comm_manager_poll() based on the dual-detection
01335  * heartbeat model (200ms implicit timeout).
01336  *
01337  * @param[in]  mgr     Pointer to initialized manager
01338  * @param[in]  channel Channel to query
01339  * @param[out] status  Receives current link status
01340  *
01341  * @return k_rx_ok on success
01342  * @return k_rx_err_invalid_arg if mgr or status is nullptr
01343  *
01344  * @since Version 1.0.0
01345  */
01346 [[nodiscard]] rx_err_t rx_comm_manager_link_status(const rx_comm_manager_t* mgr,
01347                                                    rx_comm_channel_t        channel,
01348                                                    rx_comm_link_status_t*   status);
01349
01350 /*
      ============================================================================
01351  * Status API
01352  *
      ============================================================================
01353  */
01354
01355 /**
01356  * @brief Check if channel is ready for communication
01357  *
01358  * @param[in]  mgr     Pointer to initialized manager
01359  * @param[in]  channel Channel to check
01360  * @param[out] ready   Set to true if channel is ready
01361  *
01362  * @return k_rx_ok on success
01363  * @return k_rx_err_invalid_arg if mgr or ready is nullptr
01364  */
01365 [[nodiscard]] rx_err_t
```

```
01366 rx_comm_manager_channel_ready(rx_comm_manager_t* mgr, rx_comm_channel_t channel, bool* ready);
01367
01368 /**
01369  * @brief Get channel name string
01370  *
01371  * @param[in] channel Channel identifier
01372  * @return Human-readable channel name
01373  */
01374 const char* rx_comm_manager_channel_name(rx_comm_channel_t channel);
01375
01376 /*
      ============================================================================
01377  * Retransmission Pass-Through
01378  *
      ============================================================================
01379  */
01380
01381 /**
01382  * @brief Process pending retransmissions on SPI channel
01383  *
01384  * @details
01385  * Thin pass-through to rx_spi_comm_process_retransmits() on the SPI handle.
01386  * Call this from the communication task polling loop.
01387  *
01388  * @param[in,out] mgr Communication manager
01389  * @param[in] current_time_ms Current system time in milliseconds
01390  *
01391  * @return rx_err_t Error code from rx_spi_comm_process_retransmits()
01392  * @retval k_rx_ok No action needed or retransmit sent
01393  * @retval k_rx_err_invalid_arg mgr is nullptr
01394  * @retval k_rx_err_invalid_state mgr not initialized
01395  * @retval k_rx_err_retry_limit Max retries exceeded
01396  *
01397  * @since Version 1.0.0
01398  */
01399 [[nodiscard]] rx_err_t rx_comm_manager_process_retransmits(rx_comm_manager_t* mgr,
01400                                                            uint32_t            current_time_ms);
01401
01402 /**
01403  * @brief Enable or disable automatic retransmission on SPI channel
01404  *
01405  * @details
01406  * Thin pass-through to rx_spi_comm_set_auto_retransmit() on the SPI handle.
01407  *
01408  * @param[in,out] mgr Communication manager
01409  * @param[in] enabled true to enable, false to disable
01410  * @param[in] config Retransmit config (NULL to keep current/defaults)
01411  *
01412  * @return rx_err_t Error code from rx_spi_comm_set_auto_retransmit()
01413  *
01414  * @since Version 1.0.0
01415  */
01416 [[nodiscard]] rx_err_t
01417 rx_comm_manager_set_auto_retransmit(rx_comm_manager_t*                    mgr,
01418                                     bool                                  enabled,
01419                                     const rx_spi_comm_retransmit_config_t* config);
01420
01421 #ifdef __cplusplus
01422 }
01423 #endif
```

4.3 libs/rx_comm_manager/src/rx_comm_manager.c File Reference

Unified Multi-Channel Communication Manager Implementation.

#include ”rx_comm_manager.h”
#include ”rx_check.h”
#include ”rx_frame_ascii.h”
#include ”rx_log.h”
#include ”rx_log_uart.h”
#include ”rx_simulator_config.h”
#include ”rx_time_constants.h”
#include ”tx_api.h”
Include dependency graph for rx_comm_manager.c:



Enumerations

- enum rx_comm_sim_time_t : uint32_t { k_sim_time_ms_zero = 0 }

  Get current time in milliseconds.
- enum internal_constants_t : uint32_t {
  k_receive_timeout_ms = 0 , k_frame_seq_placeholder = 0 , k_frame_crc_placeholder = 0 ,
  k_ascii_buffer_first_idx = 0 ,
  k_zero_u32 = 0 }

  Internal constants for communication manager implementation.

Functions

- static uint32_t internal_get_time_ms (void)
- RX_STATIC_TESTABLE void internal_output_decoded (rx_comm_manager_t *mgr, const rx_frame_t *frame, bool is_tx)

  Output decoded ASCII frame to USB Port 1 for debugging.
- static void internal_output_decoded_from_params (rx_comm_manager_t *mgr, const rx_comm_send_params_t *params)

  Build temporary frame from send params and output decoded ASCII (TX direction).
- RX_STATIC_TESTABLE void internal_handle_frame (rx_comm_manager_t *mgr, rx_comm_channel_t channel, const rx_frame_t *frame)

  Handle received frame by invoking user callback and optional ASCII output.
- static rx_err_t internal_poll_spi (rx_comm_manager_t *mgr)

  Poll SPI channel for incoming frames (non-blocking).
- static rx_err_t internal_poll_i2c (rx_comm_manager_t *mgr)

  Poll I2C channel for incoming frames (non-blocking).
- static rx_err_t internal_poll_uart (rx_comm_manager_t *mgr)

  Poll UART channel for incoming frames (non-blocking).
- static void internal_dequeue_event (rx_comm_manager_t *mgr, rx_comm_event_entry_t *entry)

  Process one event from the FIFO queue.
- static void internal_process_event_queue (rx_comm_manager_t *mgr)
- static void internal_check_channel_heartbeat (rx_comm_manager_t *mgr, uint8_t ch, uint32_t now_ms)

  Check heartbeat implicit timeout on all transports.
- static void internal_check_heartbeat (rx_comm_manager_t *mgr)
- rx_err_t rx_comm_manager_init (rx_comm_manager_t *mgr, const rx_comm_manager_config_t *cfg)

  Initialize communication manager with channel configuration.
- rx_err_t rx_comm_manager_deinit (rx_comm_manager_t *mgr)

  Deinitialize communication manager and release resources.
- static rx_err_t internal_poll_all_channels (rx_comm_manager_t *mgr, bool *received)

  Poll all enabled communication channels for incoming frames (non-blocking).
- rx_err_t rx_comm_manager_poll (rx_comm_manager_t *mgr)

  Poll all enabled channels for incoming frames.

- static rx_err_t internal_route_send (rx_comm_manager_t *mgr, const rx_comm_send_params_t *params)

  Send frame on specified communication channel.
- rx_err_t rx_comm_manager_send (rx_comm_manager_t *mgr, const rx_comm_send_params_t *params)

  Send frame on specific channel.
- rx_err_t rx_comm_manager_respond (rx_comm_manager_t *mgr, rx_comm_channel_t channel, const uint8_t *payload, uint32_t payload_len)

  Send response frame on specified channel (convenience wrapper).
- rx_err_t rx_comm_manager_stream_send (rx_comm_manager_t *mgr, const rx_comm_send_params_t *params)

  Send data on the stream path (fire-and-forget).
- rx_err_t rx_comm_manager_event_send (rx_comm_manager_t *mgr, const rx_comm_send_params_t *params)

  Queue data on the event path (reliable, SPI retry).
- rx_err_t rx_comm_manager_link_status (const rx_comm_manager_t *mgr, rx_comm_channel_t channel, rx_comm_link_status_t *status)

  Query link health status for a transport channel.
- rx_err_t rx_comm_manager_channel_ready (rx_comm_manager_t *mgr, rx_comm_channel_t channel, bool *ready)

  Query if communication channel is ready for send/receive operations.
- const char * rx_comm_manager_channel_name (rx_comm_channel_t channel)

  Get human-readable name for communication channel.
- rx_err_t rx_comm_manager_process_retransmits (rx_comm_manager_t *mgr, const uint32_t current_time_ms)

  Process pending retransmissions on SPI channel.
- rx_err_t rx_comm_manager_set_auto_retransmit (rx_comm_manager_t *mgr, const bool enabled, const rx_spi_comm_retransmit_config_t *config)

  Enable or disable automatic retransmission on SPI channel.

Variables

- static const char s_tag [] = "COMM_MGR"
- static const rx_comm_manager_t s_zero_mgr = {}

  Zero-initialized comm manager template for stack-safe initialization.

4.3.1 Detailed Description

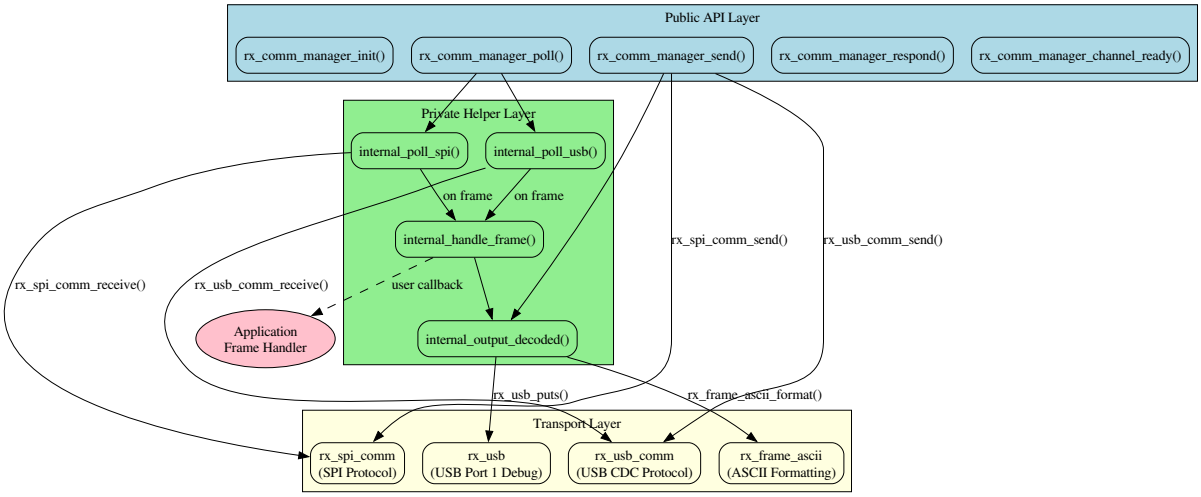
Unified Multi-Channel Communication Manager Implementation.

4.3.1.1 Overview

Production-quality implementation of unified communication manager coordinating multiple simultaneous channels (USB CDC and SPI) with automatic frame protocol handling, non-blocking polling architecture, optional decoded ASCII debugging output, and callback-based event notification.

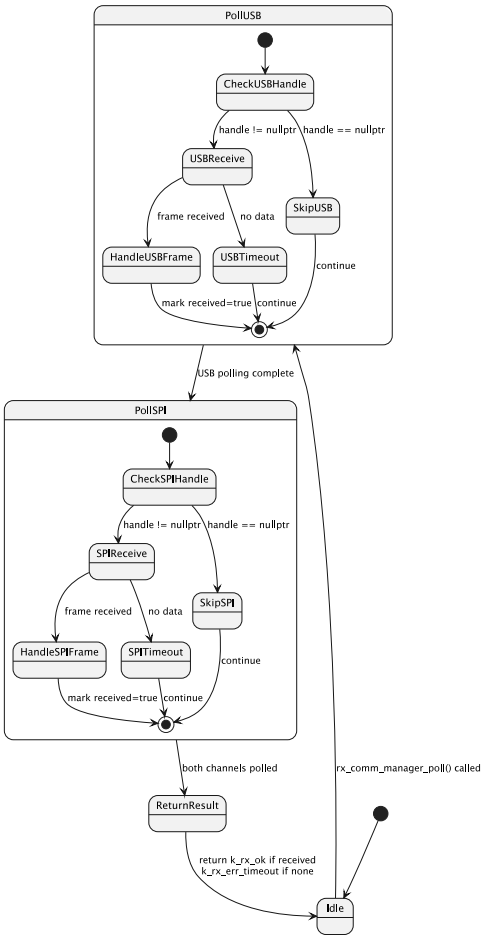
Purpose: Provide centralized, efficient coordination of independent communication channels between RX72N MCU and Raspberry Pi 5 host, abstracting low-level transport details while maintaining high performance and reliability.

4.3.1.2 Implementation Architecture



4.3.1.3 Polling State Machine

The rx_comm_manager_poll() function implements a non-blocking state machine that checks all enabled channels sequentially:



4.3.1.4 Performance Characteristics

Operation	Avg Time	Worst Case	Notes
rx_comm_manager_init()	~5 us	~10 us	memset dominates (328 bytes)
rx_comm_manager_poll()	~15-200 us	~500 us	Depends on channels enabled, frames available
rx_comm_manager_send()	~30-150 us	~300 us	Depends on payload size, channel
internal_poll_usb()	~10-100 us	~250 us	USB CDC bulk transfer latency
internal_poll_spi()	~5-50 us	~150 us	SPI hardware transfer latency
internal_handle_frame()	~2 us	~5 us	Callback invocation overhead
internal_output_decoded()	~50-150 us	~300 us	ASCII formatting + USB puts

Timing Analysis:

- Measured @ 240 MHz CPU clock with -O2 optimization
- Poll time dominated by transport layer (USB/SPI receive)
- ASCII debugging adds ~50-150 us per frame (disable in production)
- Callback execution time not included (application-dependent)

4.3.1.5 Memory Usage

Memory Type	Size	Usage
rx_comm_manager_t handle	328 bytes	Main state structure
Stack (init)	~32 bytes	Function locals
Stack (poll)	~80 bytes	rx_frame_t + locals
Stack (send)	~80 bytes	rx_frame_t + locals
ASCII buffer	256 bytes	Inside handle (for debugging)

Memory Layout (rx_comm_manager_t):

```
Offset | Size | Field              | Purpose
-------|------|--------------------|--------------------------
0x00   | 8    | usb_handle         | Pointer to USB comm handle
0x08   | 8    | spi_handle         | Pointer to SPI comm handle
0x10   | 8    | callback           | User frame callback function
0x18   | 8    | callback_ctx       | User context pointer
0x20   | 1    | enable_decoded_output | ASCII debug flag
0x21   | 1    | initialized        | Initialization flag
0x22   | 2    | (padding)          | Alignment
0x24   | 4    | (padding)          | Alignment
0x28   | 256  | ascii_buffer       | ASCII formatting buffer
-------|------|--------------------|--------------------------
Total: 328 bytes (0x148)
```

4.3.1.6 Algorithm Details

4.3.1.6.1 Polling Algorithm (rx_comm_manager_poll)

The polling algorithm maximizes throughput by checking all channels without blocking, returning immediately on the first error while continuing to poll remaining channels on timeout (no data available):

Steps:

1. Parameter validation (NULL check, initialized flag)
2. Poll USB channel:
   - If nullptr handle -> skip (k_rx_err_timeout)
   - If data available -> handle frame, mark received=true
   - If timeout -> continue to next channel
   - If error -> return error immediately (critical failure)
3. Poll SPI channel:
   - Same logic as USB
4. Return aggregated result:
   - k_rx_ok if ANY frame received on any channel
   - k_rx_err_timeout if NO frames received (normal, non-error condition)
   - Other error codes propagated immediately

Design Rationale:

- Non-blocking: Never blocks waiting for data
- Fair scheduling: Both channels polled each iteration
- Error prioritization: Critical errors returned immediately
- Timeout tolerance: Timeout is expected, not an error

4.3.1.6.2 Send Algorithm (rx_comm_manager_send)

Routing algorithm with post-send ASCII debugging support:

Steps:

1. Validate parameters (NULL checks, payload length <= k_frame_max_payload)
2. Route to channel:
   - USB: Call rx_usb_comm_send()
   - SPI: Call rx_spi_comm_send()
   - Invalid channel: Return k_rx_err_invalid_arg
3. On success, if ASCII output enabled:
   - Build temporary rx_frame_t from send parameters
   - Format as ASCII via rx_frame_ascii_format()
   - Send to USB Port 1 via rx_usb_puts()

Performance Note: ASCII formatting adds ~50-150 us overhead per send. Disable in production with cfg->enable_decoded_output = false.

4.3.1.7 Thread Safety Analysis

Conditional Thread Safety - Safe if following rules are observed:

Scenario	Thread Safety	Mitigation Required
Single-threaded	[OK] Safe	None
Multi-threaded (different channels)	[OK] Safe	Ensure each thread uses different channel
Multi-threaded (same channel)	[X] Unsafe	External mutex required
Multi-threaded (different handles)	[OK] Safe	None (independent handles)
Init/Deinit from ISR	[X] Unsafe	Call only from task context
Poll/Send from ISR	[WARN] Conditional	OK if callback is ISR-safe

Recommended Architecture:

- Dedicate one thread to poll all channels (e.g., ThreadX communication task)
- Send from any thread/ISR, but use external mutex if same channel
- Ensure user callback is reentrant if called from ISR

NASA Power of 10 Rule 5: All public functions validate preconditions with explicit checks (NULL pointers, initialized flag, parameter ranges).

4.3.1.8 Integration Example

Complete System Setup with ThreadX:

```
// Global communication manager handle
static rx_comm_manager_t g_comm_mgr;
static rx_usb_comm_handle_t g_usb_handle;
static rx_spi_comm_handle_t g_spi_handle;

// User callback for frame processing
static void on_frame_received(rx_comm_channel_t channel,
                              const rx_frame_t* frame,
                              void* ctx)
{
  (void)ctx;

  // Dispatch based on frame type
  switch (frame->header.type) {
    case k_frame_type_command:
      handle_command(channel, frame);
      break;

    case k_frame_type_request:
      handle_request(channel, frame);
      break;

    default:
      // Unknown frame type, ignore
      break;
  }
}

// ThreadX communication task
static void comm_task_entry(ULONG thread_input)
{
  (void)thread_input;

  // Initialize USB CDC communication
  rx_usb_comm_config_t usb_cfg = {
    .port = k_usb_port_proto,
  };
```

```
rx_err_t err = rx_usb_comm_init(&g_usb_handle, &usb_cfg);
if (err != k_rx_ok) {
  // Handle error
  return;
}

// Initialize SPI communication
rx_spi_comm_config_t spi_cfg = {
  .spi_channel = 0,
};
err = rx_spi_comm_init(&g_spi_handle, &spi_cfg);
if (err != k_rx_ok) {
  // Handle error
  return;
}

// Initialize communication manager
rx_comm_manager_config_t cfg = {
  .usb_handle = &g_usb_handle,
  .spi_handle = &g_spi_handle,
  .callback = on_frame_received,
  .callback_ctx = nullptr,
  .enable_decoded_output = true,  // Enable for development
};
err = rx_comm_manager_init(&g_comm_mgr, &cfg);
if (err != k_rx_ok) {
  // Handle error
  return;
}

// Main communication loop (100 Hz polling)
while (1) {
  // Poll all channels
  err = rx_comm_manager_poll(&g_comm_mgr);

  // Check for critical errors (not timeout)
  if (err != k_rx_ok && err != k_rx_err_timeout) {
    // Log error, attempt recovery
    tx_thread_sleep(100);  // 1 second backoff
  }

  // Sleep 10 ticks (100 ms @ 100 Hz tick rate)
  tx_thread_sleep(10);
}
}

// Send response from application
void send_motor_telemetry(void)
{
  uint8_t telemetry_payload[16];
  // ... fill telemetry data ...

  // Check if USB channel is ready
  bool usb_ready = false;
  rx_err_t err = rx_comm_manager_channel_ready(&g_comm_mgr,
                                               k_comm_channel_usb,
                                               &usb_ready);
  if (err == k_rx_ok && usb_ready) {
    // Send telemetry response
    err = rx_comm_manager_respond(&g_comm_mgr,
                                  k_comm_channel_usb,
                                  telemetry_payload,
                                  sizeof(telemetry_payload));
  }
}
```

4.3.1.9 NASA Power of 10 Compliance

Rule	Compliance	Implementation Notes
Rule 1: Control Flow	[OK]	No goto, setjmp, longjmp, or recursion
Rule 2: Loop Bounds	[OK]	All loops have statically provable bounds
Rule 3: No Dynamic Allocation	[OK]	Zero malloc/free - all static allocation
Rule 4: Function Size	[OK]	All functions < 60 lines
Rule 5: Assertions	[OK]	Minimum 2 checks per function (nullptr, initialized)

Rule	Compliance	Implementation Notes
Rule 6: Data Scope	[OK]	Variables declared at smallest scope
Rule 7: Check Returns	[OK]	All return values validated or cast to (void)
Rule 8: Preprocessor	[OK]	Minimal macros, typed enums for constants
Rule 9: Pointers	[WARN]	Function pointers used for callback (DIP)
Rule 10: Compiler Warnings	[OK]	-Wall -Wextra -Werror enabled

Rule 9 Deviation: Function pointer used for user callback to enable Dependency Inversion Principle (SOLID) - allows application to register frame handler without tight coupling. This is intentional and documented per project policy.

4.3.1.10 SOLID Principles

Principle	Implementation
Single Responsibility	Module responsible ONLY for channel coordination; frame parsing/transport delegated
Open/Closed	Extensible via callback; new frame types handled by application
Liskov Substitution	Channels are interchangeable (same rx_frame_t interface)
Interface Segregation	Small focused API (7 functions); send/poll/ready separated
Dependency Inversion	Depends on abstract rx_frame_t and channel handles, not concrete implementations

Key Design Decision: The manager does NOT parse frame payloads - it only routes frames to the user callback. This separation allows application-specific command handling without modifying the communication layer.

4.3.1.11 Module Dependencies

Includes:

- rx_comm_manager.h - Public API definitions
- rx_frame_ascii.h - ASCII formatting for debugging
- rx_usb.h - USB port access (Port 1 debug output)
- string.h - Standard C library (memset, memcpy)

Links Against:

- rx_usb_comm - USB CDC frame protocol implementation
- rx_spi_comm - SPI frame protocol implementation
- rx_frame_ascii - ASCII frame formatting
- rx_usb - USB hardware abstraction

Date

2026-01-27

Copyright

Since

Version 1.0.0

See also

rx_comm_manager.h Full API documentation
rx_spi_comm.h SPI protocol implementation
rx_uart_comm.h UART protocol implementation
rx_i2c_comm.h I2C peripheral protocol implementation
rx_frame_ascii.h ASCII debugging interface

Definition in file rx_comm_manager.c.

4.3.2 Enumeration Type Documentation

4.3.2.1 internal_constants_t

enum internal_constants_t : uint32_t

Internal constants for communication manager implementation.

Defines compile-time constants used internally by the communication manager for timeout values, placeholder values, and buffer indexing.

Design Notes:

- Non-blocking timeout (0 ms) ensures poll never blocks
- Placeholder values filled by transport layer (sequence, CRC)
- Explicit array index constant for bounds checking

Since

Version 1.0.0

Enumerator

k_receive_timeout_ms	Non-blocking receive timeout for polling. Set to 0 ms to ensure rx_comm_manager_poll() never blocks waiting for data. Transport layers (USB/SPI) return k_rx_err_timeout immediately if no data. Value: 0 milliseconds
k_frame_seq_placeholder	Placeholder for frame sequence number (filled by transport layer). Sequence number is managed by transport layer (rx_usb_comm/rx_spi_comm) to detect dropped frames. Manager does not manipulate sequence numbers. Value: 0
k_frame_crc_placeholder	Placeholder for frame CRC-32 (filled by transport layer). CRC-32 calculated and verified by transport layer. Manager builds temp frame with placeholder CRC for ASCII formatting only. Value: 0
k_ascii_buffer_first_idx	First index of ascii_buffer for post-condition validation. Used in rx_comm_manager_init() to verify memset properly cleared buffer. First byte must be '\0' after zero-fill. Value: 0
k_zero_u32	Typed zero constant for uint32_t comparisons. Used for payload length checks and other uint32_t zero comparisons to avoid magic number 0. Provides type safety and self-documenting code. Value: 0

Definition at line 462 of file rx_comm_manager.c.

```
00462             : uint32_t {
00463     /**
00464      * @brief Non-blocking receive timeout for polling
00465      * @details
00466      * Set to 0 ms to ensure rx_comm_manager_poll() never blocks waiting for data.
00467      * Transport layers (USB/SPI) return k_rx_err_timeout immediately if no data.
00468      * @par Value: 0 milliseconds
00469      */
00470     k_receive_timeout_ms = 0,
00471
00472     /**
00473      * @brief Placeholder for frame sequence number (filled by transport layer)
00474      * @details
00475      * Sequence number is managed by transport layer (rx_usb_comm/rx_spi_comm)
00476      * to detect dropped frames. Manager does not manipulate sequence numbers.
00477      * @par Value: 0
00478      */
00479     k_frame_seq_placeholder = 0,
```

```
00480
00481   /**
00482    * @brief Placeholder for frame CRC-32 (filled by transport layer)
00483    * @details
00484    * CRC-32 calculated and verified by transport layer. Manager builds temp
00485    * frame with placeholder CRC for ASCII formatting only.
00486    * @par Value: 0
00487    */
00488   k_frame_crc_placeholder = 0,
00489
00490   /**
00491    * @brief First index of ascii_buffer for post-condition validation
00492    * @details
00493    * Used in rx_comm_manager_init() to verify memset properly cleared buffer.
00494    * First byte must be '\0' after zero-fill.
00495    * @par Value: 0
00496    */
00497   k_ascii_buffer_first_idx = 0,
00498
00499   /**
00500    * @brief Typed zero constant for uint32_t comparisons
00501    * @details
00502    * Used for payload length checks and other uint32_t zero comparisons to avoid
00503    * magic number 0. Provides type safety and self-documenting code.
00504    * @par Value: 0
00505    */
00506   k_zero_u32 = 0,
00507 } internal_constants_t;
```

4.3.2.2 rx_comm_sim_time_t

enum rx_comm_sim_time_t : uint32_t

Get current time in milliseconds.

Returns the current system time in milliseconds. On RX72N hardware, this uses ThreadX tx_time_get() multiplied by tick period. In simulator builds, returns 0 (timing not available).

Since

Version 1.0.0

Simulator time stub value (time is unavailable in simulator)

Enumerator

k_sim_time_ms_zero	Simulator returns zero (no real-time clock)

Definition at line 429 of file rx_comm_manager.c.

```
00429             : uint32_t {
00430   k_sim_time_ms_zero = 0, /**< Simulator returns zero (no real-time clock) */
00431 } rx_comm_sim_time_t;
```

4.3.3 Function Documentation

4.3.3.1 internal_check_channel_heartbeat()

void internal_check_channel_heartbeat (
rx_comm_manager_t * mgr,
uint8_t ch,
uint32_t now_ms) [static]

Check heartbeat implicit timeout on all transports.

Evaluates the dual-detection heartbeat model. If no valid frame has been received on a transport within 200ms, the link is declared dead and the status callback is invoked.

Precondition

mgr must be non-NULL and initialized

Postcondition

Link status updated for each transport

Since

Version 1.0.0

Check heartbeat timeout on a single channel

Precondition

ch < k_comm_channel_count

Postcondition

Link status updated if timeout exceeded

Since

Version 1.0.0

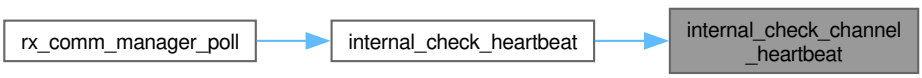
Definition at line 979 of file rx_comm_manager.c.

```
00980 {
00981   rx_comm_heartbeat_state_t* hb = &mgr->heartbeat[ch];
00982
00983   /* Skip channels that haven't received any frame yet */
00984   if (hb->status != k_link_status_healthy) {
00985     return;
00986   }
00987
00988   /* Check implicit timeout: 200ms since last valid frame */
00989   const uint32_t elapsed = now_ms - hb->last_rx_ms;
00990   if (elapsed < k_heartbeat_implicit_timeout_ms) {
00991     return;
00992   }
00993
00994   hb->status = k_link_status_dead;
00995   rx_log_warn_val(s_tag, "Link dead: no frame (ms elapsed)", elapsed);
00996
00997   if (mgr->link_status_cb != nullptr) {
00998     mgr->link_status_cb((rx_comm_channel_t)ch, k_link_status_dead, mgr->link_status_ctx);
00999   }
01000 }
```

References rx_comm_manager_t::heartbeat, k_heartbeat_implicit_timeout_ms, k_link_status_dead, k_link_status_healthy, rx_comm_heartbeat_state_t::last_rx_ms, rx_comm_manager_t::link_status_cb, rx_comm_manager_t::link_status_ctx, s_tag, and rx_comm_heartbeat_state_t::status.

Referenced by internal_check_heartbeat().

Here is the caller graph for this function:



4.3.3.2 internal_check_heartbeat()

```
void internal_check_heartbeat (
                rx_comm_manager_t * mgr)    [static]
```

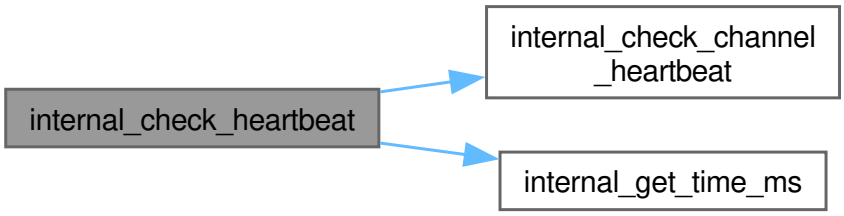
Definition at line 1002 of file rx_comm_manager.c.

```
01003 {
01004   const uint32_t now = internal_get_time_ms();
01005
01006   for (uint8_t ch = 0; ch < k_comm_channel_count; ch++) {
01007     internal_check_channel_heartbeat(mgr, ch, now);
01008   }
01009 }
```

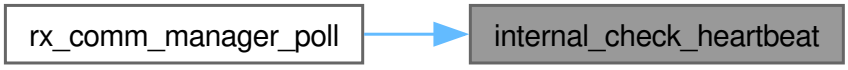
References internal_check_channel_heartbeat(), internal_get_time_ms(), and k_comm_channel_count.

Referenced by rx_comm_manager_poll().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.3 internal_dequeue_event()

```
void internal_dequeue_event (
                rx_comm_manager_t * mgr,
                rx_comm_event_entry_t * entry)    [static]
```

Process one event from the FIFO queue.

Dequeues the oldest event and attempts to send it. For SPI events, if the send fails, the event is re-enqueued with incremented retry count (up to k_event_max_retries). For USB events, failure is simply dropped (fire-and-forget).

Precondition

mgr must be non-NULL and initialized

Postcondition

One event processed (sent or dropped on max retries)

Since

Version 1.0.0

Dequeue the front event entry and advance the tail pointer

Precondition

mgr->event_queue_count > 0

Postcondition

entry cleared and tail pointer advanced

Since

Version 1.0.0

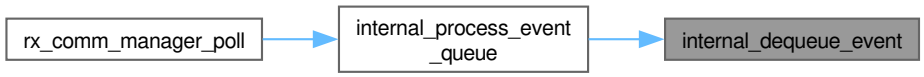
Definition at line 882 of file rx_comm_manager.c.

```
00883 {
00884   entry->occupied          = false;
00885   entry->retries           = 0;
00886   entry->next_retry_time_ms = 0;
00887   mgr->event_queue_tail     = (mgr->event_queue_tail + 1) % k_comm_event_queue_depth;
00888   mgr->event_queue_count--;
00889 }
```

References rx_comm_manager_t::event_queue_count, rx_comm_manager_t::event_queue_tail, k_comm_event_queue_depth, rx_comm_event_entry_t::next_retry_time_ms, rx_comm_event_entry_t::occupied, and rx_comm_event_entry_t::retries.

Referenced by internal_process_event_queue().

Here is the caller graph for this function:



4.3.3.4 internal_get_time_ms()

```
uint32_t internal_get_time_ms (
                void ) [static]
```

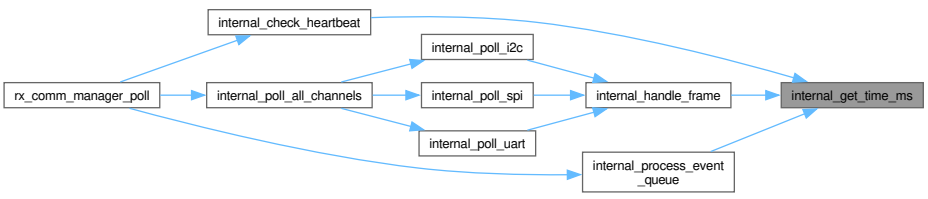
Definition at line 433 of file rx_comm_manager.c.

```
00434 {
00435 #if !RX_IS_SIMULATOR
00436   return (uint32_t)tx_time_get() * k_threadx_ms_per_tick;
00437 #else
00438   return k_sim_time_ms_zero;
00439 #endif
00440 }
```

References k_sim_time_ms_zero.

Referenced by internal_check_heartbeat(), internal_handle_frame(), and internal_process_event_queue().

Here is the caller graph for this function:



4.3.3.5 internal_handle_frame()

```
RX_STATIC_TESTABLE void internal_handle_frame (
        rx_comm_manager_t * mgr,
        rx_comm_channel_t channel,
        const rx_frame_t * frame)
```

Handle received frame by invoking user callback and optional ASCII output.

Central frame dispatch point for all channels. Performs optional ASCII debugging output, then invokes user-registered callback with frame data. This function isolates callback invocation logic from channel-specific polling.

Algorithm:

1. Validate parameters (NULL checks, channel range)
2. Output decoded ASCII to Port 1 if enabled (RX direction)
3. Invoke user callback if registered

Design Rationale: Centralizing frame handling logic ensures consistent behavior across USB and SPI channels (single point of dispatch).

Precondition

mgr must be initialized

frame must be valid and CRC-verified by transport layer

channel must be in valid range [0, k_comm_channel_count)

Postcondition

User callback invoked if registered (callback may be NULL)

ASCII output sent to Port 1 if enabled

Note

Thread safety depends on user callback implementation

Callback errors are not returned (callback should handle internally)

Warning

User callback must not block for extended periods (affects polling rate)

Performance:

- Without decoded output: ~2 us (callback overhead only)
- With decoded output: ~50-150 us (ASCII formatting + USB puts)

See also

internal_output_decoded() ASCII debugging output

rx_comm_frame_callback_t User callback type

Since

Version 1.0.0

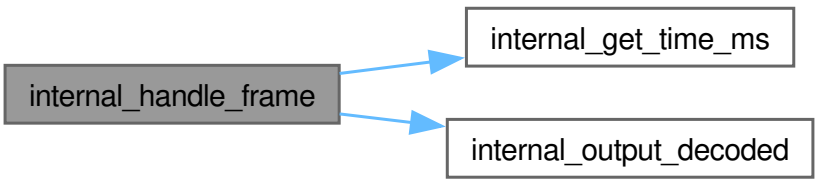
Definition at line 648 of file rx_comm_manager.c.

```
00649 {
00650   /* Update heartbeat: any valid frame resets the implicit timeout timer.
00651    * Per the dual-detection model, this is the primary link health
00652    * mechanism (200ms implicit timeout). */
00653   rx_comm_heartbeat_state_t* hb = &mgr->heartbeat[channel];
00654   hb->last_rx_ms                = internal_get_time_ms();
00655   if (hb->status != k_link_status_healthy) {
00656     rx_comm_link_status_t old_status = hb->status;
00657     hb->status                       = k_link_status_healthy;
00658     if (old_status == k_link_status_dead) {
00659       rx_log_info(s_tag, "Link recovered: frame received");
00660     } else {
00661       rx_log_info(s_tag, "Link established: first frame received");
00662     }
00663     if (mgr->link_status_cb != nullptr) {
00664       mgr->link_status_cb(channel, k_link_status_healthy, mgr->link_status_ctx);
00665     }
00666   }
00667
00668   rx_log_debug_val(s_tag, "Frame received, type", (uint8_t)frame->header.type);
00669
00670   if (frame->header.type == k_frame_type_command) {
00671     rx_log_info_val(s_tag, "CMD frame seq", frame->header.sequence);
00672     rx_log_info_val(s_tag, "CMD frame channel", (uint8_t)channel);
00673   }
00674
00675   /* Output decoded ASCII to Port 1 */
00676   internal_output_decoded(mgr, frame, false);
00677
00678   /* Invoke user callback if set */
00679   if (mgr->callback != nullptr) {
00680     mgr->callback(channel, frame, mgr->callback_ctx);
00681   }
00682 }
```

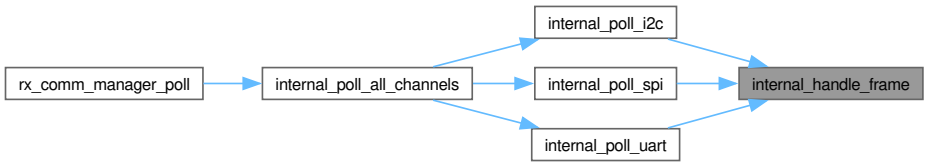
References rx_comm_manager_t::callback, rx_comm_manager_t::callback_ctx, rx_comm_manager_t::heartbeat, internal_get_time_ms(), internal_output_decoded(), k_link_status_dead, k_link_status_healthy, rx_comm_heartbeat_state_t::last_rx_ms, rx_comm_manager_t::link_status_cb, rx_comm_manager_t::link_status_tag, and rx_comm_heartbeat_state_t::status.

Referenced by internal_poll_i2c(), internal_poll_spi(), and internal_poll_uart().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.6 internal_output_decoded()

```
RX_STATIC_TESTABLE void internal_output_decoded (
        rx_comm_manager_t * mgr,
        const rx_frame_t * frame,
        bool is_tx)
```

Output decoded ASCII frame to USB Port 1 for debugging.

Formats the given frame as human-readable ASCII text and outputs to USB Port 1 (decoded debug port) if decoded output is enabled. Used for development/debugging to monitor frame traffic without binary protocol parsing.

Algorithm:

1. Validate parameters (NULL checks)
2. Check if decoded output is enabled (cfg->enable_decoded_output)
3. Format frame as ASCII via rx_frame_ascii_format()
4. Send ASCII string to USB Port 1 via rx_usb_puts()

Performance: ~50-150 us depending on frame size (ASCII formatting overhead). Disable in production for optimal performance.

Precondition

mgr must be initialized via rx_comm_manager_init()

frame must be a valid frame structure

Postcondition

mgr->ascii_buffer content is modified (temporary)

ASCII output sent to USB Port 1 if enabled and successful

Note

Not thread-safe - caller must ensure exclusive access to mgr->ascii_buffer

Errors are silently ignored (formatting/USB send failures)

USB Port 1 must be configured for ASCII output to appear

Example Output:

```
RX: [USB] Type=0x01 Flags=0x00 Len=8 Seq=42 Payload=0102030405060708 CRC=ABCD1234
TX: [SPI] Type=0x02 Flags=0x80 Len=4 Seq=43 Payload=01020304 CRC=5678CDEF
```

See also

rx_frame_ascii_format() Frame-to-ASCII conversion

rx_usb_puts() USB debug output

Since

Version 1.0.0

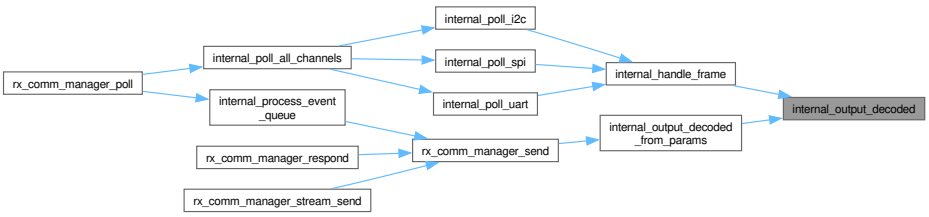
Definition at line 556 of file rx_comm_manager.c.

```
00557 {
00558   if (!mgr->enable_decoded_output) {
00559     return;
00560   }
00561
00562   uint32_t len = 0;
00563   /* rx_frame_ascii_format always succeeds and produces output for any valid frame
00564    * and adequately-sized buffer (guaranteed by ascii_buffer sizing in the manager). */
00565   (void)rx_frame_ascii_format(frame, is_tx, mgr->ascii_buffer, sizeof(mgr->ascii_buffer), &len);
00566   /* The old decoded-debug path wrote to USB CDC port 1; with USB0 gone, we
00567    * route the ASCII dump through the UART log ring so it still shows up on
00568    * the gateway prefixed with [RX72N]. */
00569   rx_log_uart_puts(mgr->ascii_buffer);
00570 }
```

References rx_comm_manager_t::ascii_buffer, and rx_comm_manager_t::enable_decoded_output.

Referenced by internal_handle_frame(), and internal_output_decoded_from_params().

Here is the caller graph for this function:



4.3.3.7 internal_output_decoded_from_params()

```
void internal_output_decoded_from_params (
            rx_comm_manager_t * mgr,
            const rx_comm_send_params_t * params)   [static]
```

Build temporary frame from send params and output decoded ASCII (TX direction).

Helper for rx_comm_manager_send() to output ASCII-formatted frame after a successful send. Constructs a temporary rx_frame_t from the send parameters and delegates to internal_output_decoded().

Precondition

mgr must be non-NULL and initialized

params must be non-NULL with valid fields

Postcondition

ASCII output sent to USB Port 1 if decoded output is enabled

Note

Not thread-safe - caller must ensure exclusive access to mgr->ascii_buffer

Since

Version 1.0.0

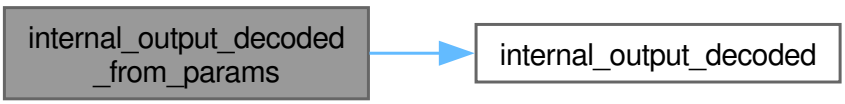
Definition at line 589 of file rx_comm_manager.c.

```
00591 {
00592   /* Build a temporary frame for ASCII formatting */
00593   rx_frame_t frame      = {}; /* Zero-initialize to clear padding/unset fields */
00594   frame.header.sequence = k_frame_seq_placeholder;
00595   frame.header.length   = (uint16_t)params->payload_len;
00596   frame.header.type     = (uint8_t)params->type;
00597   frame.header.flags    = params->flags;
00598
00599   /* payload_len <= k_frame_max_payload is guaranteed by caller validation */
00600   for (uint32_t i = 0; i < params->payload_len; i++) {
00601     frame.payload[i] = params->payload[i];
00602   }
00603   frame.crc = k_frame_crc_placeholder;
00604
00605   internal_output_decoded(mgr, &frame, true);
00606 }
```

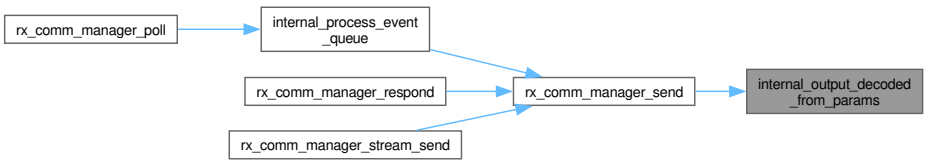
References rx_comm_send_params_t::flags, internal_output_decoded(), k_frame_crc_placeholder, k_frame_seq_placeholder, rx_comm_send_params_t::payload, rx_comm_send_params_t::payload_len, and rx_comm_send_params_t::type.

Referenced by rx_comm_manager_send().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.8 internal_poll_all_channels()

```
rx_err_t internal_poll_all_channels (
            rx_comm_manager_t * mgr,
            bool * received)   [static]
```

Poll all enabled communication channels for incoming frames (non-blocking).

Performs non-blocking poll of all enabled channels (USB and SPI) sequentially, dispatching any received frames to the registered user callback. Returns aggregate result indicating whether any frames were received across all channels.

Algorithm:

1. Validate parameters: Check mgr pointer and initialized flag (NASA Rule 5)

2. Poll USB channel via internal_poll_usb():
   - If k_rx_ok: Mark received=true, continue to SPI
   - If k_rx_err_timeout: Continue to SPI (no data is normal)
   - If other error: Return error immediately (critical failure)
3. Poll SPI channel via internal_poll_spi():
   - If k_rx_ok: Mark received=true
   - If k_rx_err_timeout: Continue (no data is normal)
   - If other error: Return error immediately (critical failure)
4. Return result:
   - k_rx_ok if ANY channel received frame
   - k_rx_err_timeout if NO channels received (normal condition)
   - Other error codes propagated from transport layers

Performance: 15-200 us typical, depends on:

- Number of enabled channels (USB/SPI/both)
- Whether frames are available (data path vs fast timeout path)
- ASCII debugging enabled (adds ~50-150 us per frame)
- User callback execution time (not included in measurement)

Design Rationale:

- Non-blocking: Never blocks waiting for data (0 ms timeout)
- Fair scheduling: Both channels polled each iteration
- Timeout tolerance: Timeout is expected, not error condition
- Error prioritization: Critical errors returned immediately
- Aggregation: k_rx_ok if ANY channel received data

Precondition

mgr must be non-NULL
mgr must be initialized
At least one channel should be enabled (usb_handle or spi_handle non-NULL)

Postcondition

User callback invoked for each successfully received frame
ASCII output sent to Port 1 if enabled

Note

Non-blocking - always returns immediately

k_rx_err_timeout is NORMAL - means no data available (not an error)

User callback execution time is NOT included in performance measurements

Warning

Call this function regularly (e.g., 100 Hz) to avoid buffer overruns

User callback must not block for extended periods

Performance:

- No frames available: ~15 us (both channels timeout, fast path)
- USB frame received: ~100-250 us (USB bulk transfer + callback)
- SPI frame received: ~50-150 us (SPI hardware transfer + callback)
- Both frames: ~150-400 us (both channels + callbacks)
- With ASCII debugging: Add ~50-150 us per frame

Typical Usage - ThreadX Communication Task:

```
static void comm_task_entry(ULONG thread_input)
{
  (void)thread_input;

  // Initialize communication manager
  // ... (see rx_comm_manager_init() documentation) ...

  // Main polling loop @ 100 Hz
  while (1) {
    // Poll all channels (non-blocking)
    rx_err_t err = rx_comm_manager_poll(&g_comm_mgr);

    // Handle critical errors (not timeout)
    if (err != k_rx_ok && err != k_rx_err_timeout) {
      // Log error, implement recovery strategy
      log_error("Comm poll failed: %d", err);
      tx_thread_sleep(100);  // 1 second backoff
      continue;
    }

    // Timeout is normal - just means no data available

    // Sleep 10 ticks (100 ms @ 100 Hz tick rate = 10 Hz polling)
    tx_thread_sleep(10);
  }
}
```

Error Handling - Distinguishing Timeout from Errors:

```
rx_err_t err = rx_comm_manager_poll(&g_comm_mgr);
if (err == k_rx_ok) {
  // At least one frame received and processed
} else if (err == k_rx_err_timeout) {
  // No frames available - this is NORMAL, not an error
} else {
  // Critical error - handle appropriately
  handle_comm_error(err);
}
```

See also

internal_poll_usb() USB channel polling implementation

internal_poll_spi() SPI channel polling implementation

internal_handle_frame() Frame dispatch

rx_comm_manager_init() Initialization with channel configuration

Since

Version 1.0.0

Poll all transport channels and aggregate results

Precondition

mgr must be non-NULL and initialized

Postcondition

∗received reflects whether any frames were received

Since

Version 1.0.0

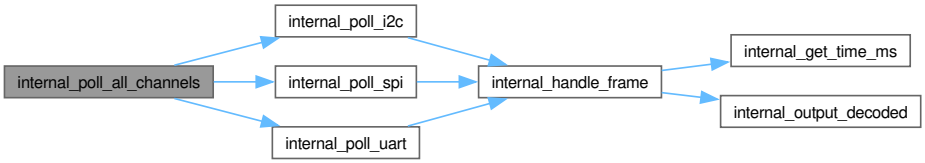
Definition at line 1351 of file rx_comm_manager.c.

```
01352 {
01353   /* Poll UART channel (primary path to the Pi5 gateway) */
01354   rx_err_t uart_err = internal_poll_uart(mgr);
01355   if (uart_err == k_rx_ok) {
01356     *received = true;
01357   } else if (uart_err != k_rx_err_timeout && uart_err != k_rx_err_no_data) {
01358     rx_log_error_val(s_tag, "UART poll error", uart_err);
01359     return uart_err;
01360   }
01361
01362   /* Poll SPI channel */
01363   rx_err_t spi_err = internal_poll_spi(mgr);
01364   if (spi_err == k_rx_ok) {
01365     *received = true;
01366   } else if (spi_err != k_rx_err_timeout) {
01367     rx_log_error_val(s_tag, "SPI poll error", spi_err);
01368     return spi_err;
01369   }
01370
01371   /* Poll I2C channel */
01372   rx_err_t i2c_err = internal_poll_i2c(mgr);
01373   if (i2c_err == k_rx_ok) {
01374     *received = true;
01375   } else if (i2c_err != k_rx_err_timeout && i2c_err != k_rx_err_no_data) {
01376     rx_log_error_val(s_tag, "I2C poll error", i2c_err);
01377     return i2c_err;
01378   }
01379
01380   return k_rx_ok;
01381 }
```

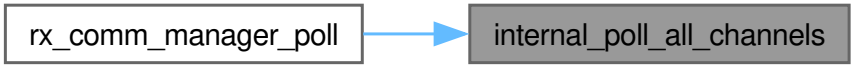
References internal_poll_i2c(), internal_poll_spi(), internal_poll_uart(), and s_tag.

Referenced by rx_comm_manager_poll().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.9 internal_poll_i2c()

rx_err_t internal_poll_i2c (
rx_comm_manager_t * mgr) [static]

Poll I2C channel for incoming frames (non-blocking).

Attempts to receive one frame from I2C peripheral channel with zero timeout. If frame is successfully received, it is dispatched via internal_handle_frame().

Precondition

mgr must be non-NULL

mgr->i2c_handle must be initialized before calling rx_comm_manager_poll()

Postcondition

On k_rx_ok: callback invoked for received frame

On k_rx_err_timeout or k_rx_err_no_data: no state change

Note

Non-blocking - always returns immediately

See also

internal_handle_frame() Frame dispatch

rx_i2c_comm_receive() I2C peripheral receive implementation

Since

Version 1.0.0

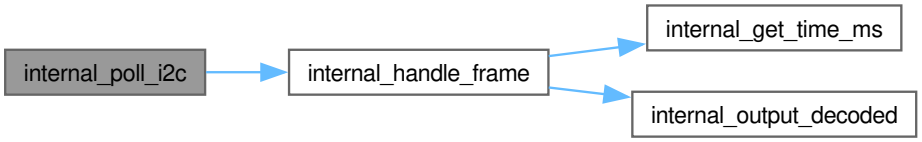
Definition at line 796 of file rx_comm_manager.c.

```
00797 {
00798   if (mgr->i2c_handle == nullptr) {
00799     return k_rx_err_timeout;
00800   }
00801
00802   rx_frame_t frame;
00803   rx_err_t   err = rx_i2c_comm_receive(mgr->i2c_handle, &frame, k_receive_timeout_ms);
00804
00805   if (err == k_rx_ok) {
00806     internal_handle_frame(mgr, k_comm_channel_i2c, &frame);
00807     return k_rx_ok;
00808   }
00809
00810   return err;
00811 }
```

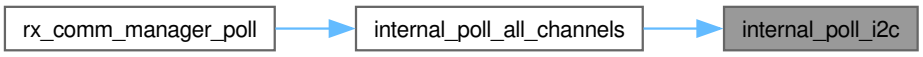
References rx_comm_manager_t::i2c_handle, internal_handle_frame(), k_comm_channel_i2c, and k_receive_timeout_ms.

Referenced by internal_poll_all_channels().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.10 internal_poll_spi()

rx_err_t internal_poll_spi (
rx_comm_manager_t * mgr) [static]

Poll SPI channel for incoming frames (non-blocking).

Attempts to receive one frame from SPI channel with zero timeout (non-blocking). If frame is successfully received, it is dispatched via internal_handle_frame(). Timeout (no data available) is expected and normal - not an error condition.

Algorithm:

1. Validate mgr pointer (NULL check)
2. Skip if SPI handle is nullptr (channel not configured)
3. Call rx_spi_comm_receive() with 0 ms timeout
4. On success: dispatch frame via internal_handle_frame()
5. On timeout: return k_rx_err_timeout (expected, not error)
6. On other errors: propagate error code

Design Note: Identical logic to internal_poll_usb() but for SPI channel. Code duplication is intentional (NASA Rule 4: keep functions short and simple).

Precondition

mgr must be non-NULL

If SPI channel enabled, spi_handle must be initialized

Postcondition

On k_rx_ok: callback invoked, ASCII output sent (if enabled)

On timeout: no side effects (no frame received)

Note

Non-blocking - always returns immediately

Timeout is NOT an error - it means no data is available

Warning

Do not call if spi_handle is uninitialized (will return timeout)

Performance:

- No data available: ~5 us (fast path)
- Frame received: ~50-150 us (SPI hardware transfer + callback)

See also

internal_handle_frame() Frame dispatch

rx_spi_comm_receive() SPI receive implementation

Since

Version 1.0.0

Definition at line 723 of file rx_comm_manager.c.

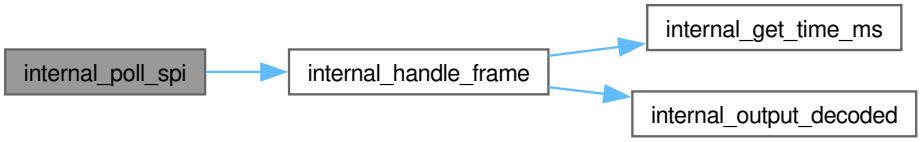
```
00724 {
00725   if (mgr->spi_handle == nullptr) {
00726     return k_rx_err_timeout;
00727   }
00728
00729   /* If SPI link layer is available, use it for HARQ decoding
00730    * NOTE: Using static buffers to avoid stack overflow (~3KB total).
00731    * Safe because comm manager poll is single-threaded (called from comm_task). */
00732   if (mgr->spi_link != nullptr) {
00733     static rx_spi_link_receive_result_t s_link_result;
00734     rx_err_t err = rx_spi_link_receive(mgr->spi_link, &s_link_result, k_receive_timeout_ms);
00735
00736     if (err == k_rx_ok) {
00737       /* Convert link result to rx_frame_t for internal_handle_frame() */
00738       static rx_frame_t s_frame1;
00739       s_frame1                = (rx_frame_t){};
00740       s_frame1.header.sequence = s_link_result.sequence;
00741       s_frame1.header.length   = (uint16_t)s_link_result.payload_len;
00742       s_frame1.header.type     = s_link_result.frame_type;
00743
00744       /* Preserve retransmit and FEC flags from link layer result */
00745       s_frame1.header.flags = k_frame_flag_none;
00746       if (s_link_result.is_retransmit) {
00747         s_frame1.header.flags |= k_frame_flag_retransmit;
00748       }
00749       if (s_link_result.fec_decoded) {
```

```
00750         s_frame1.header.flags |= k_frame_flag_fec_enabled;
00751       }
00752
00753       for (uint32_t i = 0; i < s_link_result.payload_len; i++) {
00754         s_frame1.payload[i] = s_link_result.payload[i];
00755       }
00756       internal_handle_frame(mgr, k_comm_channel_spi, &s_frame1);
00757       return k_rx_ok;
00758     }
00759
00760     return err;
00761   }
00762
00763   /* Fallback: raw SPI without HARQ */
00764   static rx_frame_t s_frame2;
00765   rx_err_t          err = rx_spi_comm_receive(mgr->spi_handle, &s_frame2, k_receive_timeout_ms);
00766
00767   if (err == k_rx_ok) {
00768     internal_handle_frame(mgr, k_comm_channel_spi, &s_frame2);
00769     return k_rx_ok;
00770   }
00771
00772   return err;
00773 }
```

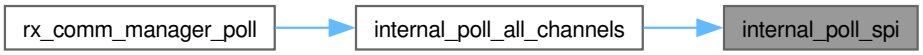
References internal_handle_frame(), k_comm_channel_spi, k_receive_timeout_ms, rx_comm_manager_t::spi_handle and rx_comm_manager_t::spi_link.

Referenced by internal_poll_all_channels().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.11 internal_poll_uart()

```
rx_err_t internal_poll_uart (
            rx_comm_manager_t * mgr)   [static]
```

Poll UART channel for incoming frames (non-blocking).

Attempts to receive one frame from UART (SCI9) channel with zero timeout. If frame is successfully received, it is dispatched via internal_handle_frame().

Precondition

mgr must be non-NULL

mgr->uart_handle must be initialized before calling rx_comm_manager_poll()

Postcondition

On k_rx_ok: callback invoked for received frame

On k_rx_err_timeout or k_rx_err_no_data: no state change

Note

Non-blocking - always returns immediately

See also

internal_handle_frame() Frame dispatch

rx_uart_comm_receive() UART receive implementation

Since

Version 1.0.0

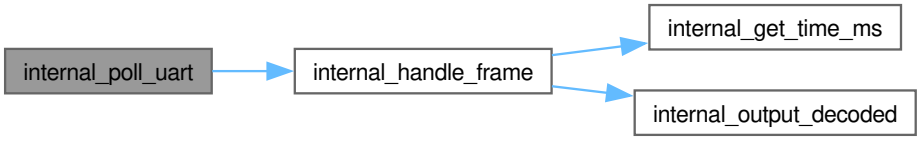
Definition at line 834 of file rx_comm_manager.c.

```
00835 {
00836   if (mgr->uart_handle == nullptr) {
00837     return k_rx_err_timeout;
00838   }
00839
00840   rx_frame_t frame;
00841   rx_err_t   err = rx_uart_comm_receive(mgr->uart_handle, &frame, k_receive_timeout_ms);
00842
00843   if (err == k_rx_ok) {
00844     if (frame.header.type == k_frame_type_command) {
00845       rx_log_info_val(s_tag, "UART RX CMD seq", frame.header.sequence);
00846     }
00847     internal_handle_frame(mgr, k_comm_channel_uart, &frame);
00848     return k_rx_ok;
00849   }
00850
00851   return err;
00852 }
```

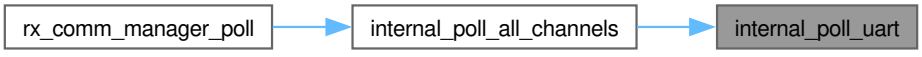
References internal_handle_frame(), k_comm_channel_uart, k_receive_timeout_ms, s_tag, and rx_comm_manager_t::uart_handle.

Referenced by internal_poll_all_channels().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.12 internal_process_event_queue()

```
void internal_process_event_queue (
                rx_comm_manager_t * mgr)    [static]
```

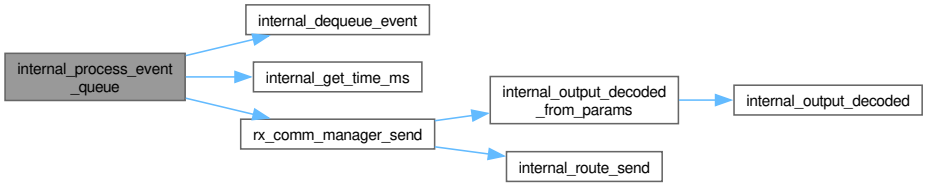
Definition at line 891 of file rx_comm_manager.c.

```
00892 {
00893   if (mgr->event_queue_count == 0) {
00894     return;
00895   }
00896
00897   rx_comm_event_entry_t* entry = &mgr->event_queue[mgr->event_queue_tail];
00898   if (!entry->occupied) {
00899     return;
00900   }
00901
00902   /* Check backoff: skip if retry time hasn't been reached yet */
00903   const uint32_t now_ms = internal_get_time_ms();
00904   if (entry->retries > 0 && now_ms < entry->next_retry_time_ms) {
00905     return; /* Backoff period not elapsed, try again later */
00906   }
00907
00908   /* Build send params from queued entry */
00909   const rx_comm_send_params_t params = {
00910     .channel     = entry->channel,
00911     .type        = entry->type,
00912     .flags       = entry->flags,
00913     .payload     = entry->payload,
00914     .payload_len = entry->payload_len,
00915   };
00916
00917   rx_err_t err = rx_comm_manager_send(mgr, &params);
00918
00919   if (err == k_rx_ok || entry->channel == k_comm_channel_uart) {
00920     /* Success, or UART fire-and-forget: dequeue.
00921      * UART has no link-layer retry (the CY7C65213 bridge + USB CDC transport
00922      * on the host provides enough reliability for the command/response path). */
00923     internal_dequeue_event(mgr, entry);
00924     if (err == k_rx_ok) {
00925       rx_log_debug_val(s_tag, "Event sent, queue depth", mgr->event_queue_count);
00926     } else {
00927       rx_log_warn_val(s_tag, "UART event dropped, queue depth", mgr->event_queue_count);
00928     }
00929     return;
00930   }
00931
00932   /* SPI send failed: retry with exponential backoff */
00933   entry->retries++;
00934   if (entry->retries >= k_event_max_retries) {
00935     rx_log_error_val(s_tag, "SPI event failed after retries", (uint8_t)entry->retries);
00936     internal_dequeue_event(mgr, entry);
00937     return;
00938   }
00939
00940   /* Compute exponential backoff: initial_ms « (retries - 1), capped at max */
00941   uint32_t backoff_ms = (uint32_t)k_event_initial_backoff_ms « (entry->retries - 1);
00942   if (backoff_ms > k_event_max_backoff_ms) {
00943     backoff_ms = k_event_max_backoff_ms;
00944   }
00945   entry->next_retry_time_ms = now_ms + backoff_ms;
00946
00947   rx_log_warn_val(s_tag, "SPI event retry attempt", (uint8_t)entry->retries);
00948   /* Entry stays in queue, will be retried after backoff period */
00949 }
```

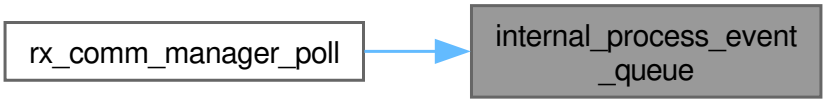
References rx_comm_event_entry_t::channel, rx_comm_manager_t::event_queue, rx_comm_manager_t::event_queu
rx_comm_manager_t::event_queue_tail, rx_comm_event_entry_t::flags, internal_dequeue_event(),
internal_get_time_ms(), k_comm_channel_uart, k_event_initial_backoff_ms, k_event_max_backoff_ms,
k_event_max_retries, rx_comm_event_entry_t::next_retry_time_ms, rx_comm_event_entry_t::occupied,
rx_comm_event_entry_t::payload, rx_comm_event_entry_t::payload_len, rx_comm_event_entry_t::retries,
rx_comm_manager_send(), s_tag, and rx_comm_event_entry_t::type.

Referenced by rx_comm_manager_poll().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.13 internal_route_send()

```
rx_err_t internal_route_send (
            rx_comm_manager_t * mgr,
            const rx_comm_send_params_t * params)    [static]
```

Send frame on specified communication channel.

Routes frame transmission to the appropriate transport layer (USB or SPI) based on specified channel. Supports configurable frame type, flags, and variable-length payload. Optionally outputs ASCII-formatted frame to USB Port 1 if debugging enabled.

Algorithm:

1. Validate parameters (NASA Rule 5):
   - mgr and params pointers non-NULL
   - Manager initialized flag
   - Payload pointer valid if payload_len > 0
   - payload_len <= k_frame_max_payload (255 bytes)
2. Route to channel:
   - USB: Verify usb_handle non-NULL, call rx_usb_comm_send()
   - SPI: Verify spi_handle non-NULL, call rx_spi_comm_send()
   - Invalid channel: Return k_rx_err_invalid_arg
3. On success, if ASCII debugging enabled:
   - Build temporary rx_frame_t from params
   - Format as ASCII via internal_output_decoded()
   - Send to USB Port 1 (TX direction)

Performance:

- USB: ~30-150 us (depends on payload size, USB bulk transfer)
- SPI: ~20-100 us (depends on payload size, SPI hardware transfer)
- ASCII debugging: Add ~50-150 us if enabled

Precondition

mgr must be non-NULL and initialized

params must be non-NULL with all valid fields

Channel handle (USB/SPI) must be initialized

If payload_len > 0, payload must point to valid data

Postcondition

Frame transmitted on specified channel if successful

ASCII output sent to Port 1 if enabled and successful

Note

Payload sequence number and CRC-32 managed by transport layer

ASCII debugging adds overhead - disable in production

Warning

Ensure channel is ready before sending (use rx_comm_manager_channel_ready)

Send Parameters Structure:

```
typedef struct {
  rx_comm_channel_t channel;      // USB or SPI
  uint8_t           type;         // Frame type
  uint8_t           flags;        // Frame flags
  const uint8_t*    payload;      // Payload data
  uint32_t          payload_len;  // Payload length [0, 255]
} rx_comm_send_params_t;
```

Example - Send Command on USB:

```
uint8_t cmd_payload[8] = {0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0x07, 0x08};

rx_comm_send_params_t params = {
  .channel = k_comm_channel_usb,
  .type = k_frame_type_command,
  .flags = k_frame_flag_none,
  .payload = cmd_payload,
  .payload_len = sizeof(cmd_payload),
};

rx_err_t err = rx_comm_manager_send(&g_comm_mgr, &params);
if (err == k_rx_ok) {
  // Command sent successfully
}
```

Example - Send Empty Frame (No Payload):

```
rx_comm_send_params_t params = {
  .channel = k_comm_channel_spi,
  .type = k_frame_type_response,
  .flags = k_frame_flag_ack,
  .payload = nullptr,         // No payload
  .payload_len = 0,       // Zero length
};

rx_err_t err = rx_comm_manager_send(&g_comm_mgr, &params);
```

See also

rx_comm_manager_respond() Convenience wrapper for response frames

rx_comm_send_params_t Send parameters structure

rx_spi_comm_send() SPI send implementation

rx_uart_comm_send() UART send implementation

rx_i2c_comm_send() I2C peripheral send implementation

Since

Version 1.0.0

Route frame to transport layer based on channel

Precondition

mgr and params must be non-NULL

params validated by caller (payload length, etc.)

Postcondition

Frame sent on specified channel transport

Since

Version 1.0.0

Definition at line 1511 of file rx_comm_manager.c.

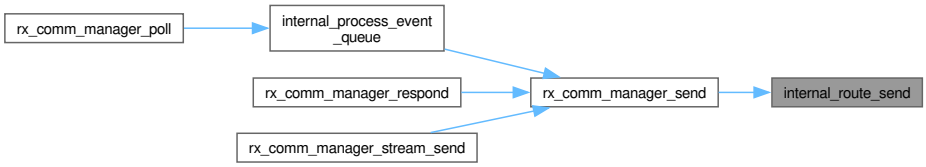
```
01512 {
01513   switch (params->channel) {
01514     case k_comm_channel_spi:
01515       if (mgr->spi_handle == nullptr) {
01516         rx_log_error(s_tag, "Send failed: SPI handle NULL");
01517         return k_rx_err_invalid_state;
01518       }
01519       /* Use HARQ link layer if available, otherwise raw SPI */
01520       if (mgr->spi_link != nullptr) {
01521         /* NOTE: params->flags intentionally NOT forwarded to rx_spi_link_send().
01522          * The link layer manages its own flags internally (k_frame_flag_requires_ack,
01523          * k_frame_flag_fec_enabled, k_frame_flag_retransmit) based on HARQ state. */
01524         return rx_spi_link_send(mgr->spi_link, params->type, params->payload, params->payload_len);
01525       }
01526       return rx_spi_comm_send(mgr->spi_handle,
01527                               params->type,
01528                               params->flags,
01529                               params->payload,
01530                               params->payload_len);
01531
01532     case k_comm_channel_i2c:
01533       if (mgr->i2c_handle == nullptr) {
01534         rx_log_error(s_tag, "Send failed: I2C handle NULL");
01535         return k_rx_err_invalid_state;
01536       }
01537       return rx_i2c_comm_send(mgr->i2c_handle,
01538                               params->type,
01539                               params->flags,
01540                               params->payload,
01541                               params->payload_len);
01542
01543     case k_comm_channel_uart:
01544       if (mgr->uart_handle == nullptr) {
01545         rx_log_error(s_tag, "Send failed: UART handle NULL");
01546         return k_rx_err_invalid_state;
01547       }
01548       return rx_uart_comm_send(mgr->uart_handle,
01549                                params->type,
```

```
01550                           params->flags,
01551                           params->payload,
01552                           params->payload_len);
01553
01554     default:
01555       rx_log_error(s_tag, "Send failed: invalid channel");
01556       return k_rx_err_invalid_arg;
01557   }
01558 }
```

References rx_comm_send_params_t::channel, rx_comm_send_params_t::flags, rx_comm_manager_t::i2c_handle, k_comm_channel_i2c, k_comm_channel_spi, k_comm_channel_uart, rx_comm_send_params_t::payload, rx_comm_send_params_t::payload_len, s_tag, rx_comm_manager_t::spi_handle, rx_comm_manager_t::spi_link, rx_comm_send_params_t::type, and rx_comm_manager_t::uart_handle.

Referenced by rx_comm_manager_send().

Here is the caller graph for this function:



4.3.3.14 rx_comm_manager_channel_name()

```
const char * rx_comm_manager_channel_name (
                rx_comm_channel_t channel)
```

Get human-readable name for communication channel.

Get channel name string.

Returns static string name for given channel enum value. Used for logging, debugging, and user interface display. Always returns a valid string (never NULL).

Returned Strings:

- k_comm_channel_usb -> "USB"
- k_comm_channel_spi -> "SPI"
- Invalid/unknown -> "UNKNOWN"

Use Cases:

- Logging channel activity
- Debugging frame traffic
- User interface display
- Error messages

Precondition

None (accepts any input)

Postcondition

Returns pointer to static string (valid for program lifetime)

Note

Returned string is static - do not free

Thread-safe (returns read-only static strings)

Never returns NULL - always safe to use in printf

Example - Logging:

```
static void on_frame_received(rx_comm_channel_t channel,
                              const rx_frame_t* frame,
                              void* ctx)
{
  printf("Frame received on %s: type=%d len=%d\n",
         rx_comm_manager_channel_name(channel),
         frame->header.type,
         frame->header.length);
}
```

Example - Error Messages:

```
rx_err_t err = rx_comm_manager_send(&g_comm_mgr, &params);
if (err != k_rx_ok) {
  fprintf(stderr, "Failed to send on %s: error %d\n",
          rx_comm_manager_channel_name(params.channel), err);
}
```

See also

rx_comm_channel_t Channel enumeration

Since

Version 1.0.0

Definition at line 1946 of file rx_comm_manager.c.

```
01947 {
01948   switch (channel) {
01949     case k_comm_channel_uart:
01950       return "UART";
01951     case k_comm_channel_spi:
01952       return "SPI";
01953     case k_comm_channel_i2c:
01954       return "I2C";
01955     default:
01956       return "UNKNOWN";
01957   }
01958 }
```

References k_comm_channel_i2c, k_comm_channel_spi, and k_comm_channel_uart.

4.3.3.15 rx_comm_manager_channel_ready()

```
rx_err_t rx_comm_manager_channel_ready (
                rx_comm_manager_t * mgr,
                rx_comm_channel_t channel,
                bool * ready)   [nodiscard]
```

Query if communication channel is ready for send/receive operations.

Check if channel is ready for communication.

Checks if specified channel is configured and ready for communication. USB channel readiness depends on USB enumeration and configuration by host. SPI channel is ready immediately if handle is configured (no enumeration required).

Readiness Criteria:

- USB: usb_handle non-NULL AND USB device enumerated and configured by host
- SPI: spi_handle non-NULL (always ready if configured)

Use Cases:

- Check readiness before sending to avoid errors
- Implement fallback logic (try USB, fall back to SPI)
- Log channel availability status

Precondition

mgr must be non-NULL and initialized
ready must be non-NULL
channel must be valid (USB or SPI)

Postcondition

*ready set to true or false based on channel state
On error, *ready set to false (safe default)

Note

USB readiness can change at runtime (host connects/disconnects)
SPI readiness is static (always ready if handle configured)
This function does NOT initiate communication - only queries state

Example - Send with Fallback:

```
uint8_t payload[16] = { ... };

// Try USB first
bool usb_ready = false;
rx_err_t err = rx_comm_manager_channel_ready(&g_comm_mgr,
                                             k_comm_channel_usb,
                                             &usb_ready);
if (err == k_rx_ok && usb_ready) {
  err = rx_comm_manager_respond(&g_comm_mgr,
                                k_comm_channel_usb,
                                payload,
                                sizeof(payload));
} else {
  // Fallback to SPI
  bool spi_ready = false;
  err = rx_comm_manager_channel_ready(&g_comm_mgr,
                                      k_comm_channel_spi,
                                      &spi_ready);
  if (err == k_rx_ok && spi_ready) {
    err = rx_comm_manager_respond(&g_comm_mgr,
                                  k_comm_channel_spi,
                                  payload,
                                  sizeof(payload));
  }
}
```

Example - Log Channel Status:

```
void log_channel_status(void)
{
  bool usb_ready = false, spi_ready = false;

  rx_comm_manager_channel_ready(&g_comm_mgr, k_comm_channel_usb, &usb_ready);
  rx_comm_manager_channel_ready(&g_comm_mgr, k_comm_channel_spi, &spi_ready);

  printf("USB: %s, SPI: %s\n",
         usb_ready ? "READY" : "NOT READY",
         spi_ready ? "READY" : "NOT READY");
}
```

See also

rx_usb_is_configured() USB enumeration status

rx_comm_manager_send() Send frame on channel

Since

Version 1.0.0

Definition at line 1857 of file rx_comm_manager.c.

```
01858 {
01859   /* Rule 5: Pre-condition validation */
01860   if (mgr == nullptr || ready == nullptr) {
01861     if (ready != nullptr) {
01862       *ready = false;
01863     }
01864     return k_rx_err_invalid_arg;
01865   }
01866   if (!mgr->initialized) {
01867     *ready = false;
01868     return k_rx_err_invalid_state;
01869   }
01870
01871   switch (channel) {
01872     case k_comm_channel_uart:
01873       *ready = (bool)(mgr->uart_handle != nullptr);
01874       break;
01875
01876     case k_comm_channel_spi:
01877       *ready = (bool)(mgr->spi_handle != nullptr);
01878       break;
```

```
01879
01880     case k_comm_channel_i2c:
01881       *ready = (bool)(mgr->i2c_handle != nullptr);
01882       break;
01883
01884     default:
01885       *ready = false;
01886       rx_log_error(s_tag, "Channel ready: invalid channel");
01887       return k_rx_err_invalid_arg;
01888   }
01889
01890   return k_rx_ok;
01891 }
```

References rx_comm_manager_t::i2c_handle, rx_comm_manager_t::initialized, k_comm_channel_i2c, k_comm_channel_spi, k_comm_channel_uart, s_tag, rx_comm_manager_t::spi_handle, and rx_comm_manager_t::uart_handle.

4.3.3.16 rx_comm_manager_deinit()

```
rx_err_t rx_comm_manager_deinit (
                rx_comm_manager_t * mgr)   [nodiscard]
```

Deinitialize communication manager and release resources.

Deinitialize communication manager.

Marks communication manager as uninitialized, preventing further operations. This function does NOT deinitialize USB or SPI transport handles - caller must separately call rx_usb_comm_deinit() and rx_spi_comm_deinit() if needed.

Algorithm:

1. Validate mgr pointer (NULL check)
2. Validate initialized flag (must be initialized to deinitialize)
3. Clear initialized flag (marks handle as invalid)

Design Note: Minimal deinitialization - no dynamic resources to free (all memory is statically allocated). Function primarily serves as safety guard to prevent use-after-deinit.

Precondition

mgr must be non-NULL

mgr must be initialized (mgr->initialized == true)

Postcondition

mgr->initialized = false

Subsequent calls to rx_comm_manager_poll/send will return k_rx_err_invalid_state

Note

Does NOT free memory (handle is statically allocated)

Does NOT deinitialize transport handles (USB/SPI)

Thread-safe if no concurrent operations on same mgr handle

Example:

```
// Deinitialize manager
rx_err_t err = rx_comm_manager_deinit(&g_comm_mgr);
if (err == k_rx_ok) {
  // Manager no longer usable

  // Separately deinitialize transports if needed
  rx_usb_comm_deinit(&g_usb_handle);
  rx_spi_comm_deinit(&g_spi_handle);
}
```

See also

rx_comm_manager_init() Initialization

rx_spi_comm_deinit() SPI transport cleanup

Since

Version 1.0.0

Definition at line 1216 of file rx_comm_manager.c.

```
01217 {
01218   /* Rule 5: Pre-condition validation */
01219   if (mgr == nullptr) {
01220     rx_log_error(s_tag, "Deinit failed: NULL mgr");
01221     return k_rx_err_invalid_arg;
01222   }
01223   if (!mgr->initialized) {
01224     rx_log_warn(s_tag, "Deinit called on uninitialized mgr");
01225     return k_rx_err_invalid_state;
01226   }
01227
01228   mgr->initialized = false;
01229
01230   rx_log_info(s_tag, "Comm manager deinitialized");
01231   return k_rx_ok;
01232 }
```

References rx_comm_manager_t::initialized, and s_tag.

4.3.3.17 rx_comm_manager_event_send()

```
rx_err_t rx_comm_manager_event_send (
                rx_comm_manager_t * mgr,
                const rx_comm_send_params_t * params)   [nodiscard]
```

Queue data on the event path (reliable, SPI retry).

Event path is for commands and other data requiring reliable delivery. The payload is copied into a static FIFO queue and processed by the poll loop. Retry behavior depends on transport:

- SPI: Up to 3 retries with exponential backoff (10ms, 20ms, 40ms)
- USB: Fire-and-forget (USB HW reliability is sufficient)

Queue is processed in FIFO order by rx_comm_manager_poll().

Parameters

in,out	mgr	Pointer to initialized manager
in	params	Send parameters (channel, type, flags, payload, payload_len)

Returns

k_rx_ok on success (queued)

k_rx_err_invalid_arg if mgr or params is nullptr

k_rx_err_invalid_state if not initialized

k_rx_err_no_mem if event queue is full

Precondition

mgr must be initialized

params must be valid

params->payload_len <= k_frame_max_payload

Postcondition

Payload copied to queue, will be sent by next poll()

Note

Payload is COPIED into queue - caller can reuse buffer immediately

Warning

Queue depth is k_comm_event_queue_depth (8). If full, returns error.

See also

rx_comm_manager_stream_send() For fire-and-forget telemetry

Since

Version 1.0.0

Definition at line 1698 of file rx_comm_manager.c.

```
01699 {
01700   if (mgr == nullptr || params == nullptr) {
01701     rx_log_error(s_tag, "Event enqueue failed: NULL arg");
01702     return k_rx_err_invalid_arg;
01703   }
01704
01705   if (!mgr->initialized) {
01706     rx_log_error(s_tag, "Event enqueue failed: not initialized");
01707     return k_rx_err_invalid_state;
01708   }
01709
01710   if (params->payload_len > k_frame_max_payload) {
```

```
01711     rx_log_error_val(s_tag, "Event enqueue failed: payload too large", params->payload_len);
01712     return k_rx_err_invalid_arg;
01713   }
01714
01715   if (params->payload_len > 0 && params->payload == nullptr) {
01716     rx_log_error(s_tag, "Event enqueue failed: NULL payload with nonzero len");
01717     return k_rx_err_invalid_arg;
01718   }
01719
01720   /* Check queue capacity */
01721   if (mgr->event_queue_count >= k_comm_event_queue_depth) {
01722     rx_log_error_val(s_tag, "Event queue full, depth", (uint8_t)mgr->event_queue_count);
01723     return k_rx_err_no_mem;
01724   }
01725
01726   /* Enqueue: copy payload into static slot */
01727   rx_comm_event_entry_t* entry = &mgr->event_queue[mgr->event_queue_head];
01728   entry->channel     = params->channel;
01729   entry->type        = params->type;
01730   entry->flags       = params->flags;
01731   entry->payload_len = params->payload_len;
01732   entry->retries     = 0;
01733   entry->occupied    = true;
01734
01735   if (params->payload != nullptr && params->payload_len > 0) {
01736     for (uint32_t i = 0; i < params->payload_len; i++) {
01737       entry->payload[i] = params->payload[i];
01738     }
01739   }
01740
01741   mgr->event_queue_head = (mgr->event_queue_head + 1) % k_comm_event_queue_depth;
01742   mgr->event_queue_count++;
01743
01744   rx_log_debug_val(s_tag, "Event enqueued, queue depth", mgr->event_queue_count);
01745   return k_rx_ok;
01746 }
```

References rx_comm_event_entry_t::channel, rx_comm_send_params_t::channel, rx_comm_manager_t::event_queu
rx_comm_manager_t::event_queue_count, rx_comm_manager_t::event_queue_head, rx_comm_event_entry_t::flag
rx_comm_send_params_t::flags, rx_comm_manager_t::initialized, k_comm_event_queue_depth,
rx_comm_event_entry_t::occupied, rx_comm_event_entry_t::payload, rx_comm_send_params_t::payload,
rx_comm_event_entry_t::payload_len, rx_comm_send_params_t::payload_len, rx_comm_event_entry_t::retries,
s_tag, rx_comm_event_entry_t::type, and rx_comm_send_params_t::type.

4.3.3.18 rx_comm_manager_init()

```
rx_err_t rx_comm_manager_init (
                rx_comm_manager_t * mgr,
                const rx_comm_manager_config_t * cfg)   [nodiscard]
```

Initialize communication manager with channel configuration.

Initialize communication manager.

Performs complete initialization of communication manager handle, configuring enabled channels, callback registration, and ASCII debugging output. This function must be called before any other communication manager operations.

Algorithm:

1. Validate parameters: Check mgr pointer for nullptr (NASA Rule 5)
2. Clear handle: Zero-fill entire structure via memset (328 bytes)
3. Apply configuration: Copy config fields if cfg is non-NULL
   - usb_handle: Pointer to initialized USB comm handle (may be NULL)
   - spi_handle: Pointer to initialized SPI comm handle (may be NULL)

- callback: User frame handler function (may be NULL)
- callback_ctx: User context pointer (may be NULL)
- enable_decoded_output: ASCII debugging flag

4. Mark initialized: Set mgr->initialized = true
5. Post-condition validation: Verify ascii_buffer properly cleared

Configuration Options:

- cfg = NULL: All channels disabled, no callback, ASCII output off
- usb_handle = NULL: USB channel disabled (poll skips USB)
- spi_handle = NULL: SPI channel disabled (poll skips SPI)
- callback = NULL: No callback invoked (silent frame consumption)
- enable_decoded_output = true: ASCII debugging enabled (Port 1)

Precondition

mgr must point to allocated memory (uninitialized content OK)
USB/SPI handles (if provided) must already be initialized
Callback function (if provided) must be valid and reentrant

Postcondition

mgr->initialized = true on success
All mgr fields set to config values or zero
mgr->ascii_buffer is zero-filled (256 bytes)

Note

This function does NOT initialize USB or SPI transports - caller must initialize rx_usb_comm and rx_spi_comm separately before passing handles to this function
On success, at least one channel should be enabled (usb_handle or spi_handle non-NULL)
Thread-safe ONLY if each thread uses a different mgr handle

Performance:

- Execution time: ~5 us @ 240 MHz
- Dominated by memset (328 bytes)

Configuration Example - Both Channels Enabled:

```
static rx_comm_manager_t g_comm_mgr;
static rx_usb_comm_handle_t g_usb_handle;
static rx_spi_comm_handle_t g_spi_handle;

// Initialize transport layers first
rx_usb_comm_config_t usb_cfg = { .port = k_usb_port_proto };
rx_err_t err = rx_usb_comm_init(&g_usb_handle, &usb_cfg);
if (err != k_rx_ok) return err;

rx_spi_comm_config_t spi_cfg = { .spi_channel = 0 };
err = rx_spi_comm_init(&g_spi_handle, &spi_cfg);
if (err != k_rx_ok) return err;

// Initialize communication manager with both channels
rx_comm_manager_config_t cfg = {
  .usb_handle = &g_usb_handle,
  .spi_handle = &g_spi_handle,
  .callback = on_frame_received,
  .callback_ctx = nullptr,
  .enable_decoded_output = true,
};
err = rx_comm_manager_init(&g_comm_mgr, &cfg);
```

Configuration Example - USB Only, No Callback:

```
rx_comm_manager_config_t cfg = {
  .usb_handle = &g_usb_handle,
  .spi_handle = nullptr,              // SPI disabled
  .callback = nullptr,                // No callback
  .callback_ctx = nullptr,
  .enable_decoded_output = false,    // No ASCII debugging
};
rx_err_t err = rx_comm_manager_init(&g_comm_mgr, &cfg);
```

See also

rx_comm_manager_deinit() Cleanup and resource release

rx_comm_manager_poll() Poll for incoming frames

rx_spi_comm_init() Initialize SPI transport

rx_uart_comm_init() Initialize UART transport

rx_i2c_comm_init() Initialize I2C peripheral transport

Since

Version 1.0.0

Definition at line 1120 of file rx_comm_manager.c.

```
01121 {
01122   /* Rule 5: Pre-condition validation - nullptr check */
01123   if (mgr == nullptr) {
01124     rx_log_error(s_tag, "Init failed: NULL mgr");
01125     return k_rx_err_invalid_arg;
01126   }
01127
01128   /* Rule 5: Pre-condition validation - SPI link consistency checks */
01129   if (cfg != nullptr && cfg->spi_link != nullptr) {
01130     /* If spi_link is provided, spi_handle must also be provided */
01131     if (cfg->spi_handle == nullptr) {
01132       rx_log_error(s_tag, "Init failed: spi_link requires non-NULL spi_handle");
01133       return k_rx_err_invalid_arg;
01134     }
01135
01136     /* If spi_link is provided, it must be initialized */
01137     if (!cfg->spi_link->initialized) {
01138       rx_log_error(s_tag, "Init failed: spi_link not initialized (call rx_spi_link_init first)");
01139       return k_rx_err_invalid_arg;
01140     }
01141
```

```
01142     /* If spi_link is provided, its spi_handle must match config spi_handle */
01143     if (cfg->spi_link->spi_handle != cfg->spi_handle) {
01144       rx_log_error(s_tag, "Init failed: spi_link->spi_handle != cfg->spi_handle");
01145       return k_rx_err_invalid_arg;
01146     }
01147   }
01148
01149   *mgr = s_zero_mgr;
01150
01151   /* Apply configuration */
01152   if (cfg != nullptr) {
01153     mgr->uart_handle          = cfg->uart_handle;
01154     mgr->spi_handle           = cfg->spi_handle;
01155     mgr->i2c_handle           = cfg->i2c_handle;
01156     mgr->spi_link             = cfg->spi_link;
01157     mgr->callback             = cfg->callback;
01158     mgr->callback_ctx         = cfg->callback_ctx;
01159     mgr->enable_decoded_output = cfg->enable_decoded_output;
01160     mgr->link_status_cb       = cfg->link_status_cb;
01161     mgr->link_status_ctx      = cfg->link_status_ctx;
01162   }
01163
01164   mgr->initialized = true;
01165
01166   rx_log_info(s_tag, "Comm manager initialized");
01167   return k_rx_ok;
01168 }
```

References rx_comm_manager_config_t::callback, rx_comm_manager_t::callback, rx_comm_manager_config_t::callb
rx_comm_manager_t::callback_ctx, rx_comm_manager_config_t::enable_decoded_output, rx_comm_manager_t::e
rx_comm_manager_config_t::i2c_handle, rx_comm_manager_t::i2c_handle, rx_comm_manager_t::initialized,
rx_comm_manager_config_t::link_status_cb, rx_comm_manager_t::link_status_cb, rx_comm_manager_config_t::
rx_comm_manager_t::link_status_ctx, s_tag, s_zero_mgr, rx_comm_manager_config_t::spi_handle,
rx_comm_manager_t::spi_handle, rx_comm_manager_config_t::spi_link, rx_comm_manager_t::spi_link,
rx_comm_manager_config_t::uart_handle, and rx_comm_manager_t::uart_handle.

4.3.3.19 rx_comm_manager_link_status()

```
rx_err_t rx_comm_manager_link_status (
                const rx_comm_manager_t * mgr,
                rx_comm_channel_t channel,
                rx_comm_link_status_t * status)   [nodiscard]
```

Query link health status for a transport channel.

Returns the current heartbeat-derived link status for the given channel. Status is updated by rx_comm_manager_poll() based on the dual-detection heartbeat model (200ms implicit timeout).

Parameters

in	mgr	Pointer to initialized manager
in	channel	Channel to query
out	status	Receives current link status

Returns

k_rx_ok on success

k_rx_err_invalid_arg if mgr or status is nullptr

Since

Version 1.0.0

Definition at line 1753 of file rx_comm_manager.c.

```
01756 {
01757   if (mgr == nullptr || status == nullptr) {
01758     rx_log_error(s_tag, "Link status query: NULL arg");
01759     return k_rx_err_invalid_arg;
01760   }
01761
01762   if (!mgr->initialized) {
01763     rx_log_error(s_tag, "Link status query: uninitialized manager");
01764     return k_rx_err_invalid_state;
01765   }
01766
01767   if (channel >= k_comm_channel_count) {
01768     rx_log_error(s_tag, "Link status query: invalid channel");
01769     return k_rx_err_invalid_arg;
01770   }
01771
01772   *status = mgr->heartbeat[channel].status;
01773   return k_rx_ok;
01774 }
```

References rx_comm_manager_t::heartbeat, rx_comm_manager_t::initialized, k_comm_channel_count, s_tag, and rx_comm_heartbeat_state_t::status.

4.3.3.20 rx_comm_manager_poll()

```
rx_err_t rx_comm_manager_poll (
            rx_comm_manager_t * mgr)   [nodiscard]
```

Poll all enabled channels for incoming frames.

Non-blocking check of USB and SPI channels. For each complete frame received:

1. If decoded output enabled, format and send ASCII to Port 1
2. Invoke user callback with frame data and channel

Call this from main loop or dedicated communication task.

Parameters

in,out	mgr	Pointer to initialized manager

Returns

k_rx_ok if at least one frame was received

k_rx_err_timeout if no frames available on any channel

k_rx_err_invalid_arg if mgr is nullptr

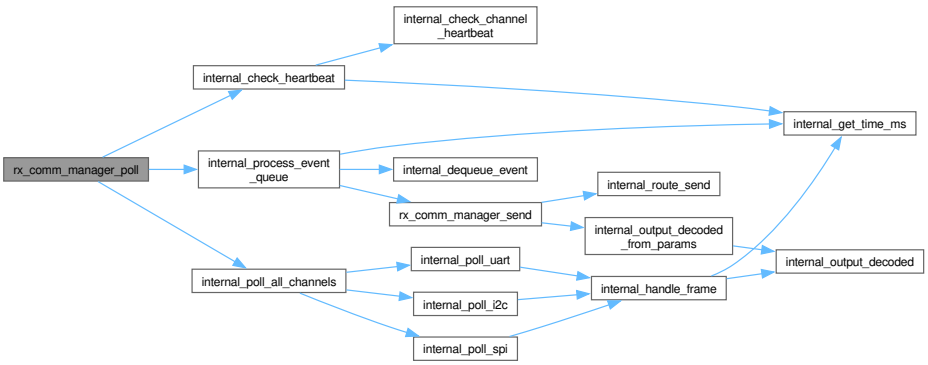
k_rx_err_invalid_state if not initialized

Definition at line 1383 of file rx_comm_manager.c.

```
01384 {
01385   /* Rule 5: Pre-condition validation */
01386   if (mgr == nullptr) {
01387     return k_rx_err_invalid_arg;
01388   }
01389   if (!mgr->initialized) {
01390     return k_rx_err_invalid_state;
01391   }
01392
01393   bool     received = false;
01394   rx_err_t poll_err = internal_poll_all_channels(mgr, &received);
01395   if (poll_err != k_rx_ok) {
01396     return poll_err;
01397   }
01398
01399   /* Process event queue: send one queued event per poll cycle */
01400   internal_process_event_queue(mgr);
01401
01402   /* Heartbeat: check implicit timeout on each transport */
01403   internal_check_heartbeat(mgr);
01404
01405   return (int)received ? k_rx_ok : k_rx_err_timeout;
01406 }
```

References rx_comm_manager_t::initialized, internal_check_heartbeat(), internal_poll_all_channels(), and internal_process_event_queue().

Here is the call graph for this function:



4.3.3.21 rx_comm_manager_process_retransmits()

```
rx_err_t rx_comm_manager_process_retransmits (
                rx_comm_manager_t * mgr,
                uint32_t current_time_ms)   [nodiscard]
```

Process pending retransmissions on SPI channel.

Thin pass-through to rx_spi_comm_process_retransmits() on the SPI handle. Call this from the communication task polling loop.

Parameters

in,out	mgr	Communication manager
in	current_time_ms	Current system time in milliseconds

Returns

rx_err_t Error code from rx_spi_comm_process_retransmits()

Return values

k_rx_ok	No action needed or retransmit sent
k_rx_err_invalid_arg	mgr is nullptr
k_rx_err_invalid_state	mgr not initialized
k_rx_err_retry_limit	Max retries exceeded

Since

Version 1.0.0

Definition at line 1965 of file rx_comm_manager.c.

```
01966 {
01967   if (mgr == nullptr) {
01968     return k_rx_err_invalid_arg;
01969   }
01970
01971   if (!mgr->initialized) {
01972     return k_rx_err_invalid_state;
01973   }
01974
01975   if (mgr->spi_handle == nullptr) {
01976     return k_rx_ok; /* No SPI channel configured */
01977   }
01978
01979   return rx_spi_comm_process_retransmits(mgr->spi_handle, current_time_ms);
01980 }
```

References rx_comm_manager_t::initialized, and rx_comm_manager_t::spi_handle.

4.3.3.22 rx_comm_manager_respond()

```
rx_err_t rx_comm_manager_respond (
                rx_comm_manager_t * mgr,
                rx_comm_channel_t channel,
                const uint8_t * payload,
                uint32_t payload_len)   [nodiscard]
```

Send response frame on specified channel (convenience wrapper).

Send response on same channel as received frame.

Convenience function for sending response frames. Automatically sets frame type to k_frame_type_response and flags to k_frame_flag_none. Simplifies common use case of responding to commands/requests.

Algorithm:

1. Build rx_comm_send_params_t with:
   - channel: User-specified
   - type: k_frame_type_response (fixed)
   - flags: k_frame_flag_none (fixed)
   - payload: User-specified

- payload_len: User-specified

2. Call rx_comm_manager_send() with constructed params
3. Return result from rx_comm_manager_send()

When to Use:

- Responding to commands with telemetry data
- Sending acknowledgments
- Replying to requests

When NOT to Use:

- Need custom frame type (use rx_comm_manager_send instead)
- Need custom flags (use rx_comm_manager_send instead)

Precondition

Same preconditions as rx_comm_manager_send()

Postcondition

Response frame transmitted with type=response, flags=none

Note

Equivalent to calling rx_comm_manager_send() with type=k_frame_type_response
This is a thin wrapper - no additional validation performed

Example - Send Telemetry Response:

```
// Received a telemetry request on USB, send response
static void on_telemetry_request(rx_comm_channel_t channel, const rx_frame_t* frame)
{
  uint8_t telemetry[16];

  // Gather telemetry data
  telemetry[0] = get_temperature_celsius();
  telemetry[1] = get_current_ma();
  // ... fill remaining fields ...

  // Send response on same channel request came from
  rx_err_t err = rx_comm_manager_respond(&g_comm_mgr,
                                         channel,
                                         telemetry,
                                         sizeof(telemetry));
  if (err != k_rx_ok) {
    // Handle send error
  }
}
```

Example - Send Empty Acknowledgment:

```
// Send empty ACK response
rx_err_t err = rx_comm_manager_respond(&g_comm_mgr,
                                       k_comm_channel_usb,
                                       NULL,    // No payload
                                       0);      // Zero length
```

See also

rx_comm_manager_send() Full send function with all parameters

rx_comm_send_params_t Send parameters structure

Since

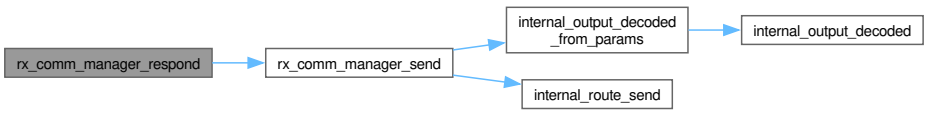
Version 1.0.0

Definition at line 1670 of file rx_comm_manager.c.

```
01674 {
01675   const rx_comm_send_params_t params = {
01676     .channel     = channel,
01677     .type        = k_frame_type_response,
01678     .flags       = k_frame_flag_none,
01679     .payload     = payload,
01680     .payload_len = payload_len,
01681   };
01682   return rx_comm_manager_send(mgr, &params);
01683 }
```

References rx_comm_manager_send().

Here is the call graph for this function:



4.3.3.23 rx_comm_manager_send()

```
rx_err_t rx_comm_manager_send (
                rx_comm_manager_t * mgr,
                const rx_comm_send_params_t * params)   [nodiscard]
```

Send frame on specific channel.

Encodes and sends a frame on the specified channel. If decoded output is enabled, also sends ASCII representation to Port 1.

Parameters

in,out	mgr	Pointer to initialized manager
in	params	Send parameters (channel, type, flags, payload, payload_len)

Returns

k_rx_ok on success

k_rx_err_invalid_arg if mgr or payload is nullptr

k_rx_err_invalid_state if not initialized or channel not enabled

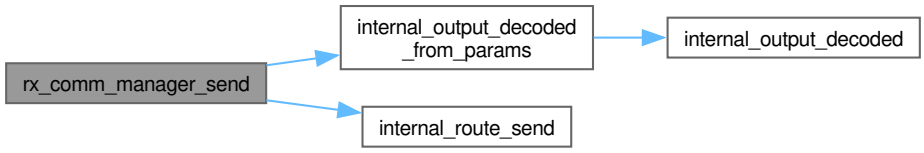
Definition at line 1560 of file rx_comm_manager.c.

```
01561 {
01562   /* Rule 5: Pre-condition validation */
01563   if (mgr == nullptr || params == nullptr) {
01564     rx_log_error(s_tag, "Send failed: NULL arg");
01565     return k_rx_err_invalid_arg;
01566   }
01567   if (!mgr->initialized) {
01568     rx_log_error(s_tag, "Send failed: not initialized");
01569     return k_rx_err_invalid_state;
01570   }
01571   if (params->payload == nullptr && params->payload_len > 0) {
01572     rx_log_error(s_tag, "Send failed: NULL payload with nonzero len");
01573     return k_rx_err_invalid_arg;
01574   }
01575   /* Validate payload length is within frame limits */
01576   if (params->payload_len > k_frame_max_payload) {
01577     rx_log_error_val(s_tag, "Send failed: payload too large", params->payload_len);
01578     return k_rx_err_invalid_arg;
01579   }
01580
01581   rx_err_t err = internal_route_send(mgr, params);
01582
01583   if (err != k_rx_ok) {
01584     rx_log_error_val(s_tag, "Transport send failed", err);
01585   }
01586
01587   /* Output decoded ASCII on success */
01588   if (err == k_rx_ok && (int)mgr->enable_decoded_output) {
01589     internal_output_decoded_from_params(mgr, params);
01590   }
01591
01592   return err;
01593 }
```

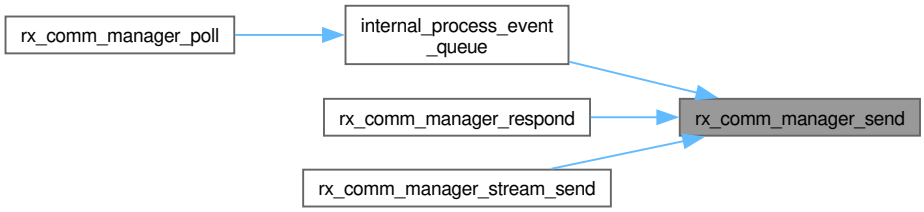
References rx_comm_manager_t::enable_decoded_output, rx_comm_manager_t::initialized, internal_output_decode
internal_route_send(), rx_comm_send_params_t::payload, rx_comm_send_params_t::payload_len,
and s_tag.

Referenced by internal_process_event_queue(), rx_comm_manager_respond(), and rx_comm_manager_stream_send(

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.24 rx_comm_manager_set_auto_retransmit()

```
rx_err_t rx_comm_manager_set_auto_retransmit (
                rx_comm_manager_t * mgr,
                bool enabled,
                const rx_spi_comm_retransmit_config_t * config)   [nodiscard]
```

Enable or disable automatic retransmission on SPI channel.

Thin pass-through to rx_spi_comm_set_auto_retransmit() on the SPI handle.

Parameters

in,out	mgr	Communication manager
in	enabled	true to enable, false to disable
in	config	Retransmit config (NULL to keep current/defaults)

Returns

rx_err_t Error code from rx_spi_comm_set_auto_retransmit()

Since

Version 1.0.0

Definition at line 1982 of file rx_comm_manager.c.

```
01985 {
01986   if (mgr == nullptr) {
01987     return k_rx_err_invalid_arg;
01988   }
01989
01990   if (!mgr->initialized) {
01991     return k_rx_err_invalid_state;
01992   }
01993
01994   if (mgr->spi_handle == nullptr) {
01995     return k_rx_err_not_supported;
01996   }
01997
01998   return rx_spi_comm_set_auto_retransmit(mgr->spi_handle, enabled, config);
01999 }
```

References rx_comm_manager_t::initialized, and rx_comm_manager_t::spi_handle.

4.3.3.25 rx_comm_manager_stream_send()

```
rx_err_t rx_comm_manager_stream_send (
                rx_comm_manager_t * mgr,
                const rx_comm_send_params_t * params)   [nodiscard]
```

Send data on the stream path (fire-and-forget).

Stream path is for time-sensitive data like telemetry where stale data has no value. Sends immediately with NO retries on any transport. If the send fails, the data is simply dropped.

- USB CDC: Direct send, no retries (USB HW provides reliability)

- SPI: Direct send, no retries (stale telemetry is worthless)

Parameters

in,out	mgr	Pointer to initialized manager
in	params	Send parameters (channel, type, flags, payload, payload_len)

Returns

k_rx_ok on success

k_rx_err_invalid_arg if mgr or params is nullptr

k_rx_err_invalid_state if not initialized or channel not enabled

Precondition

mgr must be initialized

params must be valid

Postcondition

Data sent or silently dropped on failure

Note

This is a thin wrapper over rx_comm_manager_send() for semantic clarity

See also

rx_comm_manager_event_send() For reliable command delivery

Since

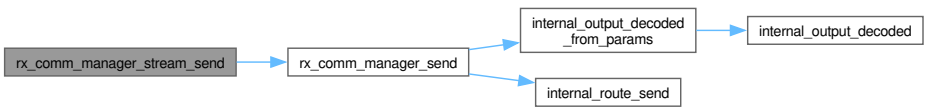
Version 1.0.0

Definition at line 1690 of file rx_comm_manager.c.

```
01691 {
01692   /* Stream path: fire-and-forget, no retries, no queuing.
01693    * Telemetry and other time-sensitive data goes here.
01694    * Stale data has no value, so failure is silently accepted. */
01695   return rx_comm_manager_send(mgr, params);
01696 }
```

References rx_comm_manager_send().

Here is the call graph for this function:



4.3.4 Variable Documentation

4.3.4.1 s_tag

const char s_tag[] = "COMM_MGR" [static]

Definition at line 410 of file rx_comm_manager.c.

Referenced by internal_check_channel_heartbeat(), internal_handle_frame(), internal_poll_all_channels(), internal_poll_uart(), internal_process_event_queue(), internal_route_send(), rx_comm_manager_channel_ready(), rx_comm_manager_deinit(), rx_comm_manager_event_send(), rx_comm_manager_init(), rx_comm_manager_link and rx_comm_manager_send().

4.3.4.2 s_zero_mgr

const rx_comm_manager_t s_zero_mgr = {} [static]

Zero-initialized comm manager template for stack-safe initialization.

Static const instance used to clear rx_comm_manager_t without creating a large compound literal on the stack. Lives in .rodata section.

Note

Read-only; never modified after static initialization

Warning

Do not remove - prevents stack overflow in init function

See also

rx_comm_manager_init() Uses this template to zero-initialize the manager handle

Since

Version 1.0.0

Definition at line 1026 of file rx_comm_manager.c.

```
01026 {};
```

Referenced by rx_comm_manager_init().

4.4 rx_comm_manager.c

Go to the documentation of this file.

```
00001 /**
00002  * @file rx_comm_manager.c
00003  * @brief Unified Multi-Channel Communication Manager Implementation
00004  *
00005  * @details
00006  * ## Overview
00007  * Production-quality implementation of unified communication manager coordinating
00008  * multiple simultaneous channels (USB CDC and SPI) with automatic frame protocol
00009  * handling, non-blocking polling architecture, optional decoded ASCII debugging
00010  * output, and callback-based event notification.
00011  *
00012  * **Purpose**: Provide centralized, efficient coordination of independent communication
00013  * channels between RX72N MCU and Raspberry Pi 5 host, abstracting low-level transport
00014  * details while maintaining high performance and reliability.
00015  *
00016  * ## Implementation Architecture
00017  *
00018  * @dot
00019  * digraph comm_manager_impl {
00020  *   rankdir=TB;
00021  *   node [shape=box, style=rounded];
00022  *
00023  *   subgraph cluster_public {
00024  *     label="Public API Layer";
00025  *     style=filled;
00026  *     fillcolor=lightblue;
00027  *     init [label="rx_comm_manager_init()"];
00028  *     poll [label="rx_comm_manager_poll()"];
00029  *     send [label="rx_comm_manager_send()"];
00030  *     respond [label="rx_comm_manager_respond()"];
00031  *     ready [label="rx_comm_manager_channel_ready()"];
00032  *   }
00033  *
00034  *   subgraph cluster_private {
00035  *     label="Private Helper Layer";
00036  *     style=filled;
00037  *     fillcolor=lightgreen;
00038  *     poll_usb [label="internal_poll_usb()"];
00039  *     poll_spi [label="internal_poll_spi()"];
00040  *     handle_frame [label="internal_handle_frame()"];
00041  *     output_decoded [label="internal_output_decoded()"];
00042  *   }
00043  *
00044  *   subgraph cluster_transport {
00045  *     label="Transport Layer";
00046  *     style=filled;
00047  *     fillcolor=lightyellow;
00048  *     usb_comm [label="rx_usb_comm\n(USB CDC Protocol)"];
00049  *     spi_comm [label="rx_spi_comm\n(SPI Protocol)"];
00050  *     usb_hw [label="rx_usb\n(USB Port 1 Debug)"];
00051  *     frame_ascii [label="rx_frame_ascii\n(ASCII Formatting)"];
00052  *   }
00053  *
00054  *   // Public API to Private Helpers
00055  *   poll -> poll_usb;
00056  *   poll -> poll_spi;
00057  *   send -> output_decoded;
00058  *
00059  *   // Private Helpers to Transport
00060  *   poll_usb -> usb_comm [label="rx_usb_comm_receive()"];
00061  *   poll_spi -> spi_comm [label="rx_spi_comm_receive()"];
00062  *   poll_usb -> handle_frame [label="on frame"];
00063  *   poll_spi -> handle_frame [label="on frame"];
00064  *   handle_frame -> output_decoded;
00065  *   output_decoded -> frame_ascii [label="rx_frame_ascii_format()"];
00066  *   output_decoded -> usb_hw [label="rx_usb_puts()"];
00067  *   send -> usb_comm [label="rx_usb_comm_send()"];
00068  *   send -> spi_comm [label="rx_spi_comm_send()"];
00069  *
00070  *   // User callback
00071  *   handle_frame -> callback [label="user callback", style=dashed];
00072  *   callback [label="Application\nFrame Handler", shape=ellipse, fillcolor=pink, style=filled];
00073  * }
00074  * @enddot
00075  *
00076  * ## Polling State Machine
00077  *
00078  * The rx_comm_manager_poll() function implements a non-blocking state machine
00079  * that checks all enabled channels sequentially:
00080  *
00081  * @startuml
00082  * [*] --> Idle
```

```
00083  * Idle --> PollUSB : rx_comm_manager_poll() called
00084  *
00085  * state PollUSB {
00086  *   [*] --> CheckUSBHandle
00087  *   CheckUSBHandle --> USBReceive : handle != nullptr
00088  *   CheckUSBHandle --> SkipUSB : handle == nullptr
00089  *   USBReceive --> HandleUSBFrame : frame received
00090  *   USBReceive --> USBTimeout : no data
00091  *   HandleUSBFrame --> [*] : mark received=true
00092  *   USBTimeout --> [*] : continue
00093  *   SkipUSB --> [*] : continue
00094  * }
00095  *
00096  * PollUSB --> PollSPI : USB polling complete
00097  *
00098  * state PollSPI {
00099  *   [*] --> CheckSPIHandle
00100  *   CheckSPIHandle --> SPIReceive : handle != nullptr
00101  *   CheckSPIHandle --> SkipSPI : handle == nullptr
00102  *   SPIReceive --> HandleSPIFrame : frame received
00103  *   SPIReceive --> SPITimeout : no data
00104  *   HandleSPIFrame --> [*] : mark received=true
00105  *   SPITimeout --> [*] : continue
00106  *   SkipSPI --> [*] : continue
00107  * }
00108  *
00109  * PollSPI --> ReturnResult : both channels polled
00110  * ReturnResult --> Idle : return k_rx_ok if received\nk_rx_err_timeout if none
00111  * @enduml
00112  *
00113  * ## Performance Characteristics
00114  *
00115  * | Operation | Avg Time | Worst Case | Notes |
00116  * |-----------|----------|------------|-------|
00117  * | **rx_comm_manager_init()** | ~5 us | ~10 us | memset dominates (328 bytes) |
00118  * | **rx_comm_manager_poll()** | ~15-200 us | ~500 us | Depends on channels enabled, frames available |
00119  * | **rx_comm_manager_send()** | ~30-150 us | ~300 us | Depends on payload size, channel |
00120  * | **internal_poll_usb()** | ~10-100 us | ~250 us | USB CDC bulk transfer latency |
00121  * | **internal_poll_spi()** | ~5-50 us | ~150 us | SPI hardware transfer latency |
00122  * | **internal_handle_frame()** | ~2 us | ~5 us | Callback invocation overhead |
00123  * | **internal_output_decoded()** | ~50-150 us | ~300 us | ASCII formatting + USB puts |
00124  *
00125  * **Timing Analysis**:
00126  * - Measured @ 240 MHz CPU clock with -O2 optimization
00127  * - Poll time dominated by transport layer (USB/SPI receive)
00128  * - ASCII debugging adds ~50-150 us per frame (disable in production)
00129  * - Callback execution time not included (application-dependent)
00130  *
00131  * ## Memory Usage
00132  *
00133  * | Memory Type | Size | Usage |
00134  * |-------------|------|-------|
00135  * | **rx_comm_manager_t handle** | 328 bytes | Main state structure |
00136  * | **Stack (init)** | ~32 bytes | Function locals |
00137  * | **Stack (poll)** | ~80 bytes | rx_frame_t + locals |
00138  * | **Stack (send)** | ~80 bytes | rx_frame_t + locals |
00139  * | **ASCII buffer** | 256 bytes | Inside handle (for debugging) |
00140  *
00141  * **Memory Layout** (rx_comm_manager_t):
00142  * ```
00143  * Offset | Size | Field               | Purpose
00144  * -------|------|---------------------|--------------------------
00145  * 0x00   | 8    | usb_handle          | Pointer to USB comm handle
00146  * 0x08   | 8    | spi_handle          | Pointer to SPI comm handle
00147  * 0x10   | 8    | callback            | User frame callback function
00148  * 0x18   | 8    | callback_ctx        | User context pointer
00149  * 0x20   | 1    | enable_decoded_output | ASCII debug flag
00150  * 0x21   | 1    | initialized         | Initialization flag
00151  * 0x22   | 2    | (padding)           | Alignment
00152  * 0x24   | 4    | (padding)           | Alignment
00153  * 0x28   | 256  | ascii_buffer        | ASCII formatting buffer
00154  * -------|------|---------------------|--------------------------
00155  * Total: 328 bytes (0x148)
00156  * ```
00157  *
00158  * ## Algorithm Details
00159  *
00160  * ### Polling Algorithm (rx_comm_manager_poll)
00161  *
00162  * The polling algorithm maximizes throughput by checking all channels without
00163  * blocking, returning immediately on the first error while continuing to poll
00164  * remaining channels on timeout (no data available):
00165  *
00166  * **Steps:**
00167  * 1. **Parameter validation** (NULL check, initialized flag)
00168  * 2. **Poll USB channel**:
00169  *    - If nullptr handle -> skip (k_rx_err_timeout)
```

```
00170 *    - If data available -> handle frame, mark received=true
00171 *    - If timeout -> continue to next channel
00172 *    - If error -> return error immediately (critical failure)
00173 * 3. **Poll SPI channel**:
00174 *    - Same logic as USB
00175 * 4. **Return aggregated result**:
00176 *    - k_rx_ok if ANY frame received on any channel
00177 *    - k_rx_err_timeout if NO frames received (normal, non-error condition)
00178 *    - Other error codes propagated immediately
00179 *
00180 * **Design Rationale**:
00181 * - Non-blocking: Never blocks waiting for data
00182 * - Fair scheduling: Both channels polled each iteration
00183 * - Error prioritization: Critical errors returned immediately
00184 * - Timeout tolerance: Timeout is expected, not an error
00185 *
00186 * ### Send Algorithm (rx_comm_manager_send)
00187 *
00188 * Routing algorithm with post-send ASCII debugging support:
00189 *
00190 * **Steps:**
00191 * 1. **Validate parameters** (NULL checks, payload length <= k_frame_max_payload)
00192 * 2. **Route to channel**:
00193 *    - USB: Call rx_usb_comm_send()
00194 *    - SPI: Call rx_spi_comm_send()
00195 *    - Invalid channel: Return k_rx_err_invalid_arg
00196 * 3. **On success, if ASCII output enabled**:
00197 *    - Build temporary rx_frame_t from send parameters
00198 *    - Format as ASCII via rx_frame_ascii_format()
00199 *    - Send to USB Port 1 via rx_usb_puts()
00200 *
00201 * **Performance Note**: ASCII formatting adds ~50-150 us overhead per send.
00202 * Disable in production with `cfg->enable_decoded_output = false`.
00203 *
00204 * ## Thread Safety Analysis
00205 *
00206 * **Conditional Thread Safety** - Safe if following rules are observed:
00207 *
00208 * | Scenario | Thread Safety | Mitigation Required |
00209 * |----------|---------------|---------------------|
00210 * | **Single-threaded** | [OK] Safe | None |
00211 * | **Multi-threaded (different channels)** | [OK] Safe | Ensure each thread uses different channel |
00212 * | **Multi-threaded (same channel)** | [X] Unsafe | External mutex required |
00213 * | **Multi-threaded (different handles)** | [OK] Safe | None (independent handles) |
00214 * | **Init/Deinit from ISR** | [X] Unsafe | Call only from task context |
00215 * | **Poll/Send from ISR** | [WARN] Conditional | OK if callback is ISR-safe |
00216 *
00217 * **Recommended Architecture**:
00218 * - Dedicate one thread to poll all channels (e.g., ThreadX communication task)
00219 * - Send from any thread/ISR, but use external mutex if same channel
00220 * - Ensure user callback is reentrant if called from ISR
00221 *
00222 * **NASA Power of 10 Rule 5**: All public functions validate preconditions with
00223 * explicit checks (NULL pointers, initialized flag, parameter ranges).
00224 *
00225 * ## Integration Example
00226 *
00227 * @par Complete System Setup with ThreadX:
00228 * @code{.c}
00229 * // Global communication manager handle
00230 * static rx_comm_manager_t g_comm_mgr;
00231 * static rx_usb_comm_handle_t g_usb_handle;
00232 * static rx_spi_comm_handle_t g_spi_handle;
00233 *
00234 * // User callback for frame processing
00235 * static void on_frame_received(rx_comm_channel_t channel,
00236 *                               const rx_frame_t* frame,
00237 *                               void* ctx)
00238 * {
00239 *   (void)ctx;
00240 *
00241 *   // Dispatch based on frame type
00242 *   switch (frame->header.type) {
00243 *     case k_frame_type_command:
00244 *       handle_command(channel, frame);
00245 *       break;
00246 *
00247 *     case k_frame_type_request:
00248 *       handle_request(channel, frame);
00249 *       break;
00250 *
00251 *     default:
00252 *       // Unknown frame type, ignore
00253 *       break;
00254 *   }
00255 * }
00256 *
```

```
00257  * // ThreadX communication task
00258  * static void comm_task_entry(ULONG thread_input)
00259  * {
00260  *   (void)thread_input;
00261  *
00262  *   // Initialize USB CDC communication
00263  *   rx_usb_comm_config_t usb_cfg = {
00264  *     .port = k_usb_port_proto,
00265  *   };
00266  *   rx_err_t err = rx_usb_comm_init(&g_usb_handle, &usb_cfg);
00267  *   if (err != k_rx_ok) {
00268  *     // Handle error
00269  *     return;
00270  *   }
00271  *
00272  *   // Initialize SPI communication
00273  *   rx_spi_comm_config_t spi_cfg = {
00274  *     .spi_channel = 0,
00275  *   };
00276  *   err = rx_spi_comm_init(&g_spi_handle, &spi_cfg);
00277  *   if (err != k_rx_ok) {
00278  *     // Handle error
00279  *     return;
00280  *   }
00281  *
00282  *   // Initialize communication manager
00283  *   rx_comm_manager_config_t cfg = {
00284  *     .usb_handle = &g_usb_handle,
00285  *     .spi_handle = &g_spi_handle,
00286  *     .callback = on_frame_received,
00287  *     .callback_ctx = nullptr,
00288  *     .enable_decoded_output = true,  // Enable for development
00289  *   };
00290  *   err = rx_comm_manager_init(&g_comm_mgr, &cfg);
00291  *   if (err != k_rx_ok) {
00292  *     // Handle error
00293  *     return;
00294  *   }
00295  *
00296  *   // Main communication loop (100 Hz polling)
00297  *   while (1) {
00298  *     // Poll all channels
00299  *     err = rx_comm_manager_poll(&g_comm_mgr);
00300  *
00301  *     // Check for critical errors (not timeout)
00302  *     if (err != k_rx_ok && err != k_rx_err_timeout) {
00303  *       // Log error, attempt recovery
00304  *       tx_thread_sleep(100);  // 1 second backoff
00305  *     }
00306  *
00307  *     // Sleep 10 ticks (100 ms @ 100 Hz tick rate)
00308  *     tx_thread_sleep(10);
00309  *   }
00310  * }
00311  *
00312  * // Send response from application
00313  * void send_motor_telemetry(void)
00314  * {
00315  *   uint8_t telemetry_payload[16];
00316  *   // ... fill telemetry data ...
00317  *
00318  *   // Check if USB channel is ready
00319  *   bool usb_ready = false;
00320  *   rx_err_t err = rx_comm_manager_channel_ready(&g_comm_mgr,
00321  *                                                k_comm_channel_usb,
00322  *                                                &usb_ready);
00323  *   if (err == k_rx_ok && usb_ready) {
00324  *     // Send telemetry response
00325  *     err = rx_comm_manager_respond(&g_comm_mgr,
00326  *                                   k_comm_channel_usb,
00327  *                                   telemetry_payload,
00328  *                                   sizeof(telemetry_payload));
00329  *   }
00330  * }
00331  * @endcode
00332  *
00333  * ## NASA Power of 10 Compliance
00334  *
00335  * | Rule | Compliance | Implementation Notes |
00336  * |------|------------|----------------------|
00337  * | **Rule 1: Control Flow** | [OK] | No goto, setjmp, longjmp, or recursion |
00338  * | **Rule 2: Loop Bounds** | [OK] | All loops have statically provable bounds |
00339  * | **Rule 3: No Dynamic Allocation** | [OK] | Zero malloc/free - all static allocation |
00340  * | **Rule 4: Function Size** | [OK] | All functions < 60 lines |
00341  * | **Rule 5: Assertions** | [OK] | Minimum 2 checks per function (nullptr, initialized) |
00342  * | **Rule 6: Data Scope** | [OK] | Variables declared at smallest scope |
00343  * | **Rule 7: Check Returns** | [OK] | All return values validated or cast to (void) |
```

```
00344  * | **Rule 8: Preprocessor** | [OK] | Minimal macros, typed enums for constants |
00345  * | **Rule 9: Pointers** | [WARN] | Function pointers used for callback (DIP) |
00346  * | **Rule 10: Compiler Warnings** | [OK] | -Wall -Wextra -Werror enabled |
00347  *
00348  * **Rule 9 Deviation**: Function pointer used for user callback to enable Dependency
00349  * Inversion Principle (SOLID) - allows application to register frame handler without
00350  * tight coupling. This is intentional and documented per project policy.
00351  *
00352  * ## SOLID Principles
00353  *
00354  * | Principle | Implementation |
00355  * |-----------|----------------|
00356  * | **Single Responsibility** | Module responsible ONLY for channel coordination; frame parsing/transport delegated |
00357  * | **Open/Closed** | Extensible via callback; new frame types handled by application |
00358  * | **Liskov Substitution** | Channels are interchangeable (same rx_frame_t interface) |
00359  * | **Interface Segregation** | Small focused API (7 functions); send/poll/ready separated |
00360  * | **Dependency Inversion** | Depends on abstract rx_frame_t and channel handles, not concrete implementations |
00361  *
00362  * **Key Design Decision**: The manager does NOT parse frame payloads - it only routes
00363  * frames to the user callback. This separation allows application-specific command
00364  * handling without modifying the communication layer.
00365  *
00366  * ## Module Dependencies
00367  *
00368  * **Includes:**
00369  * - rx_comm_manager.h - Public API definitions
00370  * - rx_frame_ascii.h - ASCII formatting for debugging
00371  * - rx_usb.h - USB port access (Port 1 debug output)
00372  * - string.h - Standard C library (memset, memcpy)
00373  *
00374  * **Links Against:**
00375  * - rx_usb_comm - USB CDC frame protocol implementation
00376  * - rx_spi_comm - SPI frame protocol implementation
00377  * - rx_frame_ascii - ASCII frame formatting
00378  * - rx_usb - USB hardware abstraction
00379  *
00380  * @date 2026-01-27
00381  * @copyright Copyright (c) 2026 Locked Inc.
00382  * SPDX-License-Identifier: MIT
00383  * @since Version 1.0.0
00384  *
00385  * @see rx_comm_manager.h Full API documentation
00386  * @see rx_spi_comm.h SPI protocol implementation
00387  * @see rx_uart_comm.h UART protocol implementation
00388  * @see rx_i2c_comm.h I2C peripheral protocol implementation
00389  * @see rx_frame_ascii.h ASCII debugging interface
00390  */
00391
00392 #include "rx_comm_manager.h"
00393
00394 #include "rx_check.h"
00395 #include "rx_frame_ascii.h"
00396 #include "rx_log.h"
00397 #include "rx_log_uart.h"
00398 #include "rx_simulator_config.h"
00399 #include "rx_time_constants.h"
00400
00401 #if !RX_IS_SIMULATOR
00402 #include "tx_api.h"
00403 #endif
00404
00405 /*
      ============================================================================
00406  * Logging
00407  *
      ============================================================================
00408  */
00409
00410 static const char s_tag[] = "COMM_MGR";
00411
00412 /*
      ============================================================================
00413  * Private Helpers
00414  *
      ============================================================================
00415  */
00416
00417 /**
00418  * @brief Get current time in milliseconds
00419  *
00420  * @details
00421  * Returns the current system time in milliseconds. On RX72N hardware, this
00422  * uses ThreadX tx_time_get() multiplied by tick period. In simulator builds,
00423  * returns 0 (timing not available).
00424  *
00425  *
00426  * @since Version 1.0.0
```

```
00427  */
00428 /** @brief Simulator time stub value (time is unavailable in simulator) */
00429 typedef enum : uint32_t {
00430   k_sim_time_ms_zero = 0, /**< Simulator returns zero (no real-time clock) */
00431 } rx_comm_sim_time_t;
00432
00433 static uint32_t internal_get_time_ms(void)
00434 {
00435 #if !RX_IS_SIMULATOR
00436   return (uint32_t)tx_time_get() * k_threadx_ms_per_tick;
00437 #else
00438   return k_sim_time_ms_zero;
00439 #endif
00440 }
00441
00442 /*
      ============================================================================
00443  * Private Constants
00444  *
      ============================================================================
00445  */
00446
00447 /**
00448  * @enum internal_constants_t
00449  * @brief Internal constants for communication manager implementation
00450  *
00451  * @details
00452  * Defines compile-time constants used internally by the communication manager
00453  * for timeout values, placeholder values, and buffer indexing.
00454  *
00455  * **Design Notes:**
00456  * - Non-blocking timeout (0 ms) ensures poll never blocks
00457  * - Placeholder values filled by transport layer (sequence, CRC)
00458  * - Explicit array index constant for bounds checking
00459  *
00460  * @since Version 1.0.0
00461  */
00462 typedef enum : uint32_t {
00463   /**
00464    * @brief Non-blocking receive timeout for polling
00465    * @details
00466    * Set to 0 ms to ensure rx_comm_manager_poll() never blocks waiting for data.
00467    * Transport layers (USB/SPI) return k_rx_err_timeout immediately if no data.
00468    * @par Value: 0 milliseconds
00469    */
00470   k_receive_timeout_ms = 0,
00471
00472   /**
00473    * @brief Placeholder for frame sequence number (filled by transport layer)
00474    * @details
00475    * Sequence number is managed by transport layer (rx_usb_comm/rx_spi_comm)
00476    * to detect dropped frames. Manager does not manipulate sequence numbers.
00477    * @par Value: 0
00478    */
00479   k_frame_seq_placeholder = 0,
00480
00481   /**
00482    * @brief Placeholder for frame CRC-32 (filled by transport layer)
00483    * @details
00484    * CRC-32 calculated and verified by transport layer. Manager builds temp
00485    * frame with placeholder CRC for ASCII formatting only.
00486    * @par Value: 0
00487    */
00488   k_frame_crc_placeholder = 0,
00489
00490   /**
00491    * @brief First index of ascii_buffer for post-condition validation
00492    * @details
00493    * Used in rx_comm_manager_init() to verify memset properly cleared buffer.
00494    * First byte must be '\0' after zero-fill.
00495    * @par Value: 0
00496    */
00497   k_ascii_buffer_first_idx = 0,
00498
00499   /**
00500    * @brief Typed zero constant for uint32_t comparisons
00501    * @details
00502    * Used for payload length checks and other uint32_t zero comparisons to avoid
00503    * magic number 0. Provides type safety and self-documenting code.
00504    * @par Value: 0
00505    */
00506   k_zero_u32 = 0,
00507 } internal_constants_t;
00508
00509 /*
      ============================================================================
00510  * Private Functions
```

```
00511  *
       ============================================================================
00512  */
00513
00514 /**
00515  * @brief Output decoded ASCII frame to USB Port 1 for debugging
00516  *
00517  * @details
00518  * Formats the given frame as human-readable ASCII text and outputs to USB Port 1
00519  * (decoded debug port) if decoded output is enabled. Used for development/debugging
00520  * to monitor frame traffic without binary protocol parsing.
00521  *
00522  * **Algorithm:**
00523  * 1. Validate parameters (NULL checks)
00524  * 2. Check if decoded output is enabled (cfg->enable_decoded_output)
00525  * 3. Format frame as ASCII via rx_frame_ascii_format()
00526  * 4. Send ASCII string to USB Port 1 via rx_usb_puts()
00527  *
00528  * **Performance**: ~50-150 us depending on frame size (ASCII formatting overhead).
00529  * Disable in production for optimal performance.
00530  *
00531  *
00532  *
00533  *
00534  *
00535  * @pre mgr must be initialized via rx_comm_manager_init()
00536  * @pre frame must be a valid frame structure
00537  * @post mgr->ascii_buffer content is modified (temporary)
00538  * @post ASCII output sent to USB Port 1 if enabled and successful
00539  *
00540  * @note Not thread-safe - caller must ensure exclusive access to mgr->ascii_buffer
00541  * @note Errors are silently ignored (formatting/USB send failures)
00542  * @note USB Port 1 must be configured for ASCII output to appear
00543  *
00544  * @par Example Output:
00545  * ```
00546  * RX: [USB] Type=0x01 Flags=0x00 Len=8 Seq=42 Payload=0102030405060708 CRC=ABCD1234
00547  * TX: [SPI] Type=0x02 Flags=0x80 Len=4 Seq=43 Payload=01020304 CRC=5678CDEF
00548  * ```
00549  *
00550  * @see rx_frame_ascii_format() Frame-to-ASCII conversion
00551  * @see rx_usb_puts() USB debug output
00552  *
00553  * @since Version 1.0.0
00554  */
00555 RX_STATIC_TESTABLE void
00556 internal_output_decoded(rx_comm_manager_t* mgr, const rx_frame_t* frame, bool is_tx)
00557 {
00558   if (!mgr->enable_decoded_output) {
00559     return;
00560   }
00561
00562   uint32_t len = 0;
00563   /* rx_frame_ascii_format always succeeds and produces output for any valid frame
00564    * and adequately-sized buffer (guaranteed by ascii_buffer sizing in the manager). */
00565   (void)rx_frame_ascii_format(frame, is_tx, mgr->ascii_buffer, sizeof(mgr->ascii_buffer), &len);
00566   /* The old decoded-debug path wrote to USB CDC port 1; with USB0 gone, we
00567    * route the ASCII dump through the UART log ring so it still shows up on
00568    * the gateway prefixed with [RX72N]. */
00569   rx_log_uart_puts(mgr->ascii_buffer);
00570 }
00571
00572 /**
00573  * @brief Build temporary frame from send params and output decoded ASCII (TX direction)
00574  *
00575  * @details
00576  * Helper for rx_comm_manager_send() to output ASCII-formatted frame after a successful
00577  * send. Constructs a temporary rx_frame_t from the send parameters and delegates to
00578  * internal_output_decoded().
00579  *
00580  *
00581  * @pre mgr must be non-NULL and initialized
00582  * @pre params must be non-NULL with valid fields
00583  * @post ASCII output sent to USB Port 1 if decoded output is enabled
00584  *
00585  * @note Not thread-safe - caller must ensure exclusive access to mgr->ascii_buffer
00586  *
00587  * @since Version 1.0.0
00588  */
00589 static void internal_output_decoded_from_params(rx_comm_manager_t*           mgr,
00590                                                 const rx_comm_send_params_t* params)
00591 {
00592   /* Build a temporary frame for ASCII formatting */
00593   rx_frame_t frame      = {}; /* Zero-initialize to clear padding/unset fields */
00594   frame.header.sequence = k_frame_seq_placeholder;
00595   frame.header.length   = (uint16_t)params->payload_len;
00596   frame.header.type     = (uint8_t)params->type;
```

```
00597   frame.header.flags    = params->flags;
00598
00599   /* payload_len <= k_frame_max_payload is guaranteed by caller validation */
00600   for (uint32_t i = 0; i < params->payload_len; i++) {
00601     frame.payload[i] = params->payload[i];
00602   }
00603   frame.crc = k_frame_crc_placeholder;
00604
00605   internal_output_decoded(mgr, &frame, true);
00606 }
00607
00608 /**
00609  * @brief Handle received frame by invoking user callback and optional ASCII output
00610  *
00611  * @details
00612  * Central frame dispatch point for all channels. Performs optional ASCII debugging
00613  * output, then invokes user-registered callback with frame data. This function
00614  * isolates callback invocation logic from channel-specific polling.
00615  *
00616  * **Algorithm:**
00617  * 1. Validate parameters (NULL checks, channel range)
00618  * 2. Output decoded ASCII to Port 1 if enabled (RX direction)
00619  * 3. Invoke user callback if registered
00620  *
00621  * **Design Rationale**: Centralizing frame handling logic ensures consistent
00622  * behavior across USB and SPI channels (single point of dispatch).
00623  *
00624  *
00625  *
00626  *
00627  *
00628  * @pre mgr must be initialized
00629  * @pre frame must be valid and CRC-verified by transport layer
00630  * @pre channel must be in valid range [0, k_comm_channel_count)
00631  * @post User callback invoked if registered (callback may be NULL)
00632  * @post ASCII output sent to Port 1 if enabled
00633  *
00634  * @note Thread safety depends on user callback implementation
00635  * @note Callback errors are not returned (callback should handle internally)
00636  * @warning User callback must not block for extended periods (affects polling rate)
00637  *
00638  * @par Performance:
00639  * - Without decoded output: ~2 us (callback overhead only)
00640  * - With decoded output: ~50-150 us (ASCII formatting + USB puts)
00641  *
00642  * @see internal_output_decoded() ASCII debugging output
00643  * @see rx_comm_frame_callback_t User callback type
00644  *
00645  * @since Version 1.0.0
00646  */
00647 RX_STATIC_TESTABLE void
00648 internal_handle_frame(rx_comm_manager_t* mgr, rx_comm_channel_t channel, const rx_frame_t* frame)
00649 {
00650   /* Update heartbeat: any valid frame resets the implicit timeout timer.
00651    * Per the dual-detection model, this is the primary link health
00652    * mechanism (200ms implicit timeout). */
00653   rx_comm_heartbeat_state_t* hb = &mgr->heartbeat[channel];
00654   hb->last_rx_ms               = internal_get_time_ms();
00655   if (hb->status != k_link_status_healthy) {
00656     rx_comm_link_status_t old_status = hb->status;
00657     hb->status                       = k_link_status_healthy;
00658     if (old_status == k_link_status_dead) {
00659       rx_log_info(s_tag, "Link recovered: frame received");
00660     } else {
00661       rx_log_info(s_tag, "Link established: first frame received");
00662     }
00663     if (mgr->link_status_cb != nullptr) {
00664       mgr->link_status_cb(channel, k_link_status_healthy, mgr->link_status_ctx);
00665     }
00666   }
00667
00668   rx_log_debug_val(s_tag, "Frame received, type", (uint8_t)frame->header.type);
00669
00670   if (frame->header.type == k_frame_type_command) {
00671     rx_log_info_val(s_tag, "CMD frame seq", frame->header.sequence);
00672     rx_log_info_val(s_tag, "CMD frame channel", (uint8_t)channel);
00673   }
00674
00675   /* Output decoded ASCII to Port 1 */
00676   internal_output_decoded(mgr, frame, false);
00677
00678   /* Invoke user callback if set */
00679   if (mgr->callback != nullptr) {
00680     mgr->callback(channel, frame, mgr->callback_ctx);
00681   }
00682 }
00683
```

```
00684 /**
00685  * @brief Poll SPI channel for incoming frames (non-blocking)
00686  *
00687  * @details
00688  * Attempts to receive one frame from SPI channel with zero timeout (non-blocking).
00689  * If frame is successfully received, it is dispatched via internal_handle_frame().
00690  * Timeout (no data available) is expected and normal - not an error condition.
00691  *
00692  * **Algorithm:**
00693  * 1. Validate mgr pointer (NULL check)
00694  * 2. Skip if SPI handle is nullptr (channel not configured)
00695  * 3. Call rx_spi_comm_receive() with 0 ms timeout
00696  * 4. On success: dispatch frame via internal_handle_frame()
00697  * 5. On timeout: return k_rx_err_timeout (expected, not error)
00698  * 6. On other errors: propagate error code
00699  *
00700  * **Design Note**: Identical logic to internal_poll_usb() but for SPI channel.
00701  * Code duplication is intentional (NASA Rule 4: keep functions short and simple).
00702  *
00703  *
00704  *
00705  * @pre mgr must be non-NULL
00706  * @pre If SPI channel enabled, spi_handle must be initialized
00707  * @post On k_rx_ok: callback invoked, ASCII output sent (if enabled)
00708  * @post On timeout: no side effects (no frame received)
00709  *
00710  * @note Non-blocking - always returns immediately
00711  * @note Timeout is NOT an error - it means no data is available
00712  * @warning Do not call if spi_handle is uninitialized (will return timeout)
00713  *
00714  * @par Performance:
00715  * - No data available: ~5 us (fast path)
00716  * - Frame received: ~50-150 us (SPI hardware transfer + callback)
00717  *
00718  * @see internal_handle_frame() Frame dispatch
00719  * @see rx_spi_comm_receive() SPI receive implementation
00720  *
00721  * @since Version 1.0.0
00722  */
00723 static rx_err_t internal_poll_spi(rx_comm_manager_t* mgr)
00724 {
00725   if (mgr->spi_handle == nullptr) {
00726     return k_rx_err_timeout;
00727   }
00728
00729   /* If SPI link layer is available, use it for HARQ decoding
00730    * NOTE: Using static buffers to avoid stack overflow (~3KB total).
00731    * Safe because comm manager poll is single-threaded (called from comm_task). */
00732   if (mgr->spi_link != nullptr) {
00733     static rx_spi_link_receive_result_t s_link_result;
00734     rx_err_t err = rx_spi_link_receive(mgr->spi_link, &s_link_result, k_receive_timeout_ms);
00735
00736     if (err == k_rx_ok) {
00737       /* Convert link result to rx_frame_t for internal_handle_frame() */
00738       static rx_frame_t s_frame1;
00739       s_frame1                = (rx_frame_t){};
00740       s_frame1.header.sequence = s_link_result.sequence;
00741       s_frame1.header.length   = (uint16_t)s_link_result.payload_len;
00742       s_frame1.header.type     = s_link_result.frame_type;
00743
00744       /* Preserve retransmit and FEC flags from link layer result */
00745       s_frame1.header.flags = k_frame_flag_none;
00746       if (s_link_result.is_retransmit) {
00747         s_frame1.header.flags |= k_frame_flag_retransmit;
00748       }
00749       if (s_link_result.fec_decoded) {
00750         s_frame1.header.flags |= k_frame_flag_fec_enabled;
00751       }
00752
00753       for (uint32_t i = 0; i < s_link_result.payload_len; i++) {
00754         s_frame1.payload[i] = s_link_result.payload[i];
00755       }
00756       internal_handle_frame(mgr, k_comm_channel_spi, &s_frame1);
00757       return k_rx_ok;
00758     }
00759
00760     return err;
00761   }
00762
00763   /* Fallback: raw SPI without HARQ */
00764   static rx_frame_t s_frame2;
00765   rx_err_t          err = rx_spi_comm_receive(mgr->spi_handle, &s_frame2, k_receive_timeout_ms);
00766
00767   if (err == k_rx_ok) {
00768     internal_handle_frame(mgr, k_comm_channel_spi, &s_frame2);
00769     return k_rx_ok;
00770   }
```

```
00771
00772   return err;
00773 }
00774
00775 /**
00776  * @brief Poll I2C channel for incoming frames (non-blocking)
00777  *
00778  * @details
00779  * Attempts to receive one frame from I2C peripheral channel with zero timeout.
00780  * If frame is successfully received, it is dispatched via internal_handle_frame().
00781  *
00782  *
00783  *
00784  * @pre mgr must be non-NULL
00785  * @pre mgr->i2c_handle must be initialized before calling rx_comm_manager_poll()
00786  * @post On k_rx_ok: callback invoked for received frame
00787  * @post On k_rx_err_timeout or k_rx_err_no_data: no state change
00788  *
00789  * @note Non-blocking - always returns immediately
00790  *
00791  * @see internal_handle_frame() Frame dispatch
00792  * @see rx_i2c_comm_receive() I2C peripheral receive implementation
00793  *
00794  * @since Version 1.0.0
00795  */
00796 static rx_err_t internal_poll_i2c(rx_comm_manager_t* mgr)
00797 {
00798   if (mgr->i2c_handle == nullptr) {
00799     return k_rx_err_timeout;
00800   }
00801
00802   rx_frame_t frame;
00803   rx_err_t   err = rx_i2c_comm_receive(mgr->i2c_handle, &frame, k_receive_timeout_ms);
00804
00805   if (err == k_rx_ok) {
00806     internal_handle_frame(mgr, k_comm_channel_i2c, &frame);
00807     return k_rx_ok;
00808   }
00809
00810   return err;
00811 }
00812
00813 /**
00814  * @brief Poll UART channel for incoming frames (non-blocking)
00815  *
00816  * @details
00817  * Attempts to receive one frame from UART (SCI9) channel with zero timeout.
00818  * If frame is successfully received, it is dispatched via internal_handle_frame().
00819  *
00820  *
00821  *
00822  * @pre mgr must be non-NULL
00823  * @pre mgr->uart_handle must be initialized before calling rx_comm_manager_poll()
00824  * @post On k_rx_ok: callback invoked for received frame
00825  * @post On k_rx_err_timeout or k_rx_err_no_data: no state change
00826  *
00827  * @note Non-blocking - always returns immediately
00828  *
00829  * @see internal_handle_frame() Frame dispatch
00830  * @see rx_uart_comm_receive() UART receive implementation
00831  *
00832  * @since Version 1.0.0
00833  */
00834 static rx_err_t internal_poll_uart(rx_comm_manager_t* mgr)
00835 {
00836   if (mgr->uart_handle == nullptr) {
00837     return k_rx_err_timeout;
00838   }
00839
00840   rx_frame_t frame;
00841   rx_err_t   err = rx_uart_comm_receive(mgr->uart_handle, &frame, k_receive_timeout_ms);
00842
00843   if (err == k_rx_ok) {
00844     if (frame.header.type == k_frame_type_command) {
00845       rx_log_info_val(s_tag, "UART RX CMD seq", frame.header.sequence);
00846     }
00847     internal_handle_frame(mgr, k_comm_channel_uart, &frame);
00848     return k_rx_ok;
00849   }
00850
00851   return err;
00852 }
00853
00854 /*
      ============================================================================
00855  * Event Queue Helpers
00856  *
```

```
00857 */
00858
00859 /**
00860  * @brief Process one event from the FIFO queue
00861  *
00862  * @details
00863  * Dequeues the oldest event and attempts to send it. For SPI events, if the
00864  * send fails, the event is re-enqueued with incremented retry count (up to
00865  * k_event_max_retries). For USB events, failure is simply dropped (fire-and-forget).
00866  *
00867  *
00868  * @pre mgr must be non-NULL and initialized
00869  * @post One event processed (sent or dropped on max retries)
00870  *
00871  * @since Version 1.0.0
00872  */
00873 /**
00874  * @brief Dequeue the front event entry and advance the tail pointer
00875  *
00876  *
00877  * @pre mgr->event_queue_count > 0
00878  * @post entry cleared and tail pointer advanced
00879  *
00880  * @since Version 1.0.0
00881  */
00882 static void internal_dequeue_event(rx_comm_manager_t* mgr, rx_comm_event_entry_t* entry)
00883 {
00884   entry->occupied           = false;
00885   entry->retries            = 0;
00886   entry->next_retry_time_ms = 0;
00887   mgr->event_queue_tail     = (mgr->event_queue_tail + 1) % k_comm_event_queue_depth;
00888   mgr->event_queue_count--;
00889 }
00890
00891 static void internal_process_event_queue(rx_comm_manager_t* mgr)
00892 {
00893   if (mgr->event_queue_count == 0) {
00894     return;
00895   }
00896
00897   rx_comm_event_entry_t* entry = &mgr->event_queue[mgr->event_queue_tail];
00898   if (!entry->occupied) {
00899     return;
00900   }
00901
00902   /* Check backoff: skip if retry time hasn't been reached yet */
00903   const uint32_t now_ms = internal_get_time_ms();
00904   if (entry->retries > 0 && now_ms < entry->next_retry_time_ms) {
00905     return; /* Backoff period not elapsed, try again later */
00906   }
00907
00908   /* Build send params from queued entry */
00909   const rx_comm_send_params_t params = {
00910     .channel     = entry->channel,
00911     .type        = entry->type,
00912     .flags       = entry->flags,
00913     .payload     = entry->payload,
00914     .payload_len = entry->payload_len,
00915   };
00916
00917   rx_err_t err = rx_comm_manager_send(mgr, &params);
00918
00919   if (err == k_rx_ok || entry->channel == k_comm_channel_uart) {
00920     /* Success, or UART fire-and-forget: dequeue.
00921      * UART has no link-layer retry (the CY7C65213 bridge + USB CDC transport
00922      * on the host provides enough reliability for the command/response path). */
00923     internal_dequeue_event(mgr, entry);
00924     if (err == k_rx_ok) {
00925       rx_log_debug_val(s_tag, "Event sent, queue depth", mgr->event_queue_count);
00926     } else {
00927       rx_log_warn_val(s_tag, "UART event dropped, queue depth", mgr->event_queue_count);
00928     }
00929     return;
00930   }
00931
00932   /* SPI send failed: retry with exponential backoff */
00933   entry->retries++;
00934   if (entry->retries >= k_event_max_retries) {
00935     rx_log_error_val(s_tag, "SPI event failed after retries", (uint8_t)entry->retries);
00936     internal_dequeue_event(mgr, entry);
00937     return;
00938   }
00939
00940   /* Compute exponential backoff: initial_ms « (retries - 1), capped at max */
00941   uint32_t backoff_ms = (uint32_t)k_event_initial_backoff_ms « (entry->retries - 1);
00942   if (backoff_ms > k_event_max_backoff_ms) {
```

```
00943     backoff_ms = k_event_max_backoff_ms;
00944   }
00945   entry->next_retry_time_ms = now_ms + backoff_ms;
00946
00947   rx_log_warn_val(s_tag, "SPI event retry attempt", (uint8_t)entry->retries);
00948   /* Entry stays in queue, will be retried after backoff period */
00949 }
00950
00951 /*
      ============================================================================
00952 * Heartbeat Helpers
00953 *
      ============================================================================
00954 */
00955
00956 /**
00957 * @brief Check heartbeat implicit timeout on all transports
00958 *
00959 * @details
00960 * Evaluates the dual-detection heartbeat model.
00961 * If no valid frame has been received on a transport within 200ms, the link
00962 * is declared dead and the status callback is invoked.
00963 *
00964 *
00965 * @pre mgr must be non-NULL and initialized
00966 * @post Link status updated for each transport
00967 *
00968 * @since Version 1.0.0
00969 */
00970 /**
00971 * @brief Check heartbeat timeout on a single channel
00972 *
00973 *
00974 * @pre ch < k_comm_channel_count
00975 * @post Link status updated if timeout exceeded
00976 *
00977 * @since Version 1.0.0
00978 */
00979 static void internal_check_channel_heartbeat(rx_comm_manager_t* mgr, uint8_t ch, uint32_t now_ms)
00980 {
00981   rx_comm_heartbeat_state_t* hb = &mgr->heartbeat[ch];
00982
00983   /* Skip channels that haven't received any frame yet */
00984   if (hb->status != k_link_status_healthy) {
00985     return;
00986   }
00987
00988   /* Check implicit timeout: 200ms since last valid frame */
00989   const uint32_t elapsed = now_ms - hb->last_rx_ms;
00990   if (elapsed < k_heartbeat_implicit_timeout_ms) {
00991     return;
00992   }
00993
00994   hb->status = k_link_status_dead;
00995   rx_log_warn_val(s_tag, "Link dead: no frame (ms elapsed)", elapsed);
00996
00997   if (mgr->link_status_cb != nullptr) {
00998     mgr->link_status_cb((rx_comm_channel_t)ch, k_link_status_dead, mgr->link_status_ctx);
00999   }
01000 }
01001
01002 static void internal_check_heartbeat(rx_comm_manager_t* mgr)
01003 {
01004   const uint32_t now = internal_get_time_ms();
01005
01006   for (uint8_t ch = 0; ch < k_comm_channel_count; ch++) {
01007     internal_check_channel_heartbeat(mgr, ch, now);
01008   }
01009 }
01010
01011 /*
      ============================================================================
01012 * Public Functions
01013 *
      ============================================================================
01014 */
01015
01016 /**
01017 * @var s_zero_mgr
01018 * @brief Zero-initialized comm manager template for stack-safe initialization
01019 * @details Static const instance used to clear rx_comm_manager_t without creating
01020 *          a large compound literal on the stack. Lives in .rodata section.
01021 * @note Read-only; never modified after static initialization
01022 * @warning Do not remove - prevents stack overflow in init function
01023 * @see rx_comm_manager_init() Uses this template to zero-initialize the manager handle
01024 * @since Version 1.0.0
01025 */
```

```
01026 static const rx_comm_manager_t s_zero_mgr = {};
01027
01028 /**
01029  * @brief Initialize communication manager with channel configuration
01030  *
01031  * @details
01032  * Performs complete initialization of communication manager handle, configuring
01033  * enabled channels, callback registration, and ASCII debugging output. This function
01034  * must be called before any other communication manager operations.
01035  *
01036  * **Algorithm:**
01037  * 1. **Validate parameters**: Check mgr pointer for nullptr (NASA Rule 5)
01038  * 2. **Clear handle**: Zero-fill entire structure via memset (328 bytes)
01039  * 3. **Apply configuration**: Copy config fields if cfg is non-NULL
01040  *    - usb_handle: Pointer to initialized USB comm handle (may be NULL)
01041  *    - spi_handle: Pointer to initialized SPI comm handle (may be NULL)
01042  *    - callback: User frame handler function (may be NULL)
01043  *    - callback_ctx: User context pointer (may be NULL)
01044  *    - enable_decoded_output: ASCII debugging flag
01045  * 4. **Mark initialized**: Set mgr->initialized = true
01046  * 5. **Post-condition validation**: Verify ascii_buffer properly cleared
01047  *
01048  * **Configuration Options**:
01049  * - **cfg = NULL**: All channels disabled, no callback, ASCII output off
01050  * - **usb_handle = NULL**: USB channel disabled (poll skips USB)
01051  * - **spi_handle = NULL**: SPI channel disabled (poll skips SPI)
01052  * - **callback = NULL**: No callback invoked (silent frame consumption)
01053  * - **enable_decoded_output = true**: ASCII debugging enabled (Port 1)
01054  *
01055  *
01056  *
01057  *
01058  * @pre mgr must point to allocated memory (uninitialized content OK)
01059  * @pre USB/SPI handles (if provided) must already be initialized
01060  * @pre Callback function (if provided) must be valid and reentrant
01061  * @post mgr->initialized = true on success
01062  * @post All mgr fields set to config values or zero
01063  * @post mgr->ascii_buffer is zero-filled (256 bytes)
01064  *
01065  * @note This function does NOT initialize USB or SPI transports - caller must initialize
01066  *       rx_usb_comm and rx_spi_comm separately before passing handles to this function
01067  * @note On success, at least one channel should be enabled (usb_handle or spi_handle non-NULL)
01068  * @note Thread-safe ONLY if each thread uses a different mgr handle
01069  *
01070  * @par Performance:
01071  * - Execution time: ~5 us @ 240 MHz
01072  * - Dominated by memset (328 bytes)
01073  *
01074  * @par Configuration Example - Both Channels Enabled:
01075  * @code{.c}
01076  * static rx_comm_manager_t g_comm_mgr;
01077  * static rx_usb_comm_handle_t g_usb_handle;
01078  * static rx_spi_comm_handle_t g_spi_handle;
01079  *
01080  * // Initialize transport layers first
01081  * rx_usb_comm_config_t usb_cfg = { .port = k_usb_port_proto };
01082  * rx_err_t err = rx_usb_comm_init(&g_usb_handle, &usb_cfg);
01083  * if (err != k_rx_ok) return err;
01084  *
01085  * rx_spi_comm_config_t spi_cfg = { .spi_channel = 0 };
01086  * err = rx_spi_comm_init(&g_spi_handle, &spi_cfg);
01087  * if (err != k_rx_ok) return err;
01088  *
01089  * // Initialize communication manager with both channels
01090  * rx_comm_manager_config_t cfg = {
01091  *   .usb_handle = &g_usb_handle,
01092  *   .spi_handle = &g_spi_handle,
01093  *   .callback = on_frame_received,
01094  *   .callback_ctx = nullptr,
01095  *   .enable_decoded_output = true,
01096  * };
01097  * err = rx_comm_manager_init(&g_comm_mgr, &cfg);
01098  * @endcode
01099  *
01100  * @par Configuration Example - USB Only, No Callback:
01101  * @code{.c}
01102  * rx_comm_manager_config_t cfg = {
01103  *   .usb_handle = &g_usb_handle,
01104  *   .spi_handle = nullptr,              // SPI disabled
01105  *   .callback = nullptr,                // No callback
01106  *   .callback_ctx = nullptr,
01107  *   .enable_decoded_output = false,     // No ASCII debugging
01108  * };
01109  * rx_err_t err = rx_comm_manager_init(&g_comm_mgr, &cfg);
01110  * @endcode
01111  *
01112  * @see rx_comm_manager_deinit() Cleanup and resource release
```

```
01113  * @see rx_comm_manager_poll() Poll for incoming frames
01114  * @see rx_spi_comm_init() Initialize SPI transport
01115  * @see rx_uart_comm_init() Initialize UART transport
01116  * @see rx_i2c_comm_init() Initialize I2C peripheral transport
01117  *
01118  * @since Version 1.0.0
01119  */
01120 rx_err_t rx_comm_manager_init(rx_comm_manager_t* mgr, const rx_comm_manager_config_t* cfg)
01121 {
01122   /* Rule 5: Pre-condition validation - nullptr check */
01123   if (mgr == nullptr) {
01124     rx_log_error(s_tag, "Init failed: NULL mgr");
01125     return k_rx_err_invalid_arg;
01126   }
01127
01128   /* Rule 5: Pre-condition validation - SPI link consistency checks */
01129   if (cfg != nullptr && cfg->spi_link != nullptr) {
01130     /* If spi_link is provided, spi_handle must also be provided */
01131     if (cfg->spi_handle == nullptr) {
01132       rx_log_error(s_tag, "Init failed: spi_link requires non-NULL spi_handle");
01133       return k_rx_err_invalid_arg;
01134     }
01135
01136     /* If spi_link is provided, it must be initialized */
01137     if (!cfg->spi_link->initialized) {
01138       rx_log_error(s_tag, "Init failed: spi_link not initialized (call rx_spi_link_init first)");
01139       return k_rx_err_invalid_arg;
01140     }
01141
01142     /* If spi_link is provided, its spi_handle must match config spi_handle */
01143     if (cfg->spi_link->spi_handle != cfg->spi_handle) {
01144       rx_log_error(s_tag, "Init failed: spi_link->spi_handle != cfg->spi_handle");
01145       return k_rx_err_invalid_arg;
01146     }
01147   }
01148
01149   *mgr = s_zero_mgr;
01150
01151   /* Apply configuration */
01152   if (cfg != nullptr) {
01153     mgr->uart_handle          = cfg->uart_handle;
01154     mgr->spi_handle           = cfg->spi_handle;
01155     mgr->i2c_handle           = cfg->i2c_handle;
01156     mgr->spi_link             = cfg->spi_link;
01157     mgr->callback             = cfg->callback;
01158     mgr->callback_ctx         = cfg->callback_ctx;
01159     mgr->enable_decoded_output = cfg->enable_decoded_output;
01160     mgr->link_status_cb       = cfg->link_status_cb;
01161     mgr->link_status_ctx      = cfg->link_status_ctx;
01162   }
01163
01164   mgr->initialized = true;
01165
01166   rx_log_info(s_tag, "Comm manager initialized");
01167   return k_rx_ok;
01168 }
01169
01170 /**
01171  * @brief Deinitialize communication manager and release resources
01172  *
01173  * @details
01174  * Marks communication manager as uninitialized, preventing further operations.
01175  * This function does NOT deinitialize USB or SPI transport handles - caller
01176  * must separately call rx_usb_comm_deinit() and rx_spi_comm_deinit() if needed.
01177  *
01178  * **Algorithm:**
01179  * 1. Validate mgr pointer (NULL check)
01180  * 2. Validate initialized flag (must be initialized to deinitialize)
01181  * 3. Clear initialized flag (marks handle as invalid)
01182  *
01183  * **Design Note**: Minimal deinitialization - no dynamic resources to free
01184  * (all memory is statically allocated). Function primarily serves as safety
01185  * guard to prevent use-after-deinit.
01186  *
01187  *
01188  *
01189  * @pre mgr must be non-NULL
01190  * @pre mgr must be initialized (mgr->initialized == true)
01191  * @post mgr->initialized = false
01192  * @post Subsequent calls to rx_comm_manager_poll/send will return k_rx_err_invalid_state
01193  *
01194  * @note Does NOT free memory (handle is statically allocated)
01195  * @note Does NOT deinitialize transport handles (USB/SPI)
01196  * @note Thread-safe if no concurrent operations on same mgr handle
01197  *
01198  * @par Example:
01199  * @code{.c}
```

```
01200  * // Deinitialize manager
01201  * rx_err_t err = rx_comm_manager_deinit(&g_comm_mgr);
01202  * if (err == k_rx_ok) {
01203  *   // Manager no longer usable
01204  *
01205  *   // Separately deinitialize transports if needed
01206  *   rx_usb_comm_deinit(&g_usb_handle);
01207  *   rx_spi_comm_deinit(&g_spi_handle);
01208  * }
01209  * @endcode
01210  *
01211  * @see rx_comm_manager_init() Initialization
01212  * @see rx_spi_comm_deinit() SPI transport cleanup
01213  *
01214  * @since Version 1.0.0
01215  */
01216 rx_err_t rx_comm_manager_deinit(rx_comm_manager_t* mgr)
01217 {
01218   /* Rule 5: Pre-condition validation */
01219   if (mgr == nullptr) {
01220     rx_log_error(s_tag, "Deinit failed: NULL mgr");
01221     return k_rx_err_invalid_arg;
01222   }
01223   if (!mgr->initialized) {
01224     rx_log_warn(s_tag, "Deinit called on uninitialized mgr");
01225     return k_rx_err_invalid_state;
01226   }
01227
01228   mgr->initialized = false;
01229
01230   rx_log_info(s_tag, "Comm manager deinitialized");
01231   return k_rx_ok;
01232 }
01233
01234 /**
01235  * @brief Poll all enabled communication channels for incoming frames (non-blocking)
01236  *
01237  * @details
01238  * Performs non-blocking poll of all enabled channels (USB and SPI) sequentially,
01239  * dispatching any received frames to the registered user callback. Returns aggregate
01240  * result indicating whether any frames were received across all channels.
01241  *
01242  * **Algorithm:**
01243  * 1. **Validate parameters**: Check mgr pointer and initialized flag (NASA Rule 5)
01244  * 2. **Poll USB channel** via internal_poll_usb():
01245  *    - If k_rx_ok: Mark received=true, continue to SPI
01246  *    - If k_rx_err_timeout: Continue to SPI (no data is normal)
01247  *    - If other error: Return error immediately (critical failure)
01248  * 3. **Poll SPI channel** via internal_poll_spi():
01249  *    - If k_rx_ok: Mark received=true
01250  *    - If k_rx_err_timeout: Continue (no data is normal)
01251  *    - If other error: Return error immediately (critical failure)
01252  * 4. **Return result**:
01253  *    - k_rx_ok if ANY channel received frame
01254  *    - k_rx_err_timeout if NO channels received (normal condition)
01255  *    - Other error codes propagated from transport layers
01256  *
01257  * **Performance**: 15-200 us typical, depends on:
01258  * - Number of enabled channels (USB/SPI/both)
01259  * - Whether frames are available (data path vs fast timeout path)
01260  * - ASCII debugging enabled (adds ~50-150 us per frame)
01261  * - User callback execution time (not included in measurement)
01262  *
01263  * **Design Rationale**:
01264  * - **Non-blocking**: Never blocks waiting for data (0 ms timeout)
01265  * - **Fair scheduling**: Both channels polled each iteration
01266  * - **Timeout tolerance**: Timeout is expected, not error condition
01267  * - **Error prioritization**: Critical errors returned immediately
01268  * - **Aggregation**: k_rx_ok if ANY channel received data
01269  *
01270  *
01271  *
01272  * @pre mgr must be non-NULL
01273  * @pre mgr must be initialized
01274  * @pre At least one channel should be enabled (usb_handle or spi_handle non-NULL)
01275  * @post User callback invoked for each successfully received frame
01276  * @post ASCII output sent to Port 1 if enabled
01277  *
01278  * @note Non-blocking - always returns immediately
01279  * @note k_rx_err_timeout is NORMAL - means no data available (not an error)
01280  * @note User callback execution time is NOT included in performance measurements
01281  * @warning Call this function regularly (e.g., 100 Hz) to avoid buffer overruns
01282  * @warning User callback must not block for extended periods
01283  *
01284  * @par Performance:
01285  * - No frames available: ~15 us (both channels timeout, fast path)
01286  * - USB frame received: ~100-250 us (USB bulk transfer + callback)
```

```
01287  * - SPI frame received: ~50-150 us (SPI hardware transfer + callback)
01288  * - Both frames: ~150-400 us (both channels + callbacks)
01289  * - With ASCII debugging: Add ~50-150 us per frame
01290  *
01291  * @par Typical Usage - ThreadX Communication Task:
01292  * @code{.c}
01293  * static void comm_task_entry(ULONG thread_input)
01294  * {
01295  *   (void)thread_input;
01296  *
01297  *   // Initialize communication manager
01298  *   // ... (see rx_comm_manager_init() documentation) ...
01299  *
01300  *   // Main polling loop @ 100 Hz
01301  *   while (1) {
01302  *     // Poll all channels (non-blocking)
01303  *     rx_err_t err = rx_comm_manager_poll(&g_comm_mgr);
01304  *
01305  *     // Handle critical errors (not timeout)
01306  *     if (err != k_rx_ok && err != k_rx_err_timeout) {
01307  *       // Log error, implement recovery strategy
01308  *       log_error("Comm poll failed: %d", err);
01309  *       tx_thread_sleep(100);  // 1 second backoff
01310  *       continue;
01311  *     }
01312  *
01313  *     // Timeout is normal - just means no data available
01314  *
01315  *     // Sleep 10 ticks (100 ms @ 100 Hz tick rate = 10 Hz polling)
01316  *     tx_thread_sleep(10);
01317  *   }
01318  * }
01319  * @endcode
01320  *
01321  * @par Error Handling - Distinguishing Timeout from Errors:
01322  * @code{.c}
01323  * rx_err_t err = rx_comm_manager_poll(&g_comm_mgr);
01324  * if (err == k_rx_ok) {
01325  *   // At least one frame received and processed
01326  * } else if (err == k_rx_err_timeout) {
01327  *   // No frames available - this is NORMAL, not an error
01328  * } else {
01329  *   // Critical error - handle appropriately
01330  *   handle_comm_error(err);
01331  * }
01332  * @endcode
01333  *
01334  * @see internal_poll_usb() USB channel polling implementation
01335  * @see internal_poll_spi() SPI channel polling implementation
01336  * @see internal_handle_frame() Frame dispatch
01337  * @see rx_comm_manager_init() Initialization with channel configuration
01338  *
01339  * @since Version 1.0.0
01340  */
01341 /**
01342  * @brief Poll all transport channels and aggregate results
01343  *
01344  *
01345  *
01346  * @pre mgr must be non-NULL and initialized
01347  * @post *received reflects whether any frames were received
01348  *
01349  * @since Version 1.0.0
01350  */
01351 static rx_err_t internal_poll_all_channels(rx_comm_manager_t* mgr, bool* received)
01352 {
01353   /* Poll UART channel (primary path to the Pi5 gateway) */
01354   rx_err_t uart_err = internal_poll_uart(mgr);
01355   if (uart_err == k_rx_ok) {
01356     *received = true;
01357   } else if (uart_err != k_rx_err_timeout && uart_err != k_rx_err_no_data) {
01358     rx_log_error_val(s_tag, "UART poll error", uart_err);
01359     return uart_err;
01360   }
01361
01362   /* Poll SPI channel */
01363   rx_err_t spi_err = internal_poll_spi(mgr);
01364   if (spi_err == k_rx_ok) {
01365     *received = true;
01366   } else if (spi_err != k_rx_err_timeout) {
01367     rx_log_error_val(s_tag, "SPI poll error", spi_err);
01368     return spi_err;
01369   }
01370
01371   /* Poll I2C channel */
01372   rx_err_t i2c_err = internal_poll_i2c(mgr);
01373   if (i2c_err == k_rx_ok) {
```

```
01374     *received = true;
01375   } else if (i2c_err != k_rx_err_timeout && i2c_err != k_rx_err_no_data) {
01376     rx_log_error_val(s_tag, "I2C poll error", i2c_err);
01377     return i2c_err;
01378   }
01379
01380   return k_rx_ok;
01381 }
01382
01383 rx_err_t rx_comm_manager_poll(rx_comm_manager_t* mgr)
01384 {
01385   /* Rule 5: Pre-condition validation */
01386   if (mgr == nullptr) {
01387     return k_rx_err_invalid_arg;
01388   }
01389   if (!mgr->initialized) {
01390     return k_rx_err_invalid_state;
01391   }
01392
01393   bool     received = false;
01394   rx_err_t poll_err = internal_poll_all_channels(mgr, &received);
01395   if (poll_err != k_rx_ok) {
01396     return poll_err;
01397   }
01398
01399   /* Process event queue: send one queued event per poll cycle */
01400   internal_process_event_queue(mgr);
01401
01402   /* Heartbeat: check implicit timeout on each transport */
01403   internal_check_heartbeat(mgr);
01404
01405   return (int)received ? k_rx_ok : k_rx_err_timeout;
01406 }
01407
01408 /**
01409  * @brief Send frame on specified communication channel
01410  *
01411  * @details
01412  * Routes frame transmission to the appropriate transport layer (USB or SPI) based
01413  * on specified channel. Supports configurable frame type, flags, and variable-length
01414  * payload. Optionally outputs ASCII-formatted frame to USB Port 1 if debugging enabled.
01415  *
01416  * **Algorithm:**
01417  * 1. **Validate parameters** (NASA Rule 5):
01418  *    - mgr and params pointers non-NULL
01419  *    - Manager initialized flag
01420  *    - Payload pointer valid if payload_len > 0
01421  *    - payload_len <= k_frame_max_payload (255 bytes)
01422  * 2. **Route to channel**:
01423  *    - USB: Verify usb_handle non-NULL, call rx_usb_comm_send()
01424  *    - SPI: Verify spi_handle non-NULL, call rx_spi_comm_send()
01425  *    - Invalid channel: Return k_rx_err_invalid_arg
01426  * 3. **On success, if ASCII debugging enabled**:
01427  *    - Build temporary rx_frame_t from params
01428  *    - Format as ASCII via internal_output_decoded()
01429  *    - Send to USB Port 1 (TX direction)
01430  *
01431  * **Performance**:
01432  * - USB: ~30-150 us (depends on payload size, USB bulk transfer)
01433  * - SPI: ~20-100 us (depends on payload size, SPI hardware transfer)
01434  * - ASCII debugging: Add ~50-150 us if enabled
01435  *
01436  *
01437  *
01438  *
01439  * @pre mgr must be non-NULL and initialized
01440  * @pre params must be non-NULL with all valid fields
01441  * @pre Channel handle (USB/SPI) must be initialized
01442  * @pre If payload_len > 0, payload must point to valid data
01443  * @post Frame transmitted on specified channel if successful
01444  * @post ASCII output sent to Port 1 if enabled and successful
01445  *
01446  * @note Payload sequence number and CRC-32 managed by transport layer
01447  * @note ASCII debugging adds overhead - disable in production
01448  * @warning Ensure channel is ready before sending (use rx_comm_manager_channel_ready)
01449  *
01450  * @par Send Parameters Structure:
01451  * @code{.c}
01452  * typedef struct {
01453  *   rx_comm_channel_t channel;      // USB or SPI
01454  *   uint8_t          type;         // Frame type
01455  *   uint8_t          flags;        // Frame flags
01456  *   const uint8_t*   payload;      // Payload data
01457  *   uint32_t         payload_len;  // Payload length [0, 255]
01458  * } rx_comm_send_params_t;
01459  * @endcode
01460  *
```

```
01461  * @par Example - Send Command on USB:
01462  * @code{.c}
01463  * uint8_t cmd_payload[8] = {0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0x07, 0x08};
01464  *
01465  * rx_comm_send_params_t params = {
01466  *   .channel = k_comm_channel_usb,
01467  *   .type = k_frame_type_command,
01468  *   .flags = k_frame_flag_none,
01469  *   .payload = cmd_payload,
01470  *   .payload_len = sizeof(cmd_payload),
01471  * };
01472  *
01473  * rx_err_t err = rx_comm_manager_send(&g_comm_mgr, &params);
01474  * if (err == k_rx_ok) {
01475  *   // Command sent successfully
01476  * }
01477  * @endcode
01478  *
01479  * @par Example - Send Empty Frame (No Payload):
01480  * @code{.c}
01481  * rx_comm_send_params_t params = {
01482  *   .channel = k_comm_channel_spi,
01483  *   .type = k_frame_type_response,
01484  *   .flags = k_frame_flag_ack,
01485  *   .payload = nullptr,          // No payload
01486  *   .payload_len = 0,        // Zero length
01487  * };
01488  *
01489  * rx_err_t err = rx_comm_manager_send(&g_comm_mgr, &params);
01490  * @endcode
01491  *
01492  * @see rx_comm_manager_respond() Convenience wrapper for response frames
01493  * @see rx_comm_send_params_t Send parameters structure
01494  * @see rx_spi_comm_send() SPI send implementation
01495  * @see rx_uart_comm_send() UART send implementation
01496  * @see rx_i2c_comm_send() I2C peripheral send implementation
01497  *
01498  * @since Version 1.0.0
01499  */
01500 /**
01501  * @brief Route frame to transport layer based on channel
01502  *
01503  *
01504  *
01505  * @pre mgr and params must be non-NULL
01506  * @pre params validated by caller (payload length, etc.)
01507  * @post Frame sent on specified channel transport
01508  *
01509  * @since Version 1.0.0
01510  */
01511 static rx_err_t internal_route_send(rx_comm_manager_t* mgr, const rx_comm_send_params_t* params)
01512 {
01513   switch (params->channel) {
01514     case k_comm_channel_spi:
01515       if (mgr->spi_handle == nullptr) {
01516         rx_log_error(s_tag, "Send failed: SPI handle NULL");
01517         return k_rx_err_invalid_state;
01518       }
01519       /* Use HARQ link layer if available, otherwise raw SPI */
01520       if (mgr->spi_link != nullptr) {
01521         /* NOTE: params->flags intentionally NOT forwarded to rx_spi_link_send().
01522          * The link layer manages its own flags internally (k_frame_flag_requires_ack,
01523          * k_frame_flag_fec_enabled, k_frame_flag_retransmit) based on HARQ state. */
01524         return rx_spi_link_send(mgr->spi_link, params->type, params->payload, params->payload_len);
01525       }
01526       return rx_spi_comm_send(mgr->spi_handle,
01527                               params->type,
01528                               params->flags,
01529                               params->payload,
01530                               params->payload_len);
01531
01532     case k_comm_channel_i2c:
01533       if (mgr->i2c_handle == nullptr) {
01534         rx_log_error(s_tag, "Send failed: I2C handle NULL");
01535         return k_rx_err_invalid_state;
01536       }
01537       return rx_i2c_comm_send(mgr->i2c_handle,
01538                               params->type,
01539                               params->flags,
01540                               params->payload,
01541                               params->payload_len);
01542
01543     case k_comm_channel_uart:
01544       if (mgr->uart_handle == nullptr) {
01545         rx_log_error(s_tag, "Send failed: UART handle NULL");
01546         return k_rx_err_invalid_state;
01547       }
```

```
01548       return rx_uart_comm_send(mgr->uart_handle,
01549                                params->type,
01550                                params->flags,
01551                                params->payload,
01552                                params->payload_len);
01553
01554     default:
01555       rx_log_error(s_tag, "Send failed: invalid channel");
01556       return k_rx_err_invalid_arg;
01557   }
01558 }
01559
01560 rx_err_t rx_comm_manager_send(rx_comm_manager_t* mgr, const rx_comm_send_params_t* params)
01561 {
01562   /* Rule 5: Pre-condition validation */
01563   if (mgr == nullptr || params == nullptr) {
01564     rx_log_error(s_tag, "Send failed: NULL arg");
01565     return k_rx_err_invalid_arg;
01566   }
01567   if (!mgr->initialized) {
01568     rx_log_error(s_tag, "Send failed: not initialized");
01569     return k_rx_err_invalid_state;
01570   }
01571   if (params->payload == nullptr && params->payload_len > 0) {
01572     rx_log_error(s_tag, "Send failed: NULL payload with nonzero len");
01573     return k_rx_err_invalid_arg;
01574   }
01575   /* Validate payload length is within frame limits */
01576   if (params->payload_len > k_frame_max_payload) {
01577     rx_log_error_val(s_tag, "Send failed: payload too large", params->payload_len);
01578     return k_rx_err_invalid_arg;
01579   }
01580
01581   rx_err_t err = internal_route_send(mgr, params);
01582
01583   if (err != k_rx_ok) {
01584     rx_log_error_val(s_tag, "Transport send failed", err);
01585   }
01586
01587   /* Output decoded ASCII on success */
01588   if (err == k_rx_ok && (int)mgr->enable_decoded_output) {
01589     internal_output_decoded_from_params(mgr, params);
01590   }
01591
01592   return err;
01593 }
01594
01595 /**
01596  * @brief Send response frame on specified channel (convenience wrapper)
01597  *
01598  * @details
01599  * Convenience function for sending response frames. Automatically sets frame type
01600  * to k_frame_type_response and flags to k_frame_flag_none. Simplifies common use
01601  * case of responding to commands/requests.
01602  *
01603  * **Algorithm:**
01604  * 1. Build rx_comm_send_params_t with:
01605  *    - channel: User-specified
01606  *    - type: k_frame_type_response (fixed)
01607  *    - flags: k_frame_flag_none (fixed)
01608  *    - payload: User-specified
01609  *    - payload_len: User-specified
01610  * 2. Call rx_comm_manager_send() with constructed params
01611  * 3. Return result from rx_comm_manager_send()
01612  *
01613  * **When to Use**:
01614  * - Responding to commands with telemetry data
01615  * - Sending acknowledgments
01616  * - Replying to requests
01617  *
01618  * **When NOT to Use**:
01619  * - Need custom frame type (use rx_comm_manager_send instead)
01620  * - Need custom flags (use rx_comm_manager_send instead)
01621  *
01622  *
01623  *
01624  *
01625  *
01626  *
01627  * @pre Same preconditions as rx_comm_manager_send()
01628  * @post Response frame transmitted with type=response, flags=none
01629  *
01630  * @note Equivalent to calling rx_comm_manager_send() with type=k_frame_type_response
01631  * @note This is a thin wrapper - no additional validation performed
01632  *
01633  * @par Example - Send Telemetry Response:
01634  * @code{.c}
```

```
01635  * // Received a telemetry request on USB, send response
01636  * static void on_telemetry_request(rx_comm_channel_t channel, const rx_frame_t* frame)
01637  * {
01638  *   uint8_t telemetry[16];
01639  *
01640  *   // Gather telemetry data
01641  *   telemetry[0] = get_temperature_celsius();
01642  *   telemetry[1] = get_current_ma();
01643  *   // ... fill remaining fields ...
01644  *
01645  *   // Send response on same channel request came from
01646  *   rx_err_t err = rx_comm_manager_respond(&g_comm_mgr,
01647  *                                         channel,
01648  *                                         telemetry,
01649  *                                         sizeof(telemetry));
01650  *   if (err != k_rx_ok) {
01651  *     // Handle send error
01652  *   }
01653  * }
01654  * @endcode
01655  *
01656  * @par Example - Send Empty Acknowledgment:
01657  * @code{.c}
01658  * // Send empty ACK response
01659  * rx_err_t err = rx_comm_manager_respond(&g_comm_mgr,
01660  *                                       k_comm_channel_usb,
01661  *                                       NULL,     // No payload
01662  *                                       0);       // Zero length
01663  * @endcode
01664  *
01665  * @see rx_comm_manager_send() Full send function with all parameters
01666  * @see rx_comm_send_params_t Send parameters structure
01667  *
01668  * @since Version 1.0.0
01669  */
01670 rx_err_t rx_comm_manager_respond(rx_comm_manager_t* mgr,
01671                                  rx_comm_channel_t  channel,
01672                                  const uint8_t*     payload,
01673                                  uint32_t           payload_len)
01674 {
01675   const rx_comm_send_params_t params = {
01676     .channel     = channel,
01677     .type        = k_frame_type_response,
01678     .flags       = k_frame_flag_none,
01679     .payload     = payload,
01680     .payload_len = payload_len,
01681   };
01682   return rx_comm_manager_send(mgr, &params);
01683 }
01684
01685 /*
      ============================================================================
01686  * Stream / Event Send API
01687  *
      ============================================================================
01688  */
01689
01690 rx_err_t rx_comm_manager_stream_send(rx_comm_manager_t* mgr, const rx_comm_send_params_t* params)
01691 {
01692   /* Stream path: fire-and-forget, no retries, no queuing.
01693    * Telemetry and other time-sensitive data goes here.
01694    * Stale data has no value, so failure is silently accepted. */
01695   return rx_comm_manager_send(mgr, params);
01696 }
01697
01698 rx_err_t rx_comm_manager_event_send(rx_comm_manager_t* mgr, const rx_comm_send_params_t* params)
01699 {
01700   if (mgr == nullptr || params == nullptr) {
01701     rx_log_error(s_tag, "Event enqueue failed: NULL arg");
01702     return k_rx_err_invalid_arg;
01703   }
01704
01705   if (!mgr->initialized) {
01706     rx_log_error(s_tag, "Event enqueue failed: not initialized");
01707     return k_rx_err_invalid_state;
01708   }
01709
01710   if (params->payload_len > k_frame_max_payload) {
01711     rx_log_error_val(s_tag, "Event enqueue failed: payload too large", params->payload_len);
01712     return k_rx_err_invalid_arg;
01713   }
01714
01715   if (params->payload_len > 0 && params->payload == nullptr) {
01716     rx_log_error(s_tag, "Event enqueue failed: NULL payload with nonzero len");
01717     return k_rx_err_invalid_arg;
01718   }
01719
```

```
01720   /* Check queue capacity */
01721   if (mgr->event_queue_count >= k_comm_event_queue_depth) {
01722     rx_log_error_val(s_tag, "Event queue full, depth", (uint8_t)mgr->event_queue_count);
01723     return k_rx_err_no_mem;
01724   }
01725
01726   /* Enqueue: copy payload into static slot */
01727   rx_comm_event_entry_t* entry = &mgr->event_queue[mgr->event_queue_head];
01728   entry->channel        = params->channel;
01729   entry->type           = params->type;
01730   entry->flags          = params->flags;
01731   entry->payload_len    = params->payload_len;
01732   entry->retries        = 0;
01733   entry->occupied       = true;
01734
01735   if (params->payload != nullptr && params->payload_len > 0) {
01736     for (uint32_t i = 0; i < params->payload_len; i++) {
01737       entry->payload[i] = params->payload[i];
01738     }
01739   }
01740
01741   mgr->event_queue_head = (mgr->event_queue_head + 1) % k_comm_event_queue_depth;
01742   mgr->event_queue_count++;
01743
01744   rx_log_debug_val(s_tag, "Event enqueued, queue depth", mgr->event_queue_count);
01745   return k_rx_ok;
01746 }
01747
01748 /*
      ============================================================================
01749  * Heartbeat API
01750  *
      ============================================================================
01751  */
01752
01753 rx_err_t rx_comm_manager_link_status(const rx_comm_manager_t* mgr,
01754                                      rx_comm_channel_t        channel,
01755                                      rx_comm_link_status_t*   status)
01756 {
01757   if (mgr == nullptr || status == nullptr) {
01758     rx_log_error(s_tag, "Link status query: NULL arg");
01759     return k_rx_err_invalid_arg;
01760   }
01761
01762   if (!mgr->initialized) {
01763     rx_log_error(s_tag, "Link status query: uninitialized manager");
01764     return k_rx_err_invalid_state;
01765   }
01766
01767   if (channel >= k_comm_channel_count) {
01768     rx_log_error(s_tag, "Link status query: invalid channel");
01769     return k_rx_err_invalid_arg;
01770   }
01771
01772   *status = mgr->heartbeat[channel].status;
01773   return k_rx_ok;
01774 }
01775
01776 /**
01777  * @brief Query if communication channel is ready for send/receive operations
01778  *
01779  * @details
01780  * Checks if specified channel is configured and ready for communication. USB channel
01781  * readiness depends on USB enumeration and configuration by host. SPI channel is
01782  * ready immediately if handle is configured (no enumeration required).
01783  *
01784  * **Readiness Criteria**:
01785  * - **USB**: usb_handle non-NULL AND USB device enumerated and configured by host
01786  * - **SPI**: spi_handle non-NULL (always ready if configured)
01787  *
01788  * **Use Cases**:
01789  * - Check readiness before sending to avoid errors
01790  * - Implement fallback logic (try USB, fall back to SPI)
01791  * - Log channel availability status
01792  *
01793  *
01794  *
01795  *
01796  *
01797  * @pre mgr must be non-NULL and initialized
01798  * @pre ready must be non-NULL
01799  * @pre channel must be valid (USB or SPI)
01800  * @post *ready set to true or false based on channel state
01801  * @post On error, *ready set to false (safe default)
01802  *
01803  * @note USB readiness can change at runtime (host connects/disconnects)
01804  * @note SPI readiness is static (always ready if handle configured)
```

```
01805  * @note This function does NOT initiate communication - only queries state
01806  *
01807  * @par Example - Send with Fallback:
01808  * @code{.c}
01809  * uint8_t payload[16] = { ... };
01810  *
01811  * // Try USB first
01812  * bool usb_ready = false;
01813  * rx_err_t err = rx_comm_manager_channel_ready(&g_comm_mgr,
01814  *                                             k_comm_channel_usb,
01815  *                                             &usb_ready);
01816  * if (err == k_rx_ok && usb_ready) {
01817  *   err = rx_comm_manager_respond(&g_comm_mgr,
01818  *                                k_comm_channel_usb,
01819  *                                payload,
01820  *                                sizeof(payload));
01821  * } else {
01822  *   // Fallback to SPI
01823  *   bool spi_ready = false;
01824  *   err = rx_comm_manager_channel_ready(&g_comm_mgr,
01825  *                                       k_comm_channel_spi,
01826  *                                       &spi_ready);
01827  *   if (err == k_rx_ok && spi_ready) {
01828  *     err = rx_comm_manager_respond(&g_comm_mgr,
01829  *                                   k_comm_channel_spi,
01830  *                                   payload,
01831  *                                   sizeof(payload));
01832  *   }
01833  * }
01834  * @endcode
01835  *
01836  * @par Example - Log Channel Status:
01837  * @code{.c}
01838  * void log_channel_status(void)
01839  * {
01840  *   bool usb_ready = false, spi_ready = false;
01841  *
01842  *   rx_comm_manager_channel_ready(&g_comm_mgr, k_comm_channel_usb, &usb_ready);
01843  *   rx_comm_manager_channel_ready(&g_comm_mgr, k_comm_channel_spi, &spi_ready);
01844  *
01845  *   printf("USB: %s, SPI: %s\n",
01846  *          usb_ready ? "READY" : "NOT READY",
01847  *          spi_ready ? "READY" : "NOT READY");
01848  * }
01849  * @endcode
01850  *
01851  * @see rx_usb_is_configured() USB enumeration status
01852  * @see rx_comm_manager_send() Send frame on channel
01853  *
01854  * @since Version 1.0.0
01855  */
01856 rx_err_t
01857 rx_comm_manager_channel_ready(rx_comm_manager_t* mgr, rx_comm_channel_t channel, bool* ready)
01858 {
01859   /* Rule 5: Pre-condition validation */
01860   if (mgr == nullptr || ready == nullptr) {
01861     if (ready != nullptr) {
01862       *ready = false;
01863     }
01864     return k_rx_err_invalid_arg;
01865   }
01866   if (!mgr->initialized) {
01867     *ready = false;
01868     return k_rx_err_invalid_state;
01869   }
01870
01871   switch (channel) {
01872     case k_comm_channel_uart:
01873       *ready = (bool)(mgr->uart_handle != nullptr);
01874       break;
01875
01876     case k_comm_channel_spi:
01877       *ready = (bool)(mgr->spi_handle != nullptr);
01878       break;
01879
01880     case k_comm_channel_i2c:
01881       *ready = (bool)(mgr->i2c_handle != nullptr);
01882       break;
01883
01884     default:
01885       *ready = false;
01886       rx_log_error(s_tag, "Channel ready: invalid channel");
01887       return k_rx_err_invalid_arg;
01888   }
01889
01890   return k_rx_ok;
01891 }
```

```
01892
01893 /**
01894  * @brief Get human-readable name for communication channel
01895  *
01896  * @details
01897  * Returns static string name for given channel enum value. Used for logging,
01898  * debugging, and user interface display. Always returns a valid string (never NULL).
01899  *
01900  * **Returned Strings**:
01901  * - k_comm_channel_usb -> "USB"
01902  * - k_comm_channel_spi -> "SPI"
01903  * - Invalid/unknown -> "UNKNOWN"
01904  *
01905  * **Use Cases**:
01906  * - Logging channel activity
01907  * - Debugging frame traffic
01908  * - User interface display
01909  * - Error messages
01910  *
01911  *
01912  *
01913  * @pre None (accepts any input)
01914  * @post Returns pointer to static string (valid for program lifetime)
01915  *
01916  * @note Returned string is static - do not free
01917  * @note Thread-safe (returns read-only static strings)
01918  * @note Never returns NULL - always safe to use in printf
01919  *
01920  * @par Example - Logging:
01921  * @code{.c}
01922  * static void on_frame_received(rx_comm_channel_t channel,
01923  *                               const rx_frame_t* frame,
01924  *                               void* ctx)
01925  * {
01926  *   printf("Frame received on %s: type=%d len=%d\n",
01927  *          rx_comm_manager_channel_name(channel),
01928  *          frame->header.type,
01929  *          frame->header.length);
01930  * }
01931  * @endcode
01932  *
01933  * @par Example - Error Messages:
01934  * @code{.c}
01935  * rx_err_t err = rx_comm_manager_send(&g_comm_mgr, &params);
01936  * if (err != k_rx_ok) {
01937  *   fprintf(stderr, "Failed to send on %s: error %d\n",
01938  *           rx_comm_manager_channel_name(params.channel), err);
01939  * }
01940  * @endcode
01941  *
01942  * @see rx_comm_channel_t Channel enumeration
01943  *
01944  * @since Version 1.0.0
01945  */
01946 const char* rx_comm_manager_channel_name(rx_comm_channel_t channel)
01947 {
01948   switch (channel) {
01949     case k_comm_channel_uart:
01950       return "UART";
01951     case k_comm_channel_spi:
01952       return "SPI";
01953     case k_comm_channel_i2c:
01954       return "I2C";
01955     default:
01956       return "UNKNOWN";
01957   }
01958 }
01959
01960 /*
      ============================================================================
01961  * Retransmission Pass-Through
01962  *
      ============================================================================
01963  */
01964
01965 rx_err_t rx_comm_manager_process_retransmits(rx_comm_manager_t* mgr, const uint32_t current_time_ms)
01966 {
01967   if (mgr == nullptr) {
01968     return k_rx_err_invalid_arg;
01969   }
01970
01971   if (!mgr->initialized) {
01972     return k_rx_err_invalid_state;
01973   }
01974
01975   if (mgr->spi_handle == nullptr) {
01976     return k_rx_ok; /* No SPI channel configured */
```

```
01977   }
01978
01979   return rx_spi_comm_process_retransmits(mgr->spi_handle, current_time_ms);
01980 }
01981
01982 rx_err_t rx_comm_manager_set_auto_retransmit(rx_comm_manager_t*                  mgr,
01983                                              const bool                          enabled,
01984                                              const rx_spi_comm_retransmit_config_t* config)
01985 {
01986   if (mgr == nullptr) {
01987     return k_rx_err_invalid_arg;
01988   }
01989
01990   if (!mgr->initialized) {
01991     return k_rx_err_invalid_state;
01992   }
01993
01994   if (mgr->spi_handle == nullptr) {
01995     return k_rx_err_not_supported;
01996   }
01997
01998   return rx_spi_comm_set_auto_retransmit(mgr->spi_handle, enabled, config);
01999 }
```